The "Depression Trap": Genetic Risk, Labor Supply, and the Case for Precision Psychiatry*

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Abstract

We analyze how genetic risk for depression influences the joint dynamics of labor supply, health, disability, and treatment decisions using a life-cycle model and longitudinal data. We find that a one standard deviation increase in genetic risk for depression substantially reduces earnings and employment, worsens health, and increases long-term sickness and disability. We identify the "depression trap", a self-reinforcing cycle of depression and non-employment, as the central mechanism linking genetic risk to persistent labor market exit: when we place every agent in the trap, the illness clears at the same speed for both genetic groups, while the employment loss does not. As publicly funded psychotherapy is free at the point of use and supply-constrained, we evaluate prioritizing access to high-risk individuals, and find that this raises employment and lowers severe depression and improves the fiscal balance at no incremental cost to the exchequer. Adding capacity also repays its cost: an additional therapist returns several times the investment in training one, though the return falls sharply as the workforce grows.

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1 Introduction

While it has long been known that genetics plays a role in mental health disorders, it has only recently become possible to use genome sequencing data to predict *who* will suffer from depression (Murray et al., 2021). Given the role of mental health in productivity and labor supply, can genetic risk predict who will be employed and how much they will earn? And as the UK National Health Service moves toward universal newborn genotyping (Davies, 2025), could that information be used to decide who receives mental health care first, and would doing so pay for itself?

The potential economic value of such a framework lies in preventing poor mental health from developing into persistent labor-market detachment. The concern is a particularly salient one among young adults. In the first quarter of 2026, an estimated 1.012 million people aged 16–24 in the UK were not in education, employment, or training (NEET), equivalent to 13.5 percent of this age group (Office for National Statistics, 2026b). The UK government summarizes the challenge as follows:

> *"With around one million 16–24-year-olds not in education, employment or training—and poor mental health a major barrier . . . "*
>
> UK Department for Education (Department for Education, 2026)

Consistent with this assessment, one in five NEET young people reports a mental health condition, more than twice the corresponding share in 2012. Among disabled NEET young people, 42.6 percent identify mental health as their main health problem (Milburn, 2026). Poor mental health and labor-market detachment plainly reinforce one another in these figures. What is not clear from them is which way the causation runs, how much of it is attributable to whom, and whether treatment delivered early enough interrupts the process.

This paper studies how genetic risk for depression affects the joint dynamics of labor supply, health and treatment decisions of working age men in the UK. It documents new facts about the effects of genetic risk for depression on earnings, employment and health. It then estimates a structural life-cycle model of labor supply and mental health investment that matches these empirical patterns. Finally, it uses the model to evaluate a policy that gives high-risk individuals priority in the queue for publicly funded psychotherapy.

We exploit within and between family variation in genetics to estimate the effects of a Polygenic Index (PGI) for depressive symptoms on health and labor market outcomes. We show that a one standard deviation increase in genetic risk raises the probability of probable major depression by 6.9 p.p., lowers the probability of employment by 3.1 p.p. and lowers labor income by 0.12 s.d. This "genetic penalty" in earnings is partially as a result of lower educational and occupational attainment, but disability-related non-employment is the main driver. Contemporaneous labor force status explains 27 percent of the income penalty, more than education and first occupation combined, and is in turn driven by worse contemporaneous physical and mental health. We explain this pattern through the mechanism of a "depression trap": a self-reinforcing cycle where poor mental health reduces labor supply, which in turn worsens mental health. We establish each leg of this cycle causally, using two independent shocks: involuntary job loss through redundancy, and the recurrence of a depressive episode. Both shocks do more damage to men at high genetic risk. Splitting the sample at the median of the index, high-risk men are roughly 70 percent more likely to enter the trap in a given year and 40 percent less likely to leave it, so that the expected duration of a spell in the trap is 2.8 years for them against 1.7 years for low-risk men.

Our model builds that feedback in. Agents choose consumption, saving, labor supply and whether to apply for psychotherapy from age 25, retire at 65 and face health-dependent mortality thereafter. Mental and physical health follow ordered processes whose transitions depend on employment and, for mental health, on realized treatment; health in turn shifts wage offers and the utility cost of working. Human capital accumulates with employment and depreciates without it, so a spell out of work lowers the wage offer that would end it. Poor health also raises benefit entitlement out of work[1], weakening the incentive to return. Two features carry the policy analysis. Treatment is rationed rather than priced: an agent who applies is treated with probability λ_τ and otherwise waits, which is how the English system allocates talking therapy. And genetic risk enters as observed ex-ante heterogeneity alongside unobserved permanent types in productivity and health, so the model can be asked what a planner who saw the genotype would do with it.

We estimate the model using a multi-step procedure with Understanding Society data. First, we identify the discrete distribution of latent individual heterogeneity by applying the clustering approach of Bonhomme et al. (2022) to outcomes that have been residualized on genetic risk for depression. This ensures that our estimated types capture unobserved productivity and health traits that are distinct from the biological predisposition measured by the PGI. Second, we estimate the physical health process outside the model by maximum likelihood, since its transitions depend only on predetermined characteristics, and we take benefits, taxes, pensions, survival and the therapy rationing probability from external sources. Third, we estimate the mental health process jointly with preferences and wage offers by the method of simulated moments. Mental health cannot be estimated outside the model in the way physical health can, because its transitions depend on realized therapy and therapy is chosen: an agent applies precisely when he expects to deteriorate, so a transition model fitted to observed treatment would confound the effect of therapy with selection into it. Estimating it inside the model lets the application rule do the work an instrument would otherwise have to.

Our model delivers several new findings. First, we find that the dynamic effect of genetic risk for depression is important in explaining life-cycle outcomes, and that these effects are not captured by other initial characteristics. We isolate the dynamic effects of genetics by simulating the same individuals with varying genetic risk for depression for a 10-year period. Switching genetic risk alone, holding every other initial condition fixed, more than doubles unemployment, raises the prevalence of severe depression by 60 percent and leaves 22 percent less wealth at retirement, differences that are largely consistent with

[1] We do not include a distinct "disabled" employment state in our model. Instead, benefits for individuals out of the labor force depend directly on their health status. Because disabled individuals predominantly fall into severe mental or physical health states, the benefit levels for these groups implicitly capture disability payments.

reduced-form evidence for the lifetime effect of genetic risk. This highlights genetics as an important source of dynamic heterogeneity, that captures not only current health or labor market status, but also future trajectories.

Second, what genetic risk governs is escape from the trap, not the course of the illness. We place every agent simultaneously in severe depression and non-employment, the two components of the trap, and simulate ten years of recovery. Severe depression returns to baseline within four years, and it does so at nearly the same speed for both genetic groups. Employment does not return. A decade after the shock, non-employment among high-risk men is still 23.9 percent above their own baseline against 6.0 percent for low-risk men, and the prevalence of the trap 24.5 against 6.9 percent. The divergence widens among agents who were employed and in healthy mental health before the shock, so who was already unwell does not explain it. What persists is the labor-market consequence of an illness that has itself resolved: human capital lost during the spell lowers the offer that would end it, non-employment feeds back into the risk of relapse, and because high-risk agents sit closer to the threshold for a severe episode, the same feedback carries more of them across it.

Third, the policy instrument that follows from this is a queue position, not a price. Talking therapy in England is free at the point of use and rationed by a queue: in 2023–24 the NHS Talking Therapies programme received 1.83 million referrals and started 1.26 million courses of treatment, so what binds is capacity (NHS England, 2024b). We therefore evaluate reordering the queue on genetic risk. Giving high-risk individuals priority while holding the number of treatment slots fixed raises employment, lowers severe depression, and improves the fiscal balance by £19 per person in present value at the highest intensity we consider, while treating no more people than the benchmark. Because nothing is added to the system, the exchequer pays nothing for it. The gain comes from targeting, not speed: accelerating everyone equally at fixed capacity returns exactly zero, as market clearing requires. Expanding capacity instead delivers a nearly identical gross return, but once the additional courses are costed it is worth substantially less than reallocation. The policy reallocates and is not a Pareto improvement: at the highest intensity the wait facing low-risk patients lengthens from around five months to over four years. We report that distributional cost alongside the gain. On a post-COVID initial distribution the returns are larger at every intensity and their source shifts from rationing low-risk access to treating high risk.

Fourth, the queue can be lengthened as well as reordered, and that also repays its cost. Raising the treatment probability uniformly, so that nobody is given priority, returns £494,000 per additional therapist in lifetime net fiscal terms at a five percent expansion, against a capitalised training investment of roughly £122,000. The return falls to £182,000 per therapist by a thirty percent expansion, since each additional course goes to a progressively less severe case, but it exceeds the cost of training at every scale we examine. Both halves of that result carry over to any costing of therapy capacity. Reduced benefit expenditure, not additional tax revenue, is what the exchequer recovers, by a factor of between 1.5 and 1.7. And expansion is progressive even though it targets nobody: severe depression falls more than five times as much among high-risk men, who apply more often and so absorb most of the new capacity unprompted.

Our paper makes three primary contributions to the health, labor and genetic psychiatry literatures. First, we establish that genetic predisposition for depression is a fundamental driver of life-cycle inequality in health and employment outcomes. The polygenic index supplies a biologically grounded measure of heterogeneity that captures latent vulnerabilities traditional observables miss. Second, we formalize and quantify the "depression trap". We provide causal evidence on each leg of the feedback loop between non-employment and mental illness, using two independent shocks, and show that both cost high-risk men more employment while doing them no more clinical damage. Finally, we provide the first structural evaluation of precision psychiatry. Proposals for genetic targeting are usually framed as changes in who is offered treatment. We show that in a capacity-constrained public system the relevant margin is instead who is treated first, that reordering the queue on genetic risk yields a fiscal return without requiring an additional therapist, and that where therapists are added, the return to each one can be set against what it costs to train them.

The paper is organized as follows. Section 2 discusses how this paper relates to the existing literature. Section 3 presents the data and our reduced-form evidence. Section 4 presents our model and Section 5 explains our multi-step identification and estimation procedure. Section 6 presents the counterfactual and policy analysis and Section 7 concludes. Various robustness checks and additional results are presented in Appendices A-H.

2 Related Literature

This paper bridges the gap between the genetic psychiatry and structural labor and health literatures by incorporating genetics, health, labor supply and endogenous treatment in a rich life-cycle labor supply model.

This paper is most closely related to the burgeoning literature incorporating mental health in structural models of labor supply. However, our model is unique in its focus on the two-way relationship between the employment and mental health and in incorporating genetics as a source of ex-ante heterogeneity. Jolivet and Postel-Vinay (2025) feature a two-way relationship between work and mental health in model of occupation choice. Stressful occupations are more costly for workers with poor mental health and feature worse health transition probabilities. Cronin et al. (2024) use a rich model of mental health treatment and labor supply to explore why psychotherapy is under-utilized relative to antidepressants despite having larger treatment effects. Abramson et al. (2024) incorporate mental health in a consumption-savings model with endogenous treatment. They focus on how the pessimism associated with poor mental health can induce individuals to under-invest in risky assets and pursue less challenging occupations, thus negatively impacting asset accumulation. These three papers allow mental health to affect labor supply, but only Jolivet and Postel-Vinay (2025) models the effect of employment on mental health. Given that their focus is on occupation choice, they do not model the persistently non-employed or mental health treatment. To model persistent non-employment in our model, we jointly model physical and mental health dynamics, and incorporate disability benefits which vary with health status.

This paper also relates to the larger literature on physical health and labor supply. A branch of the literature focuses on the effect of physical health and retirement decisions (French, 2005; French and Jones, 2011) or on long-term earnings and welfare (Hosseini et al., 2021; De Nardi et al., 2024). This literature generally does not consider the feedback effects of employment on health. Exceptions include Salvati (2025), who incorporates a bidirectional relationship between work and health for women near retirement. Models of physical health and labor supply typically do not consider mental health dynamics. Sauer et al. (2025) is an exception, modelling joint physical and mental health dynamics, as well as labor supply for women. Interestingly, they find negative effects of employment on health, in contrast with evidence from RCTs, exogenous plant closures and longitudinal analyses (Picchio and Ubaldi, 2024; Hussam et al., 2022; Sterud et al., 2025). However, the focus of these studies is on women, and findings do suggest that the protective mental and physical health effects of employment are larger for men (Picchio and Ubaldi, 2024).

We also contribute to the 'Genoeconomics' literature, which links molecular genetic data to economic surveys to study the interaction between genetic and environmental effects. We show how genetic risk for depression predicts economic outcomes and demonstrate how it reinforces health and economic shocks. Biroli et al. (2022) provides a useful introduction to this literature. The literature relies on Genome-Wide Association Studies (GWAS) to estimate the association between genetic variants and a trait of interest in a large population (Uffelmann et al., 2021). These estimated associations can then be used to generate individual-level predictors of a trait of interest, called Polygenic Scores (PGS) or Indices (PGI). Thus far, this literature has almost entirely focussed on educational attainment (EA) as a trait of interest (Okbay et al., 2022), which has been shown to be a strong predictor of asset accumulation (Barth et al., 2020), labor market outcomes (Papageorge and Thom, 2020), as well as parental investment and skill formation (Bolt et al., 2025). There have been limited attempts to incorporate genetics in structural models, and PGIs other than EA have thus far received very little attention in economics. Barth et al. (2022) incorporate the EA PGI in a consumption-savings model of asset accumulation. Houmark et al. (2024), Bolt et al. (2025) and Agostinelli and Weingarten (2025) model how genetics affect parental investment and child skill formation, with the latter incorporating genetic predictors of ADHD. Jeong et al. (2025) use reduced-form approaches to explore a genetic predictor of Alzheimer's, while De Nardi et al. (2024) briefly discusses how PGIs for educational attainment, BMI, smoking and longevity predict physical health. While Amin et al. (2019) explore gene-environment associations for a depressive symptoms PGI with education as the outcome, our is the first study to link it to labor market outcomes, mental health treatment and dynamics, and to explore the role of targeted interventions.

This paper also bridges the gap between labor economics and precision psychiatry literatures. We demonstrate that using genomic information to decide the order in which patients are treated improves both clinical and employment outcomes, and that the resulting labor supply response and reduced benefit expenditure make the intervention fiscally self-sustaining. Where capacity rather than price is the binding constraint, it requires no expenditure at all. While there is a large, long-established body of evidence for the effectiveness of psychotherapy (Baranov et al., 2020; Cuijpers et al., 2014), recent research has highlighted how PGIs can be used to identify high-risk patients and eventually develop personalized interventions (Torkamani et al., 2018). To date, there remains little evidence of how treatments could be *adapted* depending on an individual's genetics, but a recent RCT suggests that the treatment effect of an Cognitive Behavioural Therapy intervention for depressed patients does does not appear to vary depending on their depression PGI (Bäckman et al., 2025). This suggests that high-risk individuals are not 'predetermined' to be depressed, but that treatments can be effective. The take-up of psychotherapy, however remains low. Cronin et al. (2024) consider various mechanisms and conclude that the under-utilization is psychotherapy is not primarily due to financial costs or time constraints, but likely relates to stigma or difficulties opening up to a stranger. To overcome these constraints Breza et al. (2026) conduct an RCT in which they offer financial incentives to attend therapy, which increases take-up only among symptomatic patients, consistent with the broader evidence that financial incentives can increase mental health treatment engagement (Khazanov et al., 2022). Where therapy is already free at the point of use, however, price is not the margin that binds, and our policy exercise therefore works on the one that does, which is the order of the queue.

3 Empirical Insights

This section establishes four facts that motivate the structural model of Section 4. First, genetic risk for depression carries a substantial earnings penalty that particularly operates through withdrawal from the labour market into disability. Second, the penalty is dynamic: men at high genetic risk are more likely to develop depression and to leave employment. They are less likely to recover from either, and particularly struggle to leave the joint state of non-employment and depression, which we term the "depression trap". Third, each leg of that trap is causal. We identify the effect of employment on mental health from involuntary job loss, and the effect of mental health on employment from the recurrence of a depressive episode. Fourth, genetic risk exacerbates the effects of shocks to mental health and employment: individual with high risk for depression experience larger declines in mental health following redundancy and larger declines in employment following the recurrence of depression. These facts motivate our structural model, which aims to reproduce the dynamics of the depression trap and explore policy solutions.

3.1 Data and Measurement

We use the UK Household Longitudinal Study and its predecessor the British Household Panel Survey, spanning 1991–2023, which link molecular genetic data to an annual panel of health and labor market histories. Mental health is measured by the "caseness" scoring of the GHQ-12, a well-validated screen for depression (Wojujutari et al., 2024); following the psychiatric literature we use thresholds of two and four symptoms to denote mild depressive symptoms and probable major depressive disorder (Anjara et al., 2020). Of 9,220 genotyped individuals, our reduced-form sample is the roughly 2,000 men aged 25–55; model estimation extends the age range to 85. Women are excluded throughout, and not for want of data: they contribute more genotyped person-waves than men on every margin. Appendix B reports what each exercise below gives for women, in the subsection corresponding to each, and Appendix C sets out the reasons for the restriction, which turn on fertility.

3.2 Genetics Data

To measure genetic predisposition to depressive symptoms, we use a polygenic index (PGI). A PGI is an individual-level predictor constructed from genotype data. It can be interpreted as a weighted sum of genetic variants. Let $SNP_{i,j} \in \{0, 1, 2\}$ denote the number of effect alleles carried by individual i at variant j, where the allele may increase or decrease depression risk. The PGI aggregates many such variants with small individual effects into a single summary measure:

$$PGI_i = \sum_{j=1}^{J} \beta_j^{GWAS} SNP_{i,j}. \tag{1}$$

The weights β_j^{GWAS} are estimated in an external GWAS discovery sample that is separate from our UKHLS target sample. In the GWAS, the association between each variant j and the trait of interest is estimated using regressions of the form

$$Y_k = \beta_j^{GWAS} SNP_{k,j} + \gamma \mathbf{X}_k + u_{k,j}, \tag{2}$$

where k indexes individuals in the discovery sample, Y_k is depressive symptoms, and $\mathbf{X}_k$ includes standard GWAS controls (sex, age polynomials, and genetic principal components) to adjust for population structure.

In our main analysis, we use weights from Baselmans et al. (2019), which studies depressive symptoms. The GWAS discovery sample does not overlap with the UKHLS target sample. As additional controls in some specifications, we also use PGIs constructed from GWAS of educational attainment (Okbay et al., 2022) and self-reported health (Harris et al., 2017). Both the discovery and target samples are restricted to individuals of European-like White ancestry. We impose this restriction because PGIs do not transport well across ancestry groups and because non-European GWAS sample sizes remain more limited. See Price et al. (2006) for discussion of population structure and related concerns. Details on the genetic data and PGI measurement are provided in Appendix A.

Men above the median of genetic risk are worse off on every health and labor market margin (Appendix Table 9), but they do not differ in parental background: neither paternal occupation ($p = 0.31$) nor maternal education ($p = 0.26$) differs across the groups, so the index does not appear to proxy dynastic advantage. Appendix E shows that controlling for parental background and for the educational-attainment and physical-health polygenic indices, and estimating family fixed-effects models on linked relatives, leaves the estimates stable, as Okbay et al. (2022) find for depressive symptoms in a much larger sample.

3.3 The Genetic Penalty

To identify the relationship between genetic risk for depression and subsequent labor market outcomes, we estimate the following specification for individual i at time t:

$$Y_{i,t} = \alpha + \beta G_i + \mathbf{X}'_{i,t}\gamma + \mathbf{M}'_{i,t}\theta + \epsilon_{i,t}, \tag{3}$$

where Y_i is standardized gross earnings (with earnings coded as 0 for the non-employed), G_i is the standardized depression polygenic index and $\mathbf{X}_{i,t}$ collects the ten genetic principal components with age, wave and interview-year fixed effects; standard errors are clustered on the individual. To locate the penalty in the life cycle we add mediators $\mathbf{M}_{i,t}$ in four chronological blocks: educational attainment, first occupation, contemporaneous labor force status, and current occupation. We report the penalty per standard deviation of the index; Appendix B reports the median-split version, which gives the same picture in units directly comparable to the descriptive and transition tables.

Table 1 reports the ladder. A one standard deviation increase in genetic risk lowers gross labor income by 0.123 standard deviations (s.e. 0.022). Education absorbs 19 percent of that association and first occupation a further 6 percent, consistent with genetic risk impairing human capital accumulation early in life. The largest single block, however, is contemporaneous labor force status, which absorbs 27 percent on its own, more than education and first occupation combined. The penalty is therefore not primarily about what high-risk men are paid, but about whether they are working at all.

Table 1: The Genetic Earnings Penalty and Its Channels

Mediators included	Effect on labor income (s.d.)	Share of baseline (%)	Explained by this block (%)
Baseline	-0.1229***	100.0	–
	(0.0222)		
+ Education	-0.0992***	80.7	19.3
	(0.0201)		
+ First occupation	-0.0921***	75.0	5.8
	(0.0196)		
+ Labor force status	-0.0584***	47.5	27.4
	(0.0185)		
+ Current occupation	-0.0420***	34.2	13.3
	(0.0163)		

Notes: Each row is the coefficient on genetic risk in a regression of standardized gross labor income, adding one block of mediators at a time. The index is continuous and standardised, oriented so that higher values denote greater liability, so each coefficient is the effect of a one standard deviation increase in genetic risk. Appendix B reports the continuous per-standard-deviation version. All specifications control for the first ten genetic principal components and for age, survey-wave and interview-year fixed effects. Column (2) expresses each coefficient as a percentage of the baseline, column (3) the share absorbed by the block added in that row. Contemporaneous mental health is deliberately excluded: the blocks above are predetermined relative to current earnings whereas mental health is a contemporaneous outcome of genetic liability, so conditioning on it here would net out part of the effect being decomposed; its share is reported in Table 2. Standard errors clustered on the individual. N = 19,995 person-waves. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Which Margin Carries the Genetic Penalty, and How Much Runs Through Mental Health

Outcome	Effect of +1 s.d. of genetic risk	Share explained by mental health (%)	95% CI
Depressed (pp)	6.908***	–	
	(0.530)		
Employed (pp)	-3.098***	48	[33, 75]
	(0.525)		
Long-term sick or disabled (pp)	1.900***	53	[40, 81]
	(0.375)		
Labor income (s.d.)	-0.123***	23	[15, 32]
	(0.022)		
Log gross pay, if employed (log pts.)	-0.054***	7	[0, 15]
	(0.013)		
Holds a degree (pp)	-2.785**	≈ 0	
	(1.119)		

Notes: Column (1) is the effect of genetic risk on the outcome named,in the units given in the row label. The index is continuous and standardised, oriented so that higher values denote greater liability, so each coefficient is the effect of a one standard deviation increase in genetic risk. Column (2) is the percentage of that gap absorbed when contemporaneous GHQ-12 distress is added, with a cluster-bootstrap interval in column (3); ≈ 0 denotes no attenuation. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. The depression row has no share because it is the mediator itself. Appendix B reports every outcome and the continuous per-standard-deviation version. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 takes each margin as a separate outcome, and the first row fixes the scale of everything below it: a one standard deviation increase in genetic risk raises the probability of probable major depression by 6.9 percentage points (s.e. 0.5). It lowers the probability of employment by 3.1 points (s.e. 0.5), and that withdrawal is concentrated in long-term sickness and disability, at 1.9 points (s.e. 0.4). Most of the employment gap is accounted for by this one destination: these men leave the labor force through ill health rather than search for work. Conditional on working, gross pay falls by only 0.054 log points (s.e. 0.013), and the degree gap of 2.8 points (s.e. 1.1) is smaller in relative terms.

The third column reports how much of each association operates through contemporaneous mental health, and the pattern is orderly. About half of the labor-supply effects run through it: 48 percent for employment and 53 percent for long-term sickness. A fifth of the income effect does, 7 percent of the intensive margin, and essentially none of the degree effect, which is what one would expect of an outcome settled before the distress we observe. This is the pathway the structural model is built around: genetic predisposition manifests as poor mental health, which triggers withdrawal from the labor force through disability rather than through unemployment or lower wages.

The penalty is not specific to men. Estimated on women, it is if anything larger on the extensive margin, at 4.0 percentage points of employment against 2.6 for men on a common sample, with comparable mediation through mental health. The difference is where the withdrawal goes: women also move into caring for home and family, a state men essentially do not occupy, which accounts for about thirty percent of their employment penalty. Appendix B.4 reports the full comparison, and Appendix C explains why that channel, and the fertility behind it, is what keeps women out of the structural model.

3.4 The Depression Trap

The relationship between employment and mental health runs in both directions. Meta-analyses of studies exploiting exogenous variation in employment find that employment improves men's mental health by about 0.1 standard deviations (controlling for income) (Picchio and Ubaldi, 2024), and RCT evidence puts the non-pecuniary benefit at 0.23 s.d. (Hussam et al., 2022). In the other direction, Frijters et al. (2014) instrument mental health with the death of a close friend and find that a one standard deviation deterioration reduces employment by 30 percentage points, while panel estimates put the effect of disorder onset on exit from employment at 7–18 points (Mitra and Jones, 2017). Appendix D sets out the clinical channels the structural model incorporates: cognitive impairment that lowers the wage offer, fatigue and anhedonia that raise the disutility of effort, and physical-health spillovers. Both legs appear in our own panel, conditional on permanent heterogeneity and past income (Appendix Table 10), but they are associations, which is why Section 3.5 identifies each leg from a shock instead.

A relationship that runs in both directions admits a self-reinforcing state: out of work and depressed, with each condition sustaining the other. The question this paper asks is whether genetic predisposition governs how easily men enter that state and how long they remain. Table 3 addresses all three sets of transitions at once, splitting the genotyped sample at the median of the polygenic index, estimating the transition model separately within each group, and averaging both fits over the same pooled covariate distribution, so that column (3) reflects a difference in behavior rather than in composition.

The gradient appears on every margin. Men at high genetic risk are 3.5 percentage points more likely to become depressed in a given year, on a base of 9.3 percent, and 15.3 points less likely to recover if they are. They are 0.7 points more likely to leave employment, on a base of 1.6 percent, and 13.7 points less likely to re-enter it. Panel C combines the two: high-risk men are 0.7 points more likely to enter the trap, a 72 percent higher entry rate, and 24.4 points less likely to leave it, so the expected spell lasts 1.7 years at low genetic risk against 2.8 years at high genetic risk. It is the exit margin that accounts for most of this. Entry rates differ by less than a percentage point in absolute terms while persistence differs by more than a year, which is why we describe the phenomenon as a trap rather than as a difference in incidence, and it is the feature the structural model must reproduce. The gradient is not a restatement of the permanent heterogeneity that model already carries: adding the five latent types of Appendix F as controls leaves all six margins intact (Table 12).

Every one of these gradients replicates for women, and entry into the trap is more strongly graded for them than for men. Persistence is not: the exit differential is a third the size, and it is sensitive to whether spells of caring for home and family are counted as non-employment. Appendix B.3 reports the women's transitions and Appendix C argues that the same observed state means something different when it coincides with caring for a young child.

Table 3: Mental Health, Employment and "Depression Trap" Dynamics by Genetic Risk

Annual transition probability (%)	Low genetic risk (1)	High genetic risk (2)	Difference (2)−(1)
Panel A: Mental health dynamics			
Onset (healthy → depressed)	9.33	12.84	3.51***
	(0.49)	(0.61)	(0.78)
Recovery (depressed → healthy)	64.39	49.07	-15.31***
	(2.13)	(1.69)	(2.72)
Panel B: Employment dynamics			
Exit (employed → non-employed)	1.56	2.27	0.71**
	(0.18)	(0.21)	(0.28)
Entry (non-employed → employed)	38.76	25.06	-13.70***
	(3.89)	(2.19)	(4.47)
Panel C: "Depression trap" dynamics			
Entry into trap	0.96	1.65	0.69***
	(0.15)	(0.17)	(0.23)
Exit from trap	60.21	35.77	-24.44***
	(6.67)	(3.81)	(7.68)

Notes: Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit on genotyped UKHLS men aged 25–55 observed for at least five waves. Genetic risk is a median split of the depression polygenic index. The model is estimated separately within each risk group and both fits averaged over the same pooled covariate distribution, so column (3) is a difference in behaviour, not composition; its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Each cell conditions on the origin state only, so entries are marginal with respect to the other lagged state. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 4; the "depression trap" is non-employment jointly with caseness ≥ 4. Standard errors clustered on the individual. All panels use 15,206 person-waves on 1,534 individuals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Causal Evidence on the Two Legs of the Trap

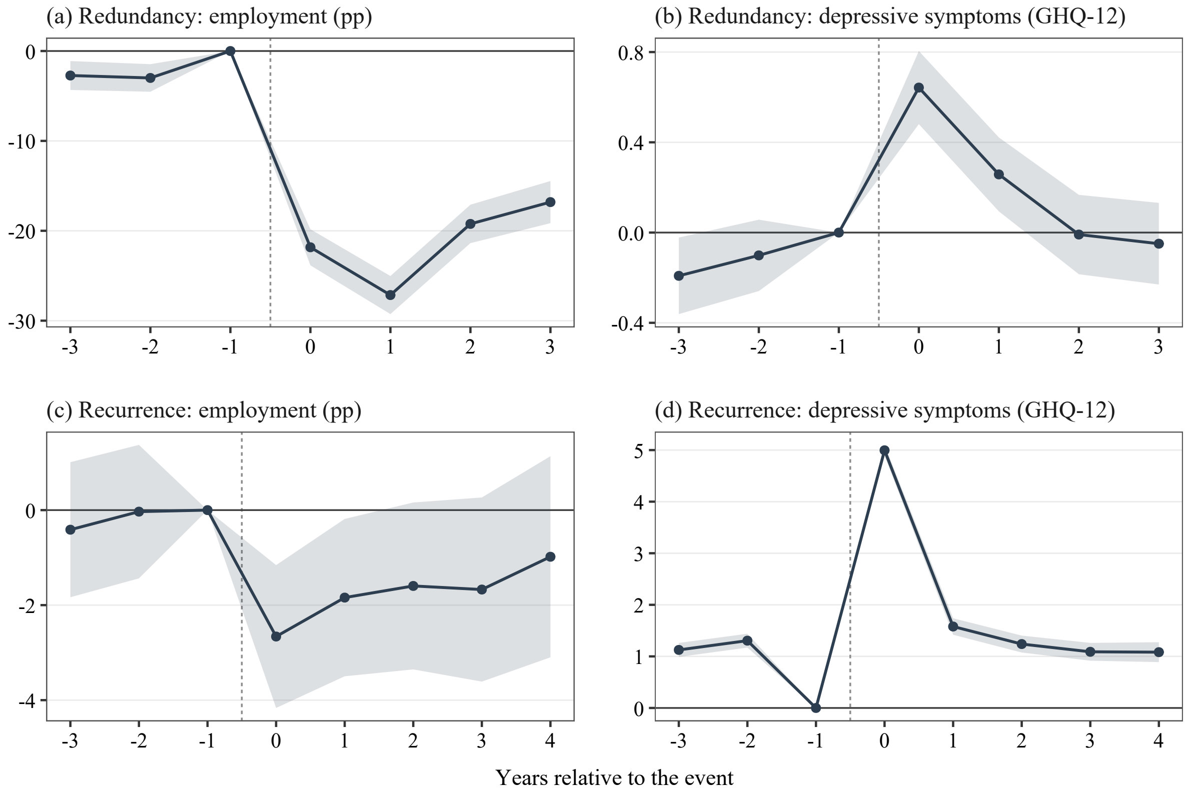
The evidence so far is associational. This subsection identifies each leg of the trap from a separate shock. Specifications, sample construction and the full battery of robustness checks are reported in Appendices B.5 and B.6.

From job loss to mental health: redundancy

To identify the effect of employment on mental health we use *redundancy*. In UK employment law a redundancy is a dismissal for which the employee is not at fault: it arises when an employer closes a site, ceases an activity, or reduces its headcount, so that the *position* is eliminated rather than the worker replaced.[2] It is therefore a firm-side event, driven by the employer's demand for labour rather than by anything about the individual worker, which is what makes it a candidate for exogenous variation in employment in the tradition of the plant-closure literature (Picchio and Ubaldi, 2024). We date it from the reason-for-job-ending questions in the two surveys. We estimate an interaction-weighted event study (Sun and Abraham, 2021) comparing men made redundant to men not yet made redundant, on 3,515 treated men aged 25–55. Employment falls by 21.8 percentage points on impact (s.e. 1.0), reaches −27.2

[2] Dismissal for poor performance, misconduct or ill health is legally distinct and is recorded separately in both surveys, as is voluntary quitting. The natural North American analogue is a permanent layoff arising from a plant closing or a firm downsizing, rather than a discharge for cause.

Figure 1: Two Shocks, Two Legs of the Depression Trap



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both use the interaction-weighted estimator of Sun and Abraham (2021), and both are estimated on men aged 25–55. Panels (a) and (b) compare men made redundant to men not yet made redundant, on 3,515 treated men observed over a ±3-year window, controlling for age, its square, and interview-year and education fixed effects. Panels (c) and (d) compare men who suffer a recurrence of a depressive episode to men who have not yet done so, on 3,114 treated men in panel (c) and 3,121 in panel (d), which does not require the employment measure; these control for age, its square, and wave fixed effects. An episode is a crossing of the GHQ-12 caseness threshold of four and a recurrence is a second crossing after an intervening spell below it. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse. In panel (d) the pre-event coefficients are positive by construction, because the reference year is mechanically a below-threshold wave whereas earlier years are not; the informative content is the size of the jump at the event and its decay. Appendices B.5 and B.6 report specifications and robustness.

a year later, and is still 16.9 points below baseline after three years; monthly earnings fall by £640 on impact and £800 a year later. GHQ-12 caseness rises by 0.64 points on impact (s.e. 0.08) and remains 0.26 points higher a year later. Panels (a) and (b) of Figure 1 plot the two paths. Both estimates are insensitive to specification: across twelve combinations of estimator, comparison group and anticipation window, the mental-health effect lies between 0.45 and 0.83 points and is significant in every one. A placebo event assigned to men never made redundant gives 0.06 (s.e. 0.07), and randomization inference over four hundred permutations of the event date gives $p < 0.001$. Appendix B.5 reports the battery.

The magnitude is in line with the literature this design speaks to. Scaled by the pre-event dispersion of GHQ-12 caseness among the treated, the impact is 0.22 standard deviations, and 0.16 to 0.22 across the heterogeneity-robust estimators. Meta-analyses of employment shocks report about 0.10 standard deviations for men (Picchio and Ubaldi, 2024) and the experimental estimate of the non-pecuniary benefit of work is 0.23 (Hussam et al., 2022); for scale, antidepressants achieve about 0.30 and psychotherapy about 0.60 (Cipriani et al., 2018; Hofmann et al., 2012).

The central threat here is reverse causality: if deteriorating mental health causes job loss, the event study attributes to redundancy what mental health caused. Panel (b) is the direct answer. The pre-event coefficients are -0.19 and -0.10 points three and two years out, small against an impact of 0.64, and almost all of the movement occurs at the event and after it. They are not jointly zero, however, and the direction is what matters: caseness is *lower* three and two years before the event than in the year immediately preceding it, so distress was rising into the job loss rather than away from it. That is the signature of anticipation of an announced redundancy, not of selection, and it makes the estimate conservative, because the reference year is already contaminated. Treating that year as treated confirms it: moving the reference back one year raises the impact from 0.64 to 0.73, and back two years to 0.83. Appendix B.5 reports the anticipation and placebo evidence in full.

The employment path does have a pre-trend, 2.7 points below the reference year three years out (s.e. 0.8), but its sign is favorable: employment was rising into the job loss, so extrapolating that trend through the event would predict a larger drop than we measure.

This design transfers to women. Estimated on 2,803 treated women, employment falls by 22.4 points on impact (s.e. 1.2) and caseness rises by 0.54 (s.e. 0.10), against 21.8 and 0.64 for men, with pre-event paths of the same size; Appendix B.5 plots the two series together.

From depression to employment: recurrence

To estimate the reverse leg, the effect of depression on employment, we exploit quasi-random variation in the timing of depression recurrence among individuals with a history of the illness. A first episode cannot supply this variation, because onset is provoked by adverse circumstances, often the very labor market shock whose effect one is trying to estimate: in a population-based twin study the odds of onset in the month of a severe life event are 5.64, and 3.58 within monozygotic pairs, which nets out shared genes and environment, and the qualifying events include job loss and marital breakdown (Kendler et al., 1999).

Recurrence is different. Under the *kindling* hypothesis (Post, 1992), a first onset typically requires a major stressor but a recurrence does not: the depressogenic effect of severe life events is far weaker for recurrences than for first onsets (Kendler et al., 2000), a pattern confirmed meta-analytically (Stroud et al., 2008), and conditional on clinical history the timing of a recurrence is close to unpredictable, with the best prognostic models discriminating at only 0.59–0.72 (Moriarty et al., 2021). Recurrence timing is as close as the natural history of depression comes to a clock that runs independently of an individual's economic circumstances.

Both parts of that premise hold in our data. The first, that recurrence is far less provoked than onset, can be tested within an individual rather than across individuals: take the waves in which a person is well, and so at risk of an onset, and compare those falling before the first episode with those falling after recovery from it, absorbing a separate fixed effect for the person in each regime. A separation or divorce raises the probability of a first onset by 28.1 percentage points (s.e. 8.2) but of a recurrence by only 3.0 (s.e. 7.6); a redundancy raises the first by 9.2 points (s.e. 1.8) and the second by 4.9 (s.e. 1.3).

Both differences are statistically significant at the five percent level ($t = 2.4$ and $t = 2.7$): the two events Kendler et al. (1999) identify as canonical triggers of an onset are substantially weaker triggers of a recurrence. The second part, that timing is unpredictable from an individual's economic position, holds directly: a redundancy in the previous year does not predict a recurrence ($p = 0.61$), and predetermined economic covariates add 0.001 to the R^2 of a recurrence model that already conditions on clinical history.

To our knowledge this is the first use of recurrence timing as identifying variation in economics. The existing literature imports exogeneity from outside the depression process, through bereavement (Frijters et al., 2014), natural disaster (Andersen et al., 2024), variation in treatment access (Biasi et al., 2021; Bütikofer and Salvanes, 2020), randomized psychotherapy (Lund et al., 2024) or Mendelian randomization (Campbell et al., 2022). Each identifies something adjacent to the object of interest: the effect of a bereavement, of a disaster, or of treatment, rather than of an episode. Recurrence timing identifies the employment cost of an episode itself, among the individuals for whom episodes recur, which is the population the structural model concerns.

We estimate an interaction-weighted event study (Sun and Abraham, 2021) on 3,114 male recurrences among men aged 25–55, the same age range as the redundancy design, comparing individuals who recur to those who have not yet recurred. A recurrence reduces employment by 2.7 percentage points on impact (s.e. 0.8); panels (c) and (d) of Figure 1 show the employment path alongside the design's first stage. Because a redundancy does remain a trigger of recurrence, the estimate must not rest on episodes that follow one, and it does not: excluding the 6.3 percent of recurrences falling within a year of a redundancy leaves the impact at 2.58 percentage points (s.e. 0.79). The estimate is also robust to alternative estimators, to how the comparison group is drawn, to sample restrictions, and to the definition of a recurrence; Appendix B.6 reports the battery.

The effect has faded by the second year, so the event study measures the contemporaneous cost of a relapse rather than its persistence, and we use it only for the former. The identifying assumption is likewise conditional rather than unconditional: recurrence timing must be unrelated to an individual's employment path given their clinical history, not random outright. The literature that supplies the kindling result also supplies the leading threat to it, since stress *sensitization* holds that the threshold falls with episode number so that minor stressors acquire triggering power (Monroe and Harkness, 2005), and the discriminating evidence favors it (Stroud et al., 2011); the design therefore trades observable severe triggers for unobservable minor ones. The checks above show that such triggers leave no economic trace, in either observed redundancies or predetermined covariates, and we therefore read the estimate as a lower bound on the contemporaneous employment cost of a relapse.

Unlike the redundancy design, this one does not transfer to women, and the reason is instructive. Estimated on women, employment is *rising* into the event, by 1.7 percentage points three waves before it and 2.3 points two waves before, both significantly different from zero, and the impact is a precise null. With 6,352 women's recurrences against 3,577 men's this is not a power problem: for women in this age range a substantial share of depressive episodes coincide with the employment interruptions around childbearing, so recurrence timing is not independent of the employment path in the way the design requires. Appendix B.6 reports the estimates and Appendix C the size of the interference.

Both shocks bite harder on the genetically vulnerable

The two designs give two independent opportunities to ask whether the same shock does more damage to men at higher genetic risk. Table 4 reports the answer at the year of impact,

Table 4: Gene-by-Environment: Effects at Impact, by Genetic Risk

Outcome at $k = 0$	High genetic risk (1)	Low genetic risk (2)	Difference (1)–(2)
Panel A: Redundancy			
Employment (pp)	-30.06	-18.79	-11.26**
	(3.68)	(3.38)	(5.00)
GHQ-12 caseness (0–12)	0.75	0.47	0.28
	(0.35)	(0.24)	(0.43)
Panel B: Recurrence of a depressive episode			
Employment (pp)	-7.69	-0.84	-6.85***
	(1.62)	(1.66)	(2.32)
GHQ-12 caseness (0–12)	5.79	5.67	0.12
	(0.16)	(0.18)	(0.24)

Notes: Effect in the year of the event relative to the year before, estimated separately within each half of the genetic-risk distribution and differenced. Both panels use Sun and Abraham (2021) and a median split of the depression polygenic index, residualised on the first ten genetic principal components; because the index is oriented towards well-being, high risk is the lower-index half. Panel A is the redundancy design on men aged 25–55; Panel B is the recurrence design on men aged 25–60. Column (3) differences two separate regressions rather than estimating an interaction, so its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Pre-event paths are close to identical across the two genetic groups in Panel A (-1.34 and -1.06 points of employment), so the employment gap is not a differential trend. The caseness rows show that the two groups experience shocks of near-identical severity, which is what makes the employment gap interpretable as a difference in response rather than in exposure. Standard errors clustered on the individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

splitting each treated sample at the median of the polygenic index.

Both panels tell the same story, and the mental-health rows are what make it legible. After redundancy the two genetic groups suffer almost the same deterioration in mental health, 0.75 against 0.47 points of GHQ-12 caseness, a difference of 0.28 that is not distinguishable from zero. After a recurrence they suffer episodes of near-identical clinical severity, 5.79 against 5.67, a difference of a tenth of a point. The shock is the same. The employment response is not. Redundancy costs high-risk men 30.1 percentage points of employment against 18.8 for low-risk men, a gap of 11.3 (s.e. 5.0); a recurrence costs them 7.7 points against 0.8, a gap of 6.9 (s.e. 2.3). The same illness, and the same job loss, cost one group far more work than the other.

What licenses the comparison differs across the two designs, and we are explicit about it. Redundancy is not selected on genotype: neither a cross-sectional linear probability model nor an annual hazard model with fixed effects finds any association between the polygenic index and being made redundant ($p = 0.58$ and $p = 0.79$), and the larger of the two point estimates is 0.52 percentage points per standard deviation. Recurrence *is* selected on genotype, because genetic risk predicts who relapses. That selection biases against the finding rather than towards it: the men who recur at low genetic risk are those with the strongest non-genetic reasons to relapse, and are therefore the more adversely selected of the two groups, so a comparison that finds the low-risk group's employment barely moving is finding it in the group where a larger effect would be expected. Appendix B.6 reports the magnitude of the selection.

Two cautions. The genetic cells in the redundancy panel are far smaller than the pooled sample, so the individual point estimates should not be read at face value; what they support is the pattern rather than any single magnitude. And the difference in each row is constructed from two separate regressions rather than estimated as an interaction. Against that, Appendix Figure 12 plots the full event-study paths by genetic group for both designs: the pre-event paths of the two genetic groups are close to identical in both, so neither gap is a differential trend, and the mental-health panels rule out the obvious alternative reading in which the high-risk group simply receives a bigger shock.

Together the two designs close the loop. Because they draw on entirely different sources of variation, one from the labor market and one from the natural history of an illness, the assumptions they require are largely non-overlapping. What the descriptive transitions could only suggest, these estimates identify: poor mental health and non-employment cause one another, and the same shocks bite harder on the genetically vulnerable across every outcome we can measure. We now turn to a structural life-cycle model in which genetic risk governs the stochastic transitions of both mental and physical health and also affects wage offers.

4 Model

We develop a partial equilibrium life-cycle model of consumption, labor supply, and health dynamics, incorporating endogenous decisions regarding psychotherapy. The life cycle commences at age $t = 25$ and terminates at $T = 80$. Agents remain active in the labor market until the exogenous retirement age $t_{ret} = 65$, after which they withdraw from the labor force and rely on pension income and accumulated savings.

State Space and Timing

Time is discrete and indexed by $t \in \{25, \ldots, T\}$. During working life, an agent is characterized by the observed state vector

$$S_t = \{x, g, t, mh_t, ph_t, hc_t, a_t, \epsilon_t\}.$$

The vector includes endogenous and predetermined variables: x captures unobserved permanent heterogeneity (type); g indicates genetic predisposition for depressive symptoms; t denotes age; mh_t and ph_t represent mental and physical health status, respectively; hc_t denotes human capital; a_t denotes assets; and ϵ_t is an idiosyncratic stochastic shock to wages. The timing of events within each period t is as follows:

Shock Realization: At the beginning of the period, stochastic shocks are realized. During working life, these include the wage shock ϵ_t. Health status at the beginning of period t is inherited from the previous period's health transition. For retired agents, survival is also realized. These realizations fully resolve the observed state vector S_t.

Decisions and Therapy Receipt: Observing S_t, the agent makes choices conditional on their life-cycle stage. Within working life, the therapy application decision and the labor supply/savings decision are resolved sequentially, in two stages:

- *Working Life* ($25 \leq t < 65$)*:* The agent first decides whether to apply for mental health therapy, $d_t \in \{0, 1\}$, by comparing the value of certain non-receipt to the expected value of applying. Conditional on applying, the agent receives therapy with probability λ_τ, which is common across individuals; let $\nu_t \in \{0, 1\}$ denote this therapy receipt shock, with $\Pr(\nu_t = 1) = \lambda_\tau$. The application decision does not mechanically imply receipt: non-applicants cannot receive therapy, while applicants receive it with probability λ_τ. Realized therapy receipt is $\tau_t = d_t \nu_t$. Once τ_t is realized, the agent chooses consumption c_t, savings a_{t+1}, and labor supply $n_t \in \{0, \frac{1}{2}, 1\}$[3] separately for whichever value of τ_t has occurred. This timing reflects that the agent cannot finance treatment-contingent consumption or adjust labor supply until the outcome of the therapy application is actually known.
- *Retirement* ($65 \leq t \leq 80$)*:* Labor supply is fixed at zero ($n_t = 0$), therapy is unavailable ($d_t = 0$ and $\tau_t = 0$), and the agent chooses only consumption c_t and savings a_{t+1}.

Dynamics: Choices made in the current period endogenously influence the state transition for period $t + 1$:

- *Health Evolution:* During the working life, health dynamics are endogenous; the transition probabilities for future mental and physical health, (mh_{t+1}, ph_{t+1}), depend on current health, genetics, labor supply n_t, and realized therapy receipt τ_t. In retirement, health transitions become exogenous, depending only on previous health status and age.
- *Human Capital:* Human capital evolves with labor supply. It accumulates when the agent works and depreciates during non-employment. This captures the dynamic channel through which health shocks can reduce employment, lower future human capital, and depress future wage offers.
- *Survival:* Survival is assumed to be certain during the working life. Upon entering retirement ($t \geq 65$), agents face an age- and health-dependent survival probability $\zeta_t(mh_t, ph_t)$. The model terminates deterministically at age $T = 80$.

Agents choose decision rules to maximize the expected present discounted value of lifetime utility, subject to the period budget constraint, therapy receipt risk, human capital accumulation, and survival risk in retirement.

Heterogeneity

We characterize agent heterogeneity along two distinct dimensions: observable genetic predisposition and unobserved permanent heterogeneity.

[3] The three labor-supply values denote non-employment, part-time employment, and full-time employment, respectively.

First, a key feature of our framework is the inclusion of an observable genetic predisposition for depressive symptoms, denoted by g. By treating this genetic endowment as observable, we directly map biological risk factors into the economic model.

Second, beyond genetic factors, we incorporate time-invariant unobserved heterogeneity to capture persistent ex-ante differences in endowments (e.g., innate ability or non-genetic health robustness). Accounting for these persistent traits is crucial for identification; neglecting them risks confounding permanent individual characteristics with state dependence in the stochastic processes. For instance, without this correction, the model might misattribute a sub-population's persistently poor health to a high probability of adverse shocks, rather than to their underlying type.

To implement this computationally, we discretize unobserved heterogeneity following the classification method of Bonhomme et al. (2022). We postulate that agents belong to a finite set of latent types, denoted by x, which govern their productivity and health trajectories.

Health Dynamics

We model the joint evolution of physical (ph) and mental (mh) health as a first-order Markov process. Each dimension is discretized into three ordered states, $h \in \{G, M, S\}$, corresponding to *Good*, *Moderate*, and *Severe* health. In our empirical application, physical health states map to Excellent/Good (G), Fair (M), and Poor/Very Poor (S), while mental health states represent the clinical progression of depression: asymptomatic (G), mild-to-moderate symptoms (M), and Major Depressive Disorder (S).

Health transitions are governed by an ordered logit framework, where the observed state h_{t+1} is determined by an underlying latent continuous variable h^*_{t+1} falling within estimated thresholds $\{\kappa_j\}$. Assuming the idiosyncratic shocks $(\iota_{t+1}, \upsilon_{t+1})$ follow a standard logistic distribution, the latent process for physical health evolves as:

$$ph^*_{t+1} = ph_t\pi_1 + mh_t\pi_2 + n_t\pi_3 + S_t^{exo}\pi_4 + \iota_{t+1}, \tag{4}$$

where ph_t and mh_t are vectors of indicators for current health status, $n_t \in \{0, 1\}$ is the labor supply choice, and S_t^{exo} collects exogenous covariates including age, unobserved type (x), and genetic predisposition (g). The latent mental health process is analogous, except that it also incorporates the endogenous therapy decision, $\tau_t \in \{0, 1\}$:

$$mh^*_{t+1} = mh_t\mu_1 + \tau_t\mu_2 + n_t\mu_3 + ph_t\mu_4 + S_t^{exo}\mu_5 + \upsilon_{t+1}, \tag{5}$$

where μ_2 captures the mental health returns to psychotherapy. Both processes are estimated as standard ordered logit models under the proportional-odds restriction: each covariate has a single coefficient that is common across health thresholds, while threshold-specific intercepts κ_j determine the cutpoints separating the three health states. The resulting probability of transitioning to state j or better is:

$$P(h_{t+1} \leq j \mid \mathbf{X}_t) = \Lambda\left(\kappa_j - \mathbf{X}_t\beta\right), \tag{6}$$

where $\Lambda(\cdot)$ denotes the logistic cumulative distribution function. The physical health process (Eq. 4) depends only on exogenous and predetermined covariates, so π_1–π_4 and its cutpoints are estimated directly from the UKHLS panel and held fixed as an input to the structural model (Panel B of Table 5). The mental health process (Eq. 5) depends on the endogenous therapy decision τ_t, so μ_1, μ_3–μ_5 and its cutpoints must instead be estimated jointly with the rest of the structural model via SMM (Panel C). We let $H(\cdot \mid S_t, n_t, \tau_t)$ denote the joint conditional distribution function of next-period health $h_{t+1} \equiv (mh_{t+1}, ph_{t+1})$ implied by these two ordered-logit processes; this is the health transition kernel used in the recursive formulation below.

Flow Utility

Agents derive utility from consumption and effective leisure, and incur a non-pecuniary cost when they receive therapy. Flow utility is

$$U(c_t, n_t, \tau_t; mh_t, ph_t) = \frac{\left(c_t^{\gamma} L_t^{1-\gamma}\right)^{1-\omega}}{1-\omega} - \psi\tau_t, \tag{7}$$

where ω is the coefficient of relative risk aversion, γ governs the consumption weight, and ψ is the disutility of therapy.

Effective leisure is

$$L_t = 1 - \ell_n n_t - \delta(mh_t, ph_t) - \ell_\tau \tau_t. \tag{8}$$

Here, ℓ_n is the time cost of work, $\delta(mh_t, ph_t)$ is the health-related time cost, and ℓ_τ is the time cost of therapy.[4]

The health time cost is

$$\delta(mh_t, ph_t) = \delta_1' mh_t + \delta_2' ph_t. \tag{9}$$

This specification captures the welfare cost of poor health even outside employment. By reducing effective leisure, poor health also increases the marginal utility cost of working.

Human Capital

Human capital evolves according to a parsimonious accumulation-depreciation process:

$$hc_{t+1} = Q(hc_t, n_t) \equiv \min\left\{10,\ (1-\delta_{hc})hc_t + n_t\right\}. \tag{10}$$

We let $Q(\cdot,\cdot)$ denote this human capital updating function. Thus, human capital depreciates during non-employment and accumulates while working up to a limit. This mechanism captures the dynamic channel through which health shocks can reduce employment, lower future human capital, and depress future wage offers.

[4] We set $\ell_n = 0.292$, corresponding to the time cost of full-time work; since $n_t \in \{0, \frac{1}{2}, 1\}$, work time is proportional to hours supplied. We set $\ell_\tau = 2/(7 \times 24) \simeq 0.012$, corresponding to two hours per week following Abramson et al. (2024).

Budget Constraint, Benefits, and Earnings

Agents face a period-by-period budget constraint:

$$c_t + a_{t+1} = (1 + r)a_t + y_t - P_\tau \tau_t, \qquad a_{t+1} \geq 0, \tag{11}$$

where r is the risk-free interest rate, P_τ is the out-of-pocket pecuniary cost of realized therapy, and borrowing is not allowed.

Disposable income y_t depends on the agent's life-cycle stage, labor supply choice, and health status:

$$y_t = \begin{cases} n_t w_t \left[1 - \mathcal{T}(n_t w_t)\right] + b(n_t, mh_t, ph_t), & \text{if } t < 65 \text{ (Working Age)}, \\ p(x), & \text{if } t \geq 65 \text{ (Retirement)}. \end{cases} \tag{12}$$

During working life, labor earnings are $n_t w_t$, and $\mathcal{T}(n_t w_t)$ denotes the effective tax rate as a function of labor earnings. Agents may also receive benefits $b(n_t, mh_t, ph_t)$, which are allowed to vary by both health and employment status.[5] After retirement, agents receive pension income $p(x)$, which may vary by permanent type.

The market wage offer w_t evolves according to a Mincerian process augmented with health, genetic traits, and human capital:

$$\ln w_t = \beta_0 + \beta_1 g + \beta_2 x + \beta_3' mh_t + \beta_4' ph_t + \beta_5 t + \beta_6 t^2 + \beta_7 hc_t + \epsilon_t. \tag{13}$$

Here, wages are determined by genetic predisposition g, permanent type x, current health status mh_t and ph_t, human capital hc_t, and a quadratic age profile. The idiosyncratic productivity shock ϵ_t follows an AR(1) process:

$$\epsilon_{t+1} = \rho \epsilon_t + \eta_{t+1}, \qquad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2), \tag{14}$$

where η_{t+1} represents a transient shock to labor productivity. We let $F(\cdot \mid \epsilon_t)$ denote the conditional cumulative distribution function of ϵ_{t+1} given ϵ_t implied by this AR(1) process, i.e. $\epsilon_{t+1} \mid \epsilon_t \sim \mathcal{N}(\rho\epsilon_t, \sigma_\eta^2)$.

Recursive Formulation

The agent enters the model at age $t = 25$ and lives up to a maximum age of $T = 80$. The life cycle is partitioned into a working stage ($t \in [25, 64]$) and a retirement stage ($t \in [65, 80]$).

Working Age ($25 \leq t < 65$***):*** In each working-age period, the agent's observed state is summarized by

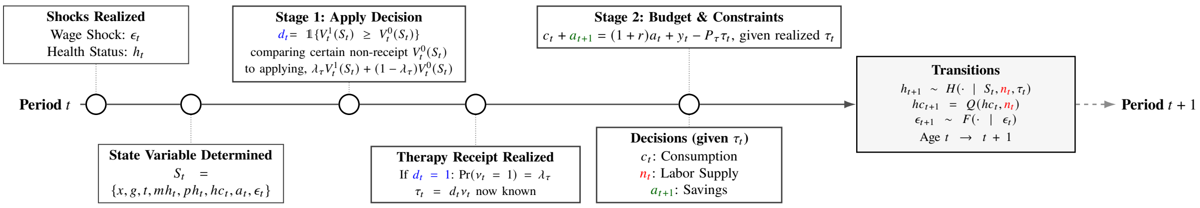
$$S_t = \{x, g, t, mh_t, ph_t, hc_t, a_t, \epsilon_t\}.$$

Given S_t, the agent first decides whether to apply for therapy, $d_t \in \{0, 1\}$. Conditional on applying, therapy receipt is uncertain: let $\nu_t \in \{0, 1\}$ denote the receipt shock, with $\Pr(\nu_t = 1) = \lambda_\tau$. Realized

[5] In particular, $b(n_t, mh_t, ph_t)$ may be positive even when $n_t > 0$, allowing employed individuals to receive benefits.

therapy receipt is $\tau_t = d_t \nu_t$. Only once τ_t is known does the agent choose labor supply $n_t \in \{0, \frac{1}{2}, 1\}$, consumption c_t, and next-period assets a_{t+1}. The within-period dynamics are illustrated in Figure 2.

Figure 2: Working Age Within-Period Dynamics



Reflecting the two-stage timing above, we first define, for each realized therapy outcome $\tau \in \{0, 1\}$, the conditional value function

$$V_t^\tau(S_t) = \max_{n_t,\, a_{t+1}} \left\{ U(c_t, n_t, \tau; mh_t, ph_t) + \beta \iint V_{t+1}(S_{t+1})\, dH(h_{t+1} \mid S_t, n_t, \tau)\, dF(\epsilon_{t+1} \mid \epsilon_t) \right\}, \quad (15)$$

subject to the period budget constraint $c_t + a_{t+1} = (1+r)a_t + y_t - P_\tau \tau$, non-negativity of assets, and the human capital law of motion. The expectation is taken over next-period health $h_{t+1} \equiv (mh_{t+1}, ph_{t+1})$, given labor supply n_t and the realized therapy outcome τ, and over next-period wage shocks.

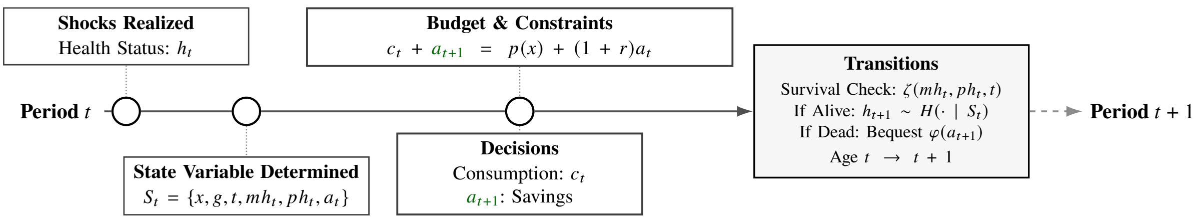
The therapy application decision d_t is then chosen to maximize the value before ν_t is realized:

$$V_t(S_t) = \max_{d_t \in \{0,1\}} \left\{ (1-d_t)\, V_t^0(S_t) + d_t \left[\lambda_\tau V_t^1(S_t) + (1-\lambda_\tau) V_t^0(S_t) \right] \right\}. \quad (16)$$

Because non-application guarantees $\tau_t = 0$, applying is weakly preferred whenever $V_t^1(S_t) \geq V_t^0(S_t)$, and the agent applies, $d_t = 1$, exactly in that case. Once ν_t is realized, $\tau_t = d_t \nu_t$ is known, and the agent implements the consumption, savings, and labor supply policy associated with $V_t^{\tau_t}(S_t)$.

***Retirement* ($t \geq 65$):** Upon reaching age 65, the agent retires exogenously. The retirement state is summarized by $S_t = \{x, g, t, mh_t, ph_t, a_t\}$, since human capital no longer affects earnings or choices. Labor supply is fixed at $n_t = 0$, therapy is unavailable ($d_t = 0$ and $\tau_t = 0$), and income consists of pension income $p(x)$ and returns on assets. Retirement utility is shaped by survival risk. At the end of each period, the agent survives with probability $\zeta(mh_t, ph_t, t)$; if the agent dies, remaining assets generate bequest utility. The retirement timing is illustrated in Figure 3.

Figure 3: Retirement Within-Period Dynamics



If the agent dies, remaining assets generate bequest utility. Following French and Jones (2011) and De Nardi (2004), the bequest motive is

$$\varphi(a_{t+1}) = \theta \frac{(a_{t+1} + \kappa)^{(1-\omega)\gamma}}{1 - \omega}, \tag{17}$$

where θ governs the strength of the bequest motive and κ controls the luxury-good nature of bequests.

The Bellman equation incorporates survival risk. Writing $\zeta_t \equiv \zeta(mh_t, ph_t, t)$ for brevity,

$$V_t(S_t) = \max_{a_{t+1}} \Big\{ U(c_t, 0, 0; mh_t, ph_t) + \beta \underbrace{\zeta_t}_{\text{Survival}} \int V_{t+1}(S_{t+1})\, dH(h_{t+1} \mid S_t) + \beta \underbrace{(1 - \zeta_t)}_{\text{Death}} \varphi(a_{t+1}) \Big\}. \tag{18}$$

subject to the retirement budget constraint and the non-negativity of assets. At the terminal age $T = 80$, the continuation value is zero.

5 Estimation and Identification

We estimate the model parameters using a three-step procedure designed to ensure transparent identification. In the first step, we calibrate a subset of standard parameters to established values from the literature. In the second step, we estimate the exogenous processes governing health transitions and institutional constraints directly from the data, outside of the structural model. In the final step, we jointly estimate the remaining behavioural parameters within the model using the simulated method of moments (SMM). This strategy allows us to anchor the model's driving forces in empirical evidence while exploiting its full structure to identify the deep parameters shaping life-cycle decisions. Table 5 summarizes the parameters and their sources of identification.

5.1 Calibrated Parameters

We calibrate three parameters using external evidence. We set the annual risk-free interest rate to $r = 1.5\%$, following the long-run estimates in Holston et al. (2017). We set the coefficient of relative risk aversion to $\omega = 3.0$, consistent with the life-cycle estimates used in De Nardi et al. (2010). Finally, identifying the causal effect of psychotherapy (μ_2) is challenging because treatment is endogenous: individuals typically seek therapy following negative mental health shocks. We therefore calibrate μ_2 using external evidence from randomized controlled trials. We take the standardized mean difference ($SMD \approx 0.6$) reported in meta-analyses and convert it into a log-odds ratio suitable for our ordered-logit health transition, using the approximation in Borenstein et al. (2009):

$$\mu_2 \approx 0.6 \times \text{SMD} \times \frac{\pi}{\sqrt{3}}. \tag{19}$$

This calibration follows the approach in recent structural work on mental health treatment (Abramson et al., 2024; Cronin et al., 2024).

Table 5: Model Parameters, Identification Strategy, and Data Sources

Parameter	Equation	Description	Targeted Empirical Moments	Source
Panel A: Calibrated Parameters				
μ_2	Eq. 5	Effectiveness of therapy	RCT effect of therapy on MH outcomes	Cronin et al. (2024)
ω	Eq. 7	Coefficient of relative risk aversion	$\omega = 3.0$	De Nardi et al. (2010)
ℓ_n	Eq. 8	Time cost of full-time work	Average weekly working hours	UKHLS
ℓ_τ	Eq. 8	Time cost of therapy	Two hours per week of therapy	Abramson et al. (2024)
δ_{hc}, ξ_{hc}	Eq. 10	Human capital depreciation and accumulation	Wage growth and labor supply histories around employment and non-employment spells	UKHLS
r	Eq. 11	Annual risk-free interest rate	$r = 1.5\%$	Holston et al. (2017)
Panel B: Parameters Estimated Outside the Model				
$\{\pi_1 - \pi_4\}$	Eq. 4	Physical health transition parameters	Physical health transition rates by current health, labor supply, age, type, and genotype	UKHLS
P_τ	Eq. 11	Pecuniary cost of therapy	Mean out-of-pocket therapy expenditure per treatment episode	UKHLS
$b(n_t, mh_t, ph_t)$	Eq. 12	Benefits schedule by employment and health status	Mean benefits by labor supply, mental health, and physical health	UKHLS
$\mathcal{T}(\cdot)$	Eq. 12	Effective tax schedule	Tax rate by labor earnings	UKHLS
λ_τ	Eq. 16	Probability of receiving therapy conditional on applying	Ratio of treatment starts to referrals in the national talking-therapies programme	NHS England (2024b)
$\zeta(mh_t, ph_t, t)$	Eq. 18	Survival probability	Survival rates by age, mental health, and physical health	UKHLS & UK Life Tables
Panel C: Parameters Estimated Within the Model				
$\{\mu_1, \mu_3 - \mu_5\}$	Eq. 5	Mental health transition parameters, including therapy effects	Mental health transition rates by current health, labor supply, therapy receipt, age, type, and genotype	UKHLS
γ	Eq. 7	Consumption share in utility	Aggregate saving ratio	UKHLS
ψ	Eq. 7	Non-pecuniary disutility of therapy	Overall therapy usage	UKHLS
$\delta(\cdot)$	Eq. 8	Health-related time cost, incurred regardless of employment status	Labor supply by mental and physical health groups	UKHLS
$\beta_0 - \beta_7$	Eq. 13	Wage offer process, including human capital returns	Mean wages by age, health, type, genotype, and labor-market history	UKHLS
ρ, σ_η	Eq. 14	Wage shock parameters	Wage dispersion by health, type, genotype, and age group	UKHLS
θ, κ	Eq. 17	Bequest parameters	Mean and median bequests	UKHLS
β	Eq. 18	Discount factor	Asset and saving profiles over the life cycle	UKHLS

Notes: This table summarizes the model parameters and their corresponding identifying moments. Panel A lists parameters calibrated to standard values or direct time-use measures. Panel B lists parameters estimated directly from the data outside the model. Panel C lists parameters estimated jointly within the structural model. The two health transition processes are treated asymmetrically: the physical health process depends only on exogenous and predetermined covariates, so it is estimated directly from the data and held fixed as an input to the model (Panel B); the mental health process depends on the endogenous therapy decision, so its ordered-logit parameters and cutoffs must instead be estimated jointly with the other structural parameters (Panel C). Therapy cost is computed as the average session cost (£130; Steele, 2026) times the average number of sessions attended (8.6; Steele, 2026) times the share of patients attending private therapy (26%, UKHLS).

5.2 Estimation Outside the Model

I. Individual Heterogeneity

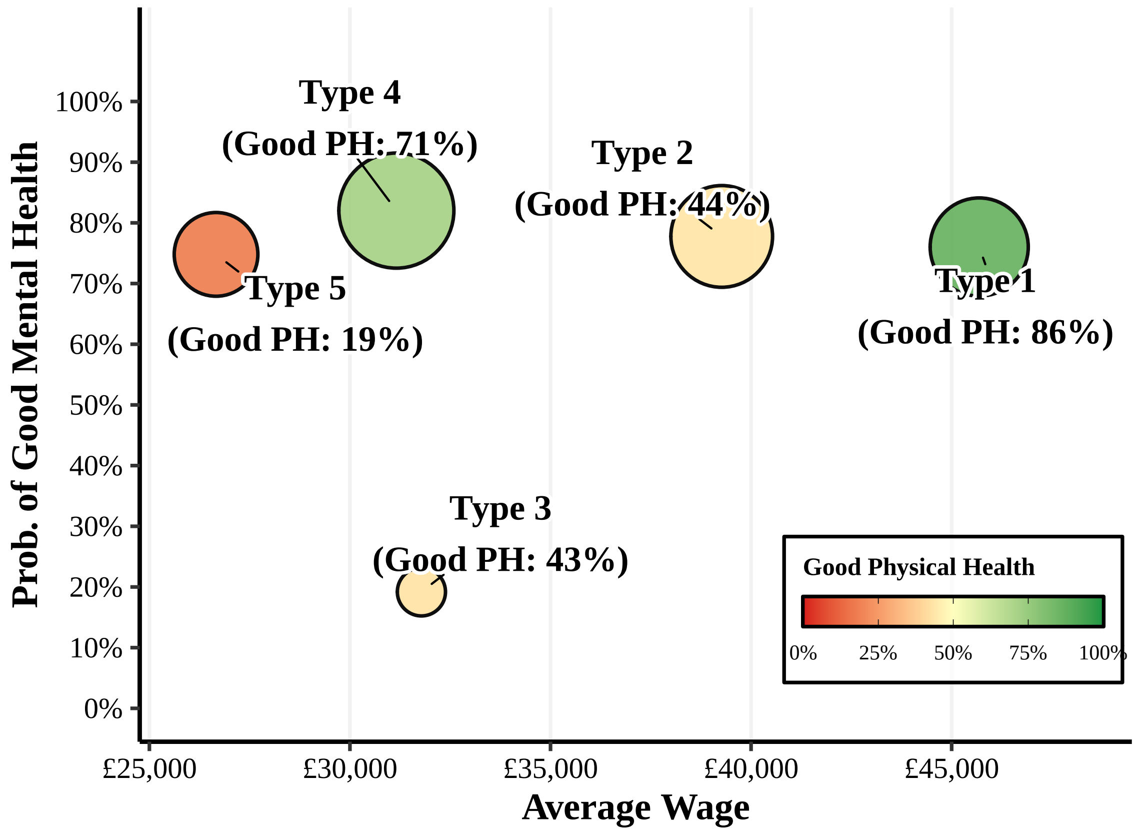
Individuals have types x that take values on a discrete set $x \in \{1, ..., K\}$. We estimate these types by building upon the approach of Bonhomme et al. (2022) (BLM) (see also Bonhomme et al. (2019). BLM estimate individual heterogeneity classes by k-means clustering on relevant moments of outcome variables, with moments calculated at an individual level. Worker have types $x \in \{1, ..., K\}$, there are individuals $i \in \{1, .., n\}$ and $\mathbf{m}_i$ is a vector of M moments of individual i's outcomes. Workers are partitioned into K classes by finding the partition that solves

$$\min_{\{x_i\} \in \{1,...,K\}^N} \sum_{i=1}^{N} \|\mathbf{m}_i - \widetilde{\mathbf{m}}(x_i)\|^2 , \tag{20}$$

The consistency of BLM's approach relies on an injective map between individual unobserved heterogeneity and the vector of moments m_i. In our application, productivity, mental health and physical health are the primary sources of individual heterogeneity. We therefore compute individual moments m using average wages, mental health status and physical health status.

Applying the clustering procedure described in Appendix F, we summarize persistent individual heterogeneity with a discrete type $x_i \in \{1, \dots, 5\}$. Figure 4 reports the five estimated types and their centroid characteristics in terms of average wages, the probability of good mental health, and physical health. Type 1 is a high-wage, high-health group. Type 2 has relatively high wages and good mental health but substantially worse physical health. Type 3 is characterized by very low mental health despite moderate wages and physical health. Type 4 combines high mental health with relatively good physical health at lower wages. Type 5 is a low-wage group with the poorest physical health. We treat x_i as a fixed state variable in the structural model.

Figure 4: Individual Heterogeneity: Estimated Types



Notes: This figure summarizes persistent individual heterogeneity using five discrete latent types, $x_i \in \{1, \dots, 5\}$, estimated via the clustering procedure described in Appendix F. Each type is represented by its centroid characteristics in average wages, the probability of good mental health, and the probability of good physical health. Type 1 is a high-wage, high-health group. Type 2 has relatively high wages and good mental health but worse physical health. Type 3 is characterized by low mental health despite moderate wages and physical health. Type 4 combines high mental health with relatively good physical health at lower wages. Type 5 is a low-wage group with the poorest physical health. Types are treated as a fixed state variable in the structural model.

II. Other Estimation

We estimate a set of parameters directly from the data outside the structural model. These objects discipline health dynamics, non-labor income, therapy costs, and survival risk. We treat them as inputs in the second-stage SMM estimation.

We estimate the mental and physical health transition processes using the ordered-logit specifications in Equation 5 and Equation 4. The parameters μ_1 and μ_3–μ_5 govern the evolution of mental health, while π_1–π_4 govern the evolution of physical health. We estimate these equations on the UKHLS panel and use the fitted probabilities to construct the health transition kernel in the model.

We also estimate the non-labor income components that enter the budget constraint. The benefits schedule $b(h_t)$ in Equation 12 is measured from UKHLS as mean benefit income by health and employment status. The pecuniary therapy cost P_τ in Equation 11 is constructed as the average out-of-pocket cost per treatment episode. We compute it as the product of the average session cost, the average number of sessions attended (8.6), and the share of patients who attend private therapy (26%, UKHLS); Table 29 reports the components. Finally, the probability that an application converts into a course of treatment, λ_τ, is set to 0.689, the ratio of treatment starts to referrals in the NHS Talking Therapies programme in 2023–24 (NHS England, 2024b). It implies that 68.9 percent of applicants begin treatment in the year they apply, an expected wait of $(1 - \lambda_\tau)/\lambda_\tau = 5.4$ months, and it is the object that the counterfactual of Section 6.3 reallocates across genetic risk groups.

Finally, we model survival risk in retirement. We estimate the survival probability $\zeta(h_t, t)$ in Equation 18 using age- and health-specific survival rates. We combine UK life tables with the model's health classification to discipline mortality risk over the life cycle. These survival probabilities pin down effective retirement horizons and are central for matching asset de-cumulation and bequest behaviour.

Details on the first-stage estimates are reported in Table 27 to Table 29.

5.3 Estimation of Core Structural Parameters via SMM

We estimate the remaining structural parameters using the Simulated Method of Moments (SMM). Identification follows a simple economic logic: each parameter is disciplined by the empirical feature it most directly governs in the model. The moments we target and their mapping to parameters are summarized in Panel C of Table 5.

Treatment demand and the labor–health interaction. The therapy disutility $\psi(n_t)$ governs the take-up decision conditional on health and prices. A higher $\psi(n_t)$ lowers treatment use, and allowing it to vary with employment shifts take-up differentially between workers and non-workers. We therefore discipline $\psi(n_t)$ using therapy participation rates by employment status. The health-dependent time tax on labor, $\delta(\cdot)$ in Equation 8, captures how poor mental and physical health reduce effective leisure when employed. A larger $\delta(\cdot)$ makes work less attractive in bad health and increases labor-force exit. We identify $\delta(\cdot)$ from employment rates across joint mental and physical health groups.

Earnings risk and the wage process. We model wages with a persistent shock component governed by (ρ, σ_η) and a deterministic component parameterized by β_0–β_6 in Equation 13. The persistence parameter ρ controls how strongly current wages predict future wages, while σ_η governs cross-sectional dispersion conditional on observables. We discipline (ρ, σ_η) using the standard deviation of wages across groups defined by health, type, and genotype, and we identify the coefficients β_0–β_6 using mean wages by age, health, type, and genotype group.

Inter-temporal choices and bequests. The discount factor β governs inter-temporal smoothing and precautionary saving. In our setting, it primarily affects the dispersion of wages over the life cycle through endogenous selection into work and the accumulation of health states. We therefore discipline β using age-group moments in wage dispersion. The consumption share parameter γ controls the relative weight on consumption in the consumption–leisure composite and shifts saving incentives. We identify γ using the aggregate saving ratio. Finally, the bequest parameters (θ, κ) govern the level and curvature of the bequest motive in Equation 17. They pin down wealth de-cumulation late in life and the mass of terminal assets. We discipline (θ, κ) using the mean and median of observed bequests.

Implementation proceeds as follows. For a given parameter vector, we solve the model and simulate life-cycle histories for a large synthetic panel. We compute the model-implied counterparts of the targeted moments and choose parameters to minimize a weighted quadratic distance between simulated and empirical moments. The identification strategy exploits the tight mapping from preferences, earnings risk, and health-dependent work costs into observed treatment, labor supply, wage, saving, and bequest patterns.

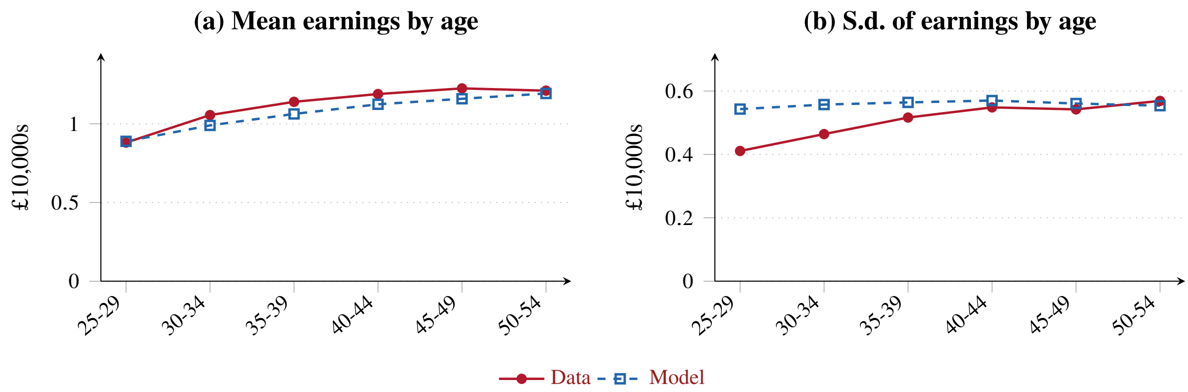
5.4 Model Fit

The model reproduces the moments used in estimation closely and, with one exception noted below, also matches moments that were never targeted. We summarize the targeted fit in Figures 5 and 6, and report untargeted validation in Figure 7.

Earnings. Figure 5 compares earnings moments in the data with their simulated counterparts over ages 25–54. The model captures the shape of the life-cycle profile, with mean earnings rising from £8,890 at ages 25–29 to £11,930 at 50–54 against £8,810 and £12,110 in the data. It also reproduces the cross-sectional gradients that identify the wage equation, which Appendix Figure 16 reports in full: low-*PGI* agents earn more than high-*PGI* agents, earnings rise with both mental and physical health, and the latent types are correctly ordered, with Types 1 and 2 at the top and Type 5 at the bottom.

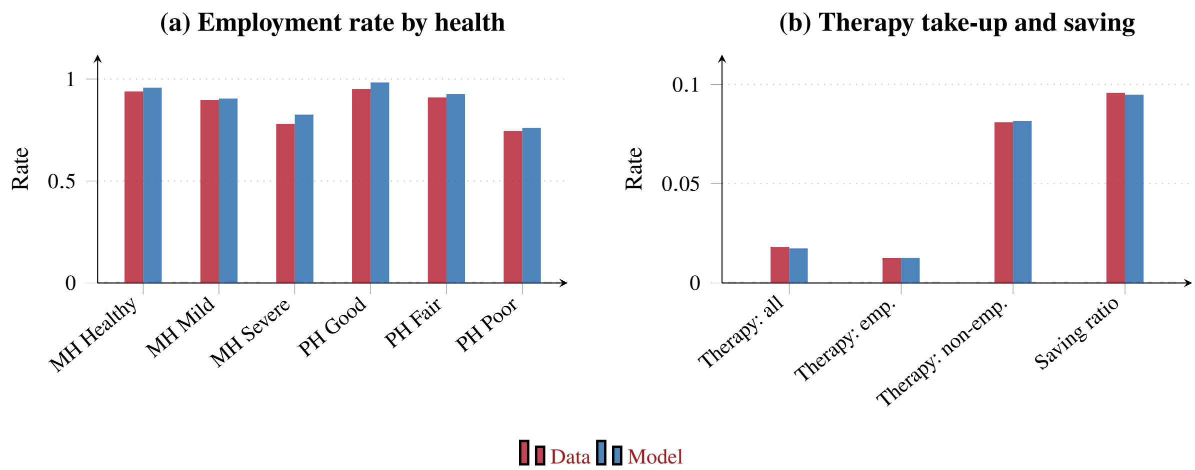
The one visible gap is in the age profile of dispersion. In the data the standard deviation of earnings rises steadily from 0.41 at ages 25–29 to 0.57 at 50–54, whereas the model holds it close to 0.55 throughout, overstating dispersion early in working life and matching it only by the early fifties. This reflects the parsimony of the wage process, which has a constant innovation variance and so cannot generate a fanning-out of the earnings distribution with age. We restrict these panels to ages 25–54 because the model has neither a partial-retirement nor an hours margin, and therefore does not generate the decline in mean

Figure 5: Model fit: earnings



Notes: Targeted moments. Data (red, solid, filled markers) against the estimated model (blue, dashed, open markers). Earnings are gross labour income in 2015 £10,000s. Both panels cover ages 25–54; beyond that the model has neither a partial-retirement nor an hours margin, so the late-career comparison is uninformative about fit. Appendix Figure 16 reports the same comparison by genetic-risk group, health state and latent permanent type.

Figure 6: Model fit: labour supply, treatment and mental-health transitions



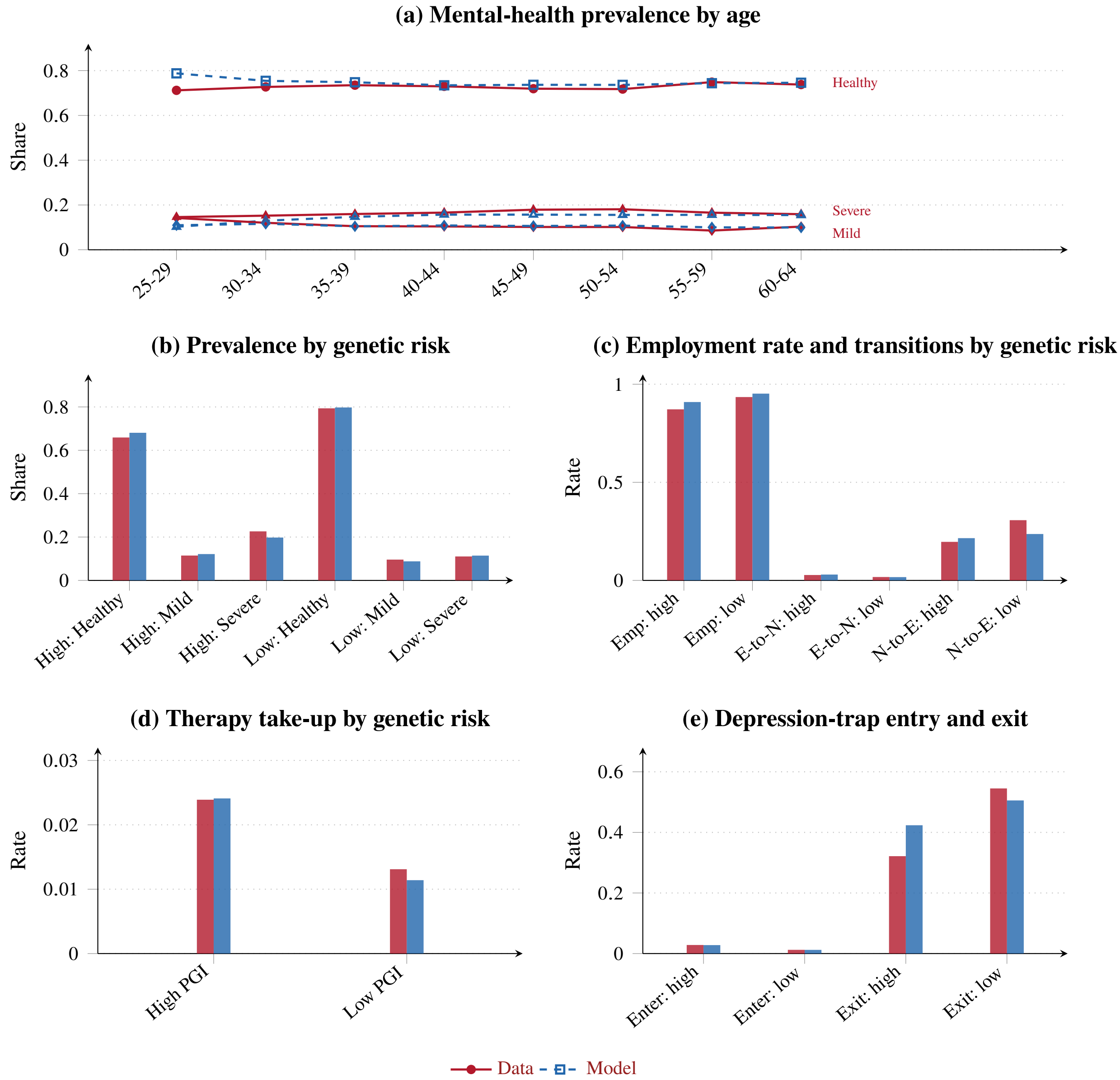
Notes: Targeted moments. Panel (a): employment rates by mental and physical health state. Panel (b): therapy take-up overall and by employment status, and the aggregate saving ratio. The one-year mental-health transition probabilities, the third block of targeted moments, are reported in Appendix Figure 17: across all 45 cells the largest absolute error is 0.013 and the mean absolute error is 0.004.

earnings observed after age 55; that late-career gap reflects an omitted institutional margin rather than a failure of the estimated wage process, and it lies outside the ages at which our counterfactuals operate.

Labor supply, treatment and mental-health dynamics. Figure 6 shows the remaining targeted moments. The model matches employment rates by health state, though it is uniformly a little optimistic: employment among the severely depressed is 82.6% against 78.0% in the data, and among those in poor physical health 76.0% against 74.5%. The health-dependent cost of work therefore captures the ordering and most of the level, but slightly understates how sharply poor health depresses participation.

Therapy take-up is matched almost exactly – 1.27% among the employed against 1.27% in the data, and 8.15% among the non-employed against 8.09% – which disciplines the out-of-pocket price and the two therapy disutility parameters. The aggregate saving ratio is 9.48% against 9.57%.

Figure 7: Model fit: untargeted moments



Notes: Untargeted moments – none entered the SMM objective, so these are out-of-sample validation. Panel (a): share of the working-age population in each mental-health state by age (circles Healthy, diamonds Mild, triangles Severe). Panel (b): mental-health prevalence by genetic risk. Panel (c): employment rate, and annual employment-to-non-employment and non-employment-to-employment transition rates, by genetic risk. Panel (d): therapy take-up by genetic risk. Panel (e): annual probability of entering and exiting the depression trap (depressed and out of work) by genetic risk. High and Low PGI denote the two depression polygenic-index groups.

The mental-health transition probabilities are the moments the therapy technology acts on, and the model reproduces them almost exactly. Appendix Figure 17 plots all 45 cells against their empirical counterparts. The largest absolute error is 0.013, the mean absolute error is 0.004, and the correlation between simulated and empirical cells is 0.9999.

Untargeted moments. Figure 7 reports moments that never entered the SMM objective. The model reproduces the mental-health prevalence profile by age and the gradient by genetic risk: among high-*PGI* agents the simulated share in severe depression is 19.8% against 22.6% in the data, and among low-*PGI* agents 11.4% against 11.0%. It matches therapy take-up by genetic risk (2.41% against 2.39% for high-*PGI*) and, notably, the rate of *entry* into the depression trap – 2.79% against 2.83% for high-*PGI* agents and 1.22% against 1.23% for low-*PGI* agents – despite neither moment being targeted. The employment gradient by genetic risk is reproduced with the same mild optimism as the health gradient (90.9% against 87.2% for high-*PGI* agents).

One untargeted dimension the model does not match deserves emphasis, and we state it explicitly because it bears on how the counterfactuals should be read. The model inverts the mental-health ordering of returns to work. In the data the annual non-employment-to-employment rate falls from 34.2% among the mentally healthy to 8.5% among the severely depressed; the model generates 18.8% and 26.6% respectively. It likewise understates job exit among the severely depressed (3.1% against 6.1%), and exit from the depression trap is too fast for high-*PGI* agents (42.4% against 32.2%). Taken together, the model reproduces the *stock* of depressed non-employment well – prevalence, trap entry and the employment gradient by genetic risk are all close – while generating too much churn among the severely depressed and too little among the healthy.

The direction of this error is informative. A model in which the severely depressed return to work too readily will, if anything, *overstate* the employment response to a policy that improves mental health, because the marginal treated agent re-enters employment more easily than his empirical counterpart. The employment and fiscal gains we report should therefore be read as upper bounds on that margin. The mental-health effects, which rest on the transition process in Appendix Figure 17, are not subject to this concern.

Details on the estimated structural parameters are reported in Appendix Table 30.

6 Results

This section presents the main findings from our structural model, focusing on the role of genetic risk for depression in shaping individual life-cycle trajectories and determining the success of clinical policy interventions. Our framework allows us to disentangle latent genetic risk from observable initial conditions, providing a controlled environment to measure how genetic endowments interact with exogenous shocks and endogenous labor supply decisions.

The section moves from individual mechanisms to aggregate policy. Section 6.1 establishes the lifetime imprint of genetic risk by swapping endowments between otherwise identical agents. Section 6.2 pushes every agent into the depression trap and watches them climb out; the illness clears at the same speed for both genetic groups, while the employment loss does not. Section 6.3 takes that asymmetry to the policy question the English system actually poses. Talking therapy there is free at the point of use and rationed by a queue, so the instrument available to a planner is not the price of treatment but its order, and we ask what happens when the queue is reordered by genetic risk.

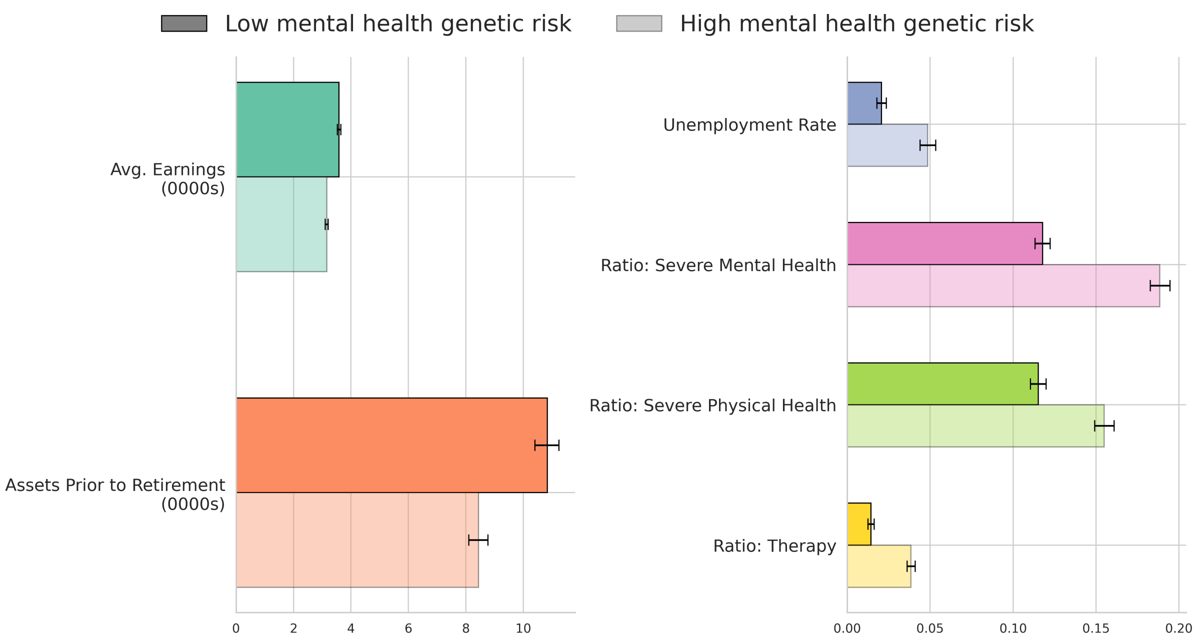
6.1 The Impact of Genetic Risk for Depression on Lifetime Outcomes

Genetic risk for depression generates large and persistent differences in life-cycle trajectories, even when individuals start from the same non-genetic conditions. To isolate the role of genetics, we simulate individuals over a 10-year period holding all non-genetic initial conditions fixed, and counterfactually switch each individual's genetic risk between high and low. By shutting down selection on initial health, earnings and assets, any resulting gaps reflect the dynamic impact of genetic risk beyond baseline mental health. Figure 8 reports the outcomes.

That single change more than doubles unemployment, from 2.0 percent of person-years at low genetic risk to 4.8 percent at high risk. Average earnings fall by 11 percent, and because these labor-market disadvantages compound over time the high-risk group arrives at retirement with 22 percent less in accumulated assets, despite starting the counterfactual with identical wealth.

Health and behavioural outcomes show a similarly large gap. Severe mental health states are 60 percent more prevalent at high genetic risk, 18.8 against 11.7 percent of person-years. Physical health is worse too, 15.5 against 11.5 percent in the severe state, since the two health processes are linked in the model. Therapy use differs more sharply still, at 3.8 against 1.4 percent, because high-risk agents reach the state in which applying is worthwhile more often.

Figure 8: Lifetime Outcomes by Genetic Risk for Depression



Notes: This figure reports counterfactual life-cycle outcomes implied by switching each individual's genetic risk for depression (high ↔ low) while holding all other initial characteristics fixed at their empirical values. For each individual and horizon, the model computes the implied outcome under each genetic endowment, which is then aggregated to obtain the levels shown. The sample includes individuals first observed between ages 25–45, and outcomes are simulated over the subsequent 10 years. All metrics are computed from this 10-year window, except pre-retirement assets, which are obtained by simulating each individual forward to age 65 and aggregating assets accumulated prior to retirement. Earnings and assets are reported in tens of thousands of pounds at 2015 prices; the remaining outcomes are shares of person-years. Whiskers denote uncertainty across simulation draws.

Taken together, these findings are consistent with reduced-form evidence on the lifetime impact of genetic risk. Genetic risk for depression is not simply a proxy for initial health status; it drives health, labor and wealth dynamics over the whole life cycle.

6.2 Escape from the Depression Trap

Section 3.5 identified each leg of the trap on its own, one from a job loss and one from a relapse. Neither design can say what happens when both legs are engaged at once, because no observable event switches them on together. The model can. We put every agent into the trap and watch them climb out.

The experiment begins from the empirical initial distribution and keeps individuals first observed between ages 25 and 45. At the start of the first year we place each of them in severe depression and out of work simultaneously. Nothing else is imposed: from that point on every choice and every transition is the agent's own. We simulate ten years and compare the shocked economy with the benchmark, separately for the high- and low-genetic-risk groups. Because the two groups start from different levels, we report each group's gap as a fraction of its own benchmark, so that a given percentage means the same thing for both. Figure 9 plots the three series.

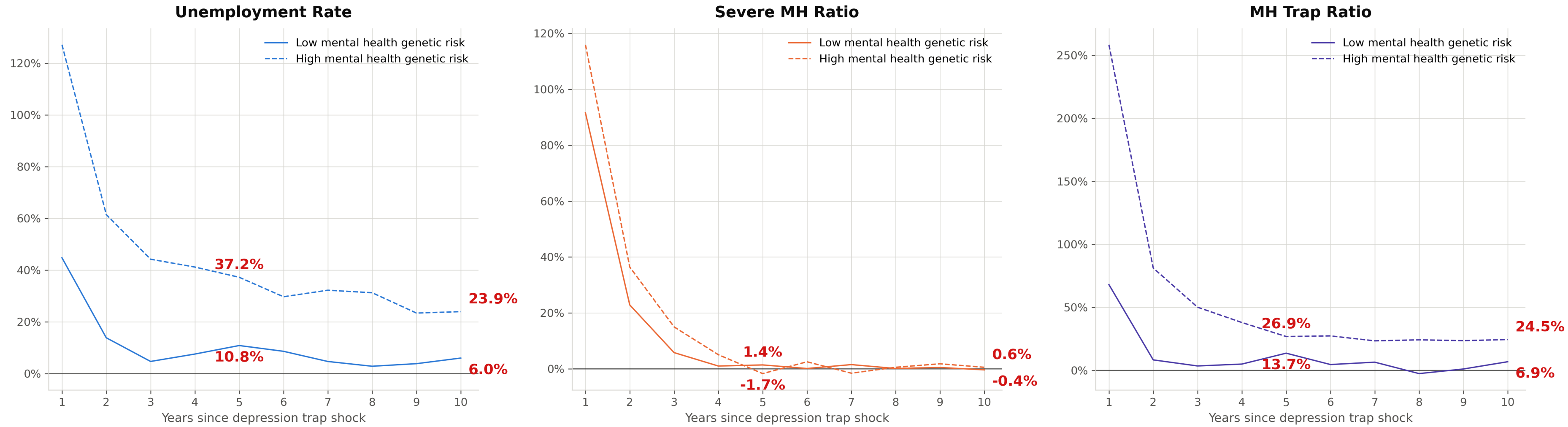
The illness clears, and it clears for both groups. Severe depression is 92 percent above baseline for the low-risk group in the year of the shock and 115 percent above for the high-risk group; by the fourth year both are back within two percent of where they began and they stay there. At ten years the two series are indistinguishable from zero and from each other. On the margin a clinician would watch, genetic risk delays recovery by something under two years and does nothing after that.

The job does not come back. In Figure 9, unemployment is 128 percent above baseline for the high-risk group on impact against 45 percent for the low-risk group, and that gap does not close as the illness resolves: 37.2 against 10.8 percent at five years, 23.9 against 6.0 percent at ten. Prevalence of the trap itself traces the same path, 26.9 against 13.7 percent at five years and 24.5 against 6.9 at ten. A decade after an episode that both groups have long since recovered from, the high-risk group is still four times as far above its own rate of non-employment as the low-risk group, and three and a half times as far above its own prevalence of the trap. Since the high-risk group also begins from a worse baseline, the gap in levels is wider than the gap in ratios.

The persistence comes from the interaction of two model mechanisms with a single nonlinearity. Human capital depreciates during the spell out of work, so the wage offer facing a recovered agent is worse than the one he left, and returning to work is correspondingly less attractive; and non-employment enters the mental health transition directly, so the longer the spell the higher the risk of a further episode. Genetic risk does not enter that loop as an interaction. It shifts the latent index of the ordered logit governing mental health by a constant. But high-risk agents sit closer to the severe threshold to begin with, so the same shift carries far more of them across it. The loop closes more tightly for them at every point, and a shock that both groups clear clinically leaves one group in work and the other out of it.

The divergence survives conditioning on the pre-shock state. Appendix Figure 18 repeats the experiment on agents who were employed in the year before the shock and, separately, on agents who were in healthy mental health before it. Restricting attention to men who looked well and attached beforehand widens the gap, to 29.9 against 7.6 percent of baseline unemployment at ten years among the pre-shock employed and 31.9 against 7.4 percent among those in healthy mental health. Genetic risk thus separates who will and will not recover economically among men who are, on both of the margins a clinician or a caseworker can see, identical beforehand. That is what makes it a candidate for targeting care before an episode instead of after one (Collins and Varmus, 2015; Torkamani et al., 2018), and it is the policy we turn to next.

Figure 9: Recovery from the Depression Trap, by Genetic Risk



Notes: We start from the empirical initial distribution and keep individuals first observed between ages 25–45. At the start of year 1 every agent is placed simultaneously in the severe mental health state and in non-employment, the two components of the depression trap, with all other characteristics at their benchmark values. No further intervention is applied: consumption, saving, labor supply and the therapy application are chosen by the agent in every subsequent period, and health evolves under the estimated transition processes. Each series is the difference between the shocked and benchmark economies, expressed as a percentage of that group's own benchmark level in the same year, so that groups with different baselines are on a common scale. High genetic risk denotes the group with the lower polygenic index, since the index is oriented towards good mental health. The mental health trap is the joint probability of non-employment and severe depression. Labelled values are the fifth and tenth years after the shock. Appendix Figure 18 repeats the exercise conditioning on the pre-shock state.

6.3 Priority Access to Therapy by Genetic Risk

Talking therapy in England is free at the point of use and rationed by a queue. In 2023–24 the NHS Talking Therapies programme received 1.83 million referrals and began a course of treatment for 1.26 million people, so a referral converts into treatment with probability 0.689 (NHS England, 2024b). It is the object the model calls λ_τ, and the value to which λ_τ is calibrated. What binds in this system is capacity rather than price, so the instrument a planner actually holds is the order of the queue and not the cost of a session. This section asks what happens when that order is set by genetic risk.

The policy works on the order of the queue, which the model represents through the treatment probability. An individual who applies is treated within the year with probability λ, independently across years, so the expected time spent waiting before treatment begins is $(1-\lambda)/\lambda$ years. At the calibrated $\lambda = 0.689$, 68.9 percent of applicants begin treatment in the year they apply, and the expected wait is 5.4 months.[6] Write λ_L and λ_H for the treatment probabilities facing low and high genetic risk. A policy that gives high risk priority of intensity X scales the expected number of applications per course, $1/\lambda$, down by $1-X$:

$$\frac{1}{\lambda_H} = (1-X)\frac{1}{\lambda_L} \qquad \Longleftrightarrow \qquad \lambda_H = \frac{\lambda_L}{1-X}. \tag{21}$$

Following the tables, we refer to X as the wait gap.[7] Equation (21) leaves one free parameter, which is pinned down by the supply side. We consider two polar regimes, and we solve them in a specific order so that they are directly comparable.

Under *fixed capacity*, which we solve first, the number of treatment slots is held at its benchmark level. Prioritising high risk must then be paid for by low risk: λ_L falls until the market clears, with λ_H tied to it by (21). This is pure reallocation. No additional therapy is delivered, so the policy has no incremental provision cost.

Under *elastic supply*, λ_L is restored to its estimated value and λ_H is held at *the value the fixed-capacity solution pinned down*. Capacity expands to accommodate the extra demand, and the additional courses are charged to the government at the difference between the market price of a course, £841, and the patient's out-of-pocket contribution, giving £488 per employed course and £841 per non-employed course, discounted to labor-market entry.

Solving in this order is what makes the comparison clean. At a given X the two regimes deliver the *identical* treatment probability to the high-risk group and differ only in who pays for it: low-risk patients, whose wait lengthens, or the exchequer, which buys more capacity. Because clearing drives λ_L down, the inherited λ_H stays below one even at $X = 80\%$, so the probability ceiling that would otherwise constrain the elastic arm never binds (Appendix H.2). Since λ_L is unchanged in the elastic arm, its *realised* wait gap is smaller than the target; both are reported.

[6] Waits count completed years before the year in which treatment begins. Counted inclusively of that year, the expected number of annual applications per course begun is $1/\lambda = 1.45$.

[7] Measured as expected time spent waiting, $(1-\lambda)/\lambda$, the implied high-risk reductions are larger than X.

Table 6 and Figure 10 report the targeted policy against the no-priority benchmark, for wait gaps of 20 to 80 percent. The figure traces the policy and its effects against the high-risk access rate λ_H, which the two regimes share at each intensity; panel A of the figure shows what delivering that access rate costs the low-risk group under fixed capacity, a trade-off that is close to linear at roughly 1.9 percentage points of low-risk access per point of high-risk access.

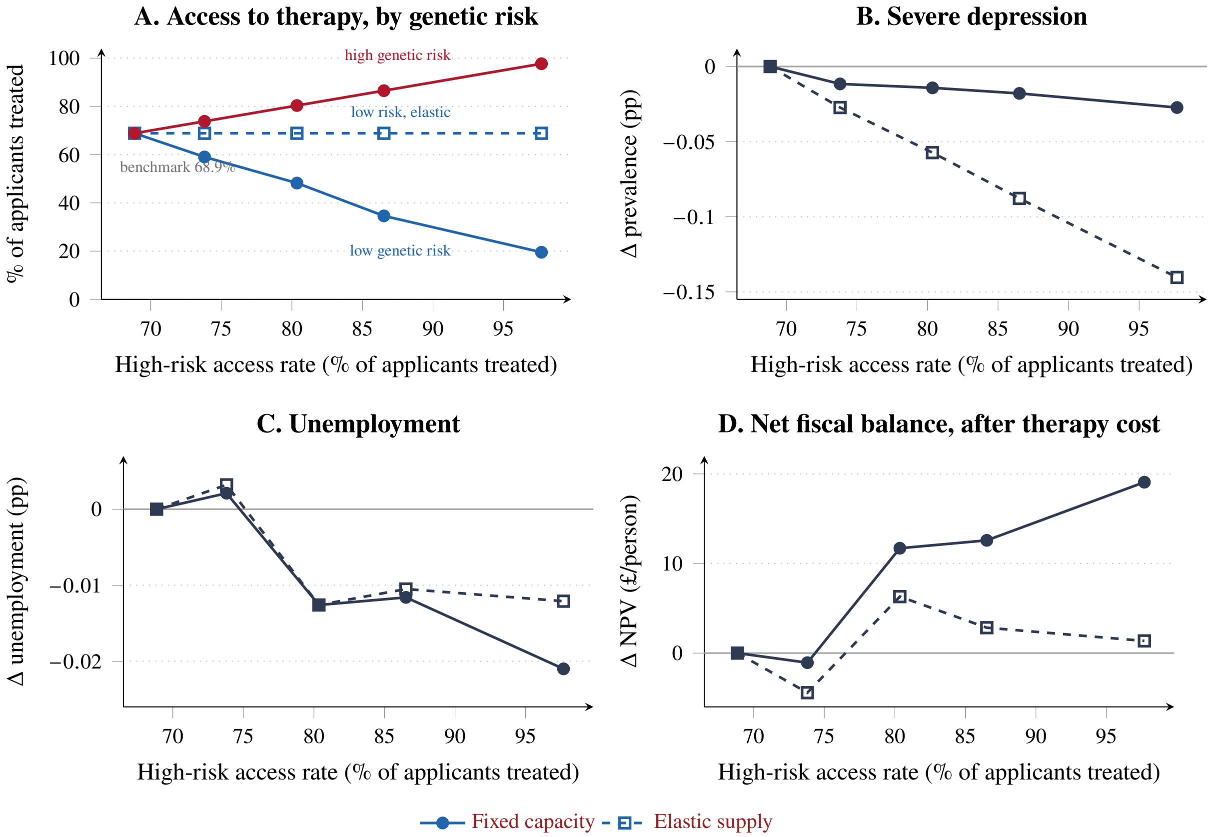
Reallocation dominates capacity expansion. At every intensity the fixed-capacity regime returns more than elastic supply, and the margin widens with the gap: at $X = 80\%$ the fixed policy is worth £19.1 per person against £1.4 under elastic supply. Because the two regimes grant the same λ_H, their *gross* returns are nearly identical, at £20.5 and £18.4 respectively. The entire difference is the provision bill: expanding capacity requires 21.8% more treatment volume, costing £17.0 per person, whereas reallocation delivers its gain with 0.1% more volume and essentially no incremental cost. The clinical ranking runs the other way, as it should, since the elastic arm treats more people: at $X = 80\%$ it cuts severe depression prevalence by 0.140 percentage points against 0.027 under fixed capacity. A planner who values health directly, and not only through the budget constraint, should prefer expansion; a planner constrained by the size of the therapist workforce loses nothing fiscally by reallocating instead.

In aggregate the magnitudes are small, and the reason is arithmetic rather than economic. Fewer than two percent of working-age person-years involve a course of therapy, so reordering the queue touches a small share of the population in any year. What makes the result worth having is not its size but its price: under fixed capacity, the exchequer is paid to reorder the queue and pays nothing to do it.

Decomposing the change by genetic group is instructive, and uncomfortable. In the elastic arm the low-risk group is untouched by construction. Since λ_L is unchanged, and this is a partial-equilibrium environment, their tax, benefit and provision flows are numerically identical to the benchmark, so the entire elastic return is the high-risk contribution. Under fixed capacity that same high-risk contribution is present, but it is joined by a large low-risk term: at $X = 80\%$ the low-risk group contributes £17.71 of the £19.07 total, of which £15.63 is simply the money the state saves by *not* treating them. The high-risk group contributes only £1.36. Pre-COVID, in other words, the fixed-capacity policy pays for itself mainly by rationing low-risk access, and only secondarily through the returns to treating high risk.

The distributional consequences are the other side of that result. The fixed-capacity policy is a reallocation, not a Pareto improvement, and at high intensities the reallocation is severe. At $X = 80\%$ the low-risk treatment probability falls from 0.689 to 0.195, lengthening the low-risk expected wait from 5.4 months to 4.1 years and cutting low-risk treatment volume by 67%. Their severe-depression prevalence rises by 0.215 percentage points. That deterioration is fiscally invisible, and indeed mildly favourable: losing access to therapy, which is free out of work and costly in work, slightly raises their employment. It is nonetheless a real welfare cost borne by identifiable people, and the aggregate figures in Table 6 should not be read without it. This is also the sharpest form of the equity objection to genetic targeting: a polygenic index is not a diagnosis, and rationing on it makes an unchosen endowment the basis of a claim

Figure 10: Priority access by genetic risk, pre-COVID initial distribution



Notes: Pre-COVID initial distribution; targeted policy, so only high genetic risk receives priority. The horizontal axis is the share of high-risk applicants treated in the period they apply, λ_H; both supply regimes deliver the same λ_H at each intensity, so it indexes policy intensity for both. The benchmark is λ = 68.9%, calibrated to the 1.26 million treatments begun from 1.83 million referrals in NHS Talking Therapies in 2023–24. Panels B–D are anchored at the benchmark, where all effects are zero by construction. In panel A the high-risk series lies on the 45-degree line by construction; what the panel shows is how far fixed capacity must push low-risk access down to deliver a given high-risk access rate. Under fixed capacity the trade-off is close to linear: each additional percentage point of high-risk access costs 1.9 points of low-risk access, the ratio of the two groups' applicant volumes in the market-clearing condition. Under elastic supply low-risk access is unchanged by construction. Panels B–D are population aggregates in a single tone, with line style carrying the supply regime throughout. A negative entry in panel C means the policy moves people into work.

on public care. The people who wait longer are not at low risk of depression in any absolute sense, only at lower risk than the group given priority. We report the trade-off rather than resolve it.

The gain comes from targeting and not from speed. In an earlier specification we also simulated a universal policy raising λ for both groups by the same factor, leaving the gap in (21) at zero. Under fixed capacity this returned exactly zero on every margin: the market-clearing solution returns λ to its benchmark value and the fiscal balance is unchanged to the fourth decimal. With a fixed number of slots, accelerating everyone equally is a no-op. The effects in Table 6 are therefore attributable to *who* receives treatment, and not to how quickly the queue moves.

A final question is whether the planner can see what the policy conditions on. Throughout, both the planner and the individual observe the genetic-risk group. Neither does today. Polygenic indices are not held in administrative data, are not returned to patients, and play no part in how talking therapy is rationed. Two things can close that gap. The first is direct measurement: the NHS has announced whole genome

Table 6: Priority access by genetic risk, pre-COVID initial distribution

Wait gap	λ_L	λ_H	Take-up	Δ Unemp. (pp)	Δ Severe MH (pp)	Δ MH trap (pp)	Δ NPV (£/person)
Panel A: fixed capacity (reallocation, no incremental cost)							
Benchmark	0.6885	0.6885	0.01822	—	—	—	—
20%	0.5905	0.7381	0.01822	+0.002	−0.012	−0.001	−1.1
40%	0.4821	0.8036	0.01822	−0.013	−0.014	−0.011	+11.7
60%	0.3461	0.8652	0.01822	−0.012	−0.018	−0.007	+12.6
80%	0.1954	0.9768	0.01822	−0.021	−0.027	−0.018	+19.1
Panel B: elastic supply (net of therapy provision cost)							
20%	0.6885	0.7381	0.01891	+0.003	−0.027	+0.000	−4.4
40%	0.6885	0.8036	0.01974	−0.013	−0.057	−0.012	+6.3
60%	0.6885	0.8652	0.02075	−0.011	−0.088	−0.009	+2.8
80%	0.6885	0.9768	0.02220	−0.012	−0.140	−0.014	+1.4

Notes. Only high genetic risk receives priority. The wait gap is the target reduction in the high-risk expected wait relative to low risk, implemented through (21). Panel A solves λ_L so that total treatments equal the benchmark; Panel B restores λ_L and holds λ_H at the value Panel A pinned down, so both panels grant the high-risk group the same access and differ only in who funds it. Panel B's realised gap is therefore smaller than the target: 6.7, 14.3, 20.4 and 29.5 percent respectively. Take-up is treatments per working-age person-year; in Panel A it is pinned by construction, clearing to within 6×10^{-5} of the benchmark. The MH trap is the share of person-years spent both depressed and out of work. Δ NPV is the change in discounted tax revenue less benefit expenditure less therapy provision cost, per person, discounted to labour-market entry at $r = 0.046$; provision is unchanged in Panel A by construction and charged at market price less the patient's contribution in Panel B. At the benchmark the expected wait is 5.4 months for both groups, unemployment 6.47 percent, severe depression 15.83 percent, the MH trap 3.49 percent, the fiscal balance £58,449 per person and therapy provision £74.0 per person. Effects are population aggregates over 10,000 agents, with common random numbers across scenarios.

sequencing for every newborn over the coming decade (Davies, 2025), which would place the index in the record for exactly the cohorts the model follows through working life. The second requires genotyping nobody, and asks whether the same targeting can be recovered from an individual's observable trajectory O_i, comprising age and the histories of mental and physical health, employment, wages and therapy receipt. Written as a sufficient-statistic condition, the question is whether

$$g_i \perp (\text{policy-relevant outcomes}) \,\big|\, O_i. \tag{22}$$

It can be answered inside the model, where the true g_i of every simulated agent is known. We summarize each agent's history up to age A into the observables a clinician or a caseworker would plausibly hold, and predict the binary genetic-risk group by gradient boosting with five-fold cross-validation, so that every figure is out of sample. Observables become a near-sufficient statistic for genetic risk early in working life. By age 30, on a handful of observed years, the cross-validated area under the ROC curve is 0.974; from age 35 it sits between 0.992 and 0.994 and is flat thereafter. Further history adds nothing after 35, so the statistic has already saturated by then. Appendix H.5 reports the predictor and the full set of estimates.

That has two implications. The informational assumption the model imposes does little work from age 30 onward: our agents are given a genotype nobody in the data observes, but a learner restricted to what is observed classifies them almost as well, so the assumption grants them little they could not have inferred for themselves. Genetic targeting also turns out not to require genotyping. An agency holding a decade of ordinary records could rank individuals by genetic risk accurately enough to implement close to the allocation studied here without sequencing anyone, which makes the obstacle to the policy an administrative and political one.

One qualification limits how far the second conclusion travels. The exercise characterises the model, not the world. In the simulation g enters only the wage equation and the mental-health transition process, and the sole remaining source of persistent unobserved heterogeneity is the permanent type, so a long history is far more informative about g than it could be in a population where genetic risk competes with everything the model abstracts from. What the exercise establishes is that our informational assumption is close to innocuous inside the model. Whether observed histories substitute for a genotype in practice is an empirical question this design cannot settle.

Appendix H.2 repeats the whole exercise on an initial distribution constructed from post-pandemic waves. The ranking of the two regimes is unchanged, but the gains are larger throughout, and positive at every intensity in both supply regimes. Their source also shifts: pre-COVID the fixed-capacity policy pays for itself mainly by rationing low-risk access, while post-COVID treating the high-risk group pays for itself. The appendix reports the magnitudes and the caveats that attach to the smaller post-COVID sample.

6.4 The Return to an Additional Therapist

Section 6.3 held the therapist workforce fixed and asked who should be treated first. The complementary question is what that workforce is worth at the margin: whether training one more therapist repays what it costs to produce. This section raises λ uniformly, so that every wait falls by the same proportion and no group is given priority, and prices the resulting expansion against the cost of the therapists who deliver it.

The exercise follows each individual over the remainder of the life cycle and discounts tax revenue, benefit expenditure and the cost of the additional courses to age 25 at the model's interest rate of 4.6 percent. Because applications fall as treated people recover, raising course volume by one percent requires λ to rise by about 1.3 percent. We convert the per-person figures to a per-therapist basis using the therapist density implied by the baseline caseload, one therapist per 3,923 people of working age. Table 7 reports the result.

Expansion repays its cost at every scale we examine. Five percent of additional capacity, roughly 565 therapists, returns £494,000 per therapist in lifetime net fiscal terms against a capitalised training investment of about £122,000, a benefit–cost ratio of 4.1. At a thirty percent expansion the ratio has fallen to 1.5, and it remains above one even when training is valued at the deliberately conservative £161,000 upper bound that charges the whole of a trainee's final-year salary to training. Appendix H.4 sets out where these training figures come from.

What the exchequer recovers is mostly benefit expenditure. Reduced benefit payments exceed additional tax revenue by a factor of between 1.5 and 1.7 at every scale, so the fiscal case rests on people leaving health-related benefits; higher earnings among those already in work contribute the smaller part of it.

Table 7: The Fiscal Return to Expanding Therapy Capacity

Expansion	Extra therapists	Δ tax	Δ benefits	Δ provision	Net NPV	Per therapist	Benefit–cost ratio Central	Benefit–cost ratio Upper
		(£ per person, lifetime)						
5%	565	+3.19	−5.03	+1.93	+6.29	£493,612	4.05	3.07
10%	1,092	+4.43	−6.50	+4.01	+6.92	£271,328	2.22	1.69
15%	1,620	+5.56	−9.37	+6.70	+8.23	£215,245	1.76	1.34
20%	2,185	+7.41	−11.96	+8.67	+10.69	£209,684	1.72	1.30
30%	3,277	+10.93	−18.10	+15.14	+13.88	£181,535	1.49	1.13

Notes: Untargeted expansion: λ is raised uniformly, so no group is given priority. Raising course volume by one percent requires λ to rise by about 1.3 percent, since applications fall as treated people recover. Each individual is followed over the remainder of the life cycle and all flows are discounted to age 25 at $r = 0.046$. Provision is the cost of the additional courses at market price less the patient's contribution, and Net NPV is Δ tax less Δ benefits less Δ provision. The per-therapist column applies the therapist density implied by the baseline caseload, one per 3,923 people of working age, and so scales inversely with the assumed 115 courses per therapist per year. Benefit–cost ratios divide that return by the capitalised cost of training a High Intensity CBT therapist, £122,000 centrally and £161,000 at the conservative upper bound (Appendix H.4). Appendix H.3 reports the health and labor-market outcomes.

The returns diminish sharply. The gain per person rises from £6.29 to £13.88 across the range, but the workforce required rises almost sixfold, so the return per therapist falls by nearly two-thirds. Provision cost rises almost eightfold while the net gain barely doubles, because each additional course costs the same to deliver but goes to a progressively less severe case. The first therapists hired are worth several times the last, which is the same logic that made reordering the queue in Section 6.3 the better buy per pound: reallocation moves treatment towards the cases where it converts into employment without paying for any additional courses at all.

The aggregate health effects are correspondingly modest. Against a benchmark in which 14.50 percent of working-age person-years are spent in severe depression, 6.77 percent in unemployment and 3.43 percent in the depression trap, a thirty percent expansion lowers severe depression by 0.111 percentage points, unemployment by 0.031 and trap prevalence by 0.030. All three improve monotonically in capacity. The large fiscal totals come from the size of the population; the effect on any one individual is small.

Although the expansion targets nobody, its benefits fall unevenly. At a thirty percent expansion severe depression falls by 0.199 percentage points among the high-risk group against 0.035 among the low-risk group, a ratio of 5.7, even though λ rises equally for both. High-risk individuals apply for treatment more often, so they absorb a disproportionate share of the new capacity with no rule directing it to them. The self-selection that Section 6.3 exploits deliberately is already at work here by default, so untargeted expansion turns out to be progressive by accident of who applies.

Appendix H.3 states two caveats alongside the full outcome table, and both bound how hard these numbers can be pushed. The per-therapist conversion rests on an assumed caseload of 115 courses per therapist per year. Every per-therapist figure scales inversely with it, so a published full-time-equivalent workforce statistic should replace it, and until one does the level of the return is less firmly established than its ranking across scenarios. The second caveat concerns discounting. We discount each individual's flows to their own age 25, which is a lifetime-welfare convention; a Treasury costing would discount to the present instead, and doing so raises the returns by a factor of between 1.1 and 1.6. On that margin

Table 7 is conservative. None of these figures measures welfare. They are flows to the exchequer, and since therapy is publicly funded its provision cost never enters household utility, so the welfare return exceeds the fiscal one reported here.

7 Conclusion

In this paper, we show that genetic predisposition to depressive symptoms strongly predicts lifetime mental health and economic outcomes. A one standard deviation increase in genetic risk for depression raises the probability of probable major depression by 6.9 p.p., lowers the probability of employment by 3.1 p.p. and lowers labor income by 0.12 s.d. These results are robust to the inclusion of controls for socioeconomic background and broadly consistent with within-family estimates. We identify the "depression trap", a self-reinforcing cycle of depression and non-employment, as a key mechanism explaining this disparity in labor market outcomes. Men with a high genetic risk for depression are more likely to enter the trap and less likely to leave it. Each leg of that cycle is identified from a shock of its own, involuntary job loss and the recurrence of a depressive episode, and both shocks cost high-risk men more employment despite doing them no more clinical damage.

We then estimate a life-cycle model of mental health, benefits, treatment and labor supply, in which therapy is rationed rather than priced. Holding initial conditions fixed, genetic endowment drives inequalities in lifetime earnings, employment, asset accumulation and health. Its most consequential role, though, is not in how ill people become but in whether they come back. Placing every agent in the depression trap and simulating recovery, severe depression returns to baseline within four years at nearly the same speed for both genetic groups, while a decade later non-employment among high-risk men remains four times as far above its own baseline. Because talking therapy in England is free at the point of use and rationed by a queue, the instrument a planner holds is the order of that queue. Giving high-risk individuals priority raises employment, lowers severe depression and improves the fiscal balance, and under fixed capacity it does so without treating anyone extra and so without costing the exchequer anything. The gain comes from who is treated rather than from how quickly the queue moves: accelerating everyone equally returns exactly zero. The policy is nonetheless a reallocation and not a Pareto improvement: at high intensities the wait facing lower-risk patients lengthens from months to years, and that cost has to be weighed against the gain rather than netted out of it.

Adding to the queue's capacity, instead of reordering it, also repays its cost and gives a way of valuing the therapist workforce itself. Raising the treatment probability uniformly returns £494,000 per additional therapist at a five percent expansion against a capitalised training investment of about £122,000, and the return stays above the cost of training at every scale we examine, though it falls by two-thirds as capacity grows and each additional course reaches a less severe case. Most of what the exchequer recovers is reduced benefit spending, not added tax revenue, and the gains accrue disproportionately to high-risk men even when no rule directs treatment towards them, because they are the ones who apply.

Our analysis builds on recent advances in psychiatric genetics (Baselmans et al., 2019). It highlights the untapped potential of Polygenic Indices to inform cost-effective precision public health strategies. Currently, PGIs explain only about 5% of the variation in depressive symptoms within our sample. Because of this, our estimates likely represent a lower bound of the true impact of genetic risk. Yet, even at this lower bound, the life-cycle effects and policy responses we uncover are non-trivial.

Furthermore, the predictive power of GWAS is rapidly increasing. As larger-scale databases and higher-quality sequencing become available, genetic predictors could theoretically capture up to 40% of the variation in depression (Sullivan et al., 2000; Wainschtein et al., 2025). The UK National Health Service already plans to offer whole-genome sequencing to all newborns over the next decade (Davies, 2025). As these predictive tools mature, using ex-ante genetic screening to keep vulnerable individuals out of the "depression trap" will become a highly practical policy option.

The data and scope of this paper limits the applicability of our findings: we focus on the working-age men as we have very few observations below age 25 and to avoid modelling fertility decisions. Future research could use data focussed on early childhood development to explore in more detail the mechanisms by which genetic risk for depression affects the development of mental health and human capital. Early-life, targeted mental health interventions could plausibly be even more cost-effective, insofar as they could improve educational and occupational attainment. Follow-up research could also examine how genetic risk for depression shapes the career and mental health outcomes of women: women suffer from depression at higher rates than men (IHME, 2025) and have more elastic labor supply (Keane, 2011). Appendix B shows that the descriptive evidence of Section 3 replicates for women, and often more strongly, so what stands in the way is not the reduced form but the modelling; Appendix C sets out what would be required. Genetic risk predicts women's completed fertility as well as their health and labor supply, about a third of their employment penalty runs into caring for home and family rather than into disability, and the recurrence design that identifies the mental-health-to-employment leg of the trap fails for them because episodes coincide with the employment interruptions around childbearing. Extending the model to women therefore means treating fertility, labor supply and health as jointly determined, rather than re-estimating the present model on a different sample. Furthermore, our analysis focusses on white men with european-like ancestry, despite the fact that ethnic minorities suffer from mental health issues at higher rates in the UK (Nazroo and Iley, 2011). This is due to data limitations, both in our discovery and target samples and due to the limited portability of PGIs across ancestry groups (Kim and Milliken, 2018). As more data becomes available, and better methods are developed for improving the portability of PGIs, exploring the role of genetic risk for depression in shaping the mental health and labor market outcomes of minority ethnic groups will be of increasing research interest.

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Appendices

A Genetic Data and Polygenic Indices

We summarize genetic predisposition using polygenic indices (PGIs) constructed from common genetic variants. The raw genotype data record variation at single-nucleotide polymorphisms (SNPs), which are genomic locations where individuals differ by a single DNA base. At each SNP j, an individual carries two alleles, one inherited from each biological parent. We code the genotype as an allele count $SNP_{i,j} \in \{0, 1, 2\}$, which equals the number of effect alleles carried by individual i at SNP j.

GWAS provide the weights used to aggregate SNPs into a scalar index. In a GWAS discovery sample, indexed by k, the outcome of interest Y_k is regressed on each SNP separately:

$$Y_k = \beta_j^{GWAS} SNP_{k,j} + \gamma \mathbf{X}_k + u_{k,j}, \tag{23}$$

where $\mathbf{X}_k$ includes standard controls such as sex, age polynomials, and genetic principal components to address population structure. The estimated SNP coefficients $\widehat{\beta}_j^{GWAS}$ are then applied to individuals in our UKHLS analysis (target) sample, indexed by i, to form the PGI:

$$PGI_i = \sum_{j=1}^{J} \widehat{\beta}_j^{GWAS} SNP_{i,j}. \tag{24}$$

Consistent with evidence that predictive performance improves when many variants are included, we construct the index using genome-wide SNP information rather than restricting to genome-wide significant hits (Okbay et al., 2016).

The UKHLS genetic subsample includes blood and saliva genotypes for 9,961 respondents. The released data contain 24,214,238 SNPs (hg38 build). After standard quality control, 9,920 individuals remain for analysis. We construct PGIs using PLINK2 and a standard workflow following Choi et al. (2020), combining UKHLS genotypes with publicly available GWAS summary statistics. In addition to the depressive-symptoms PGI used in the main analysis, we also construct PGIs for a broader set of traits for robustness and controls, including educational attainment, health, and related outcomes.

A.1 UKHLS Genetic Sample

We use data from Understanding Society: the UK Household Longitudinal Study (UKHLS). UKHLS is a nationally representative panel with rich information on income, employment, education, health, and well-being. A genetic subsample collected blood and saliva from 9,961 respondents. Genotypes are available on 24,214,238 SNPs (hg38 build). After standard quality control, 9,920 individuals remain.

We construct PGIs using PLINK2 and a standard pipeline following Choi et al. (2020). We combine UKHLS genotypes with publicly available GWAS summary statistics. In addition to the depressive-symptoms PGI used in the main analysis, we construct PGIs for a wider set of traits for robustness and controls, including education, health, and related outcomes.

The genetic data can be linked to family members within UKHLS. This feature allows us to characterize genetic relationships within households and, when needed, study patterns consistent with assortative mating or intergenerational links. Table 8 reports sample sizes for the main variables and the linked-relative subsamples after quality control.

Table 8: Sample Size by Information Availability

Sample Category	N
Core Analysis Sample	
Full genetic sample (post-QC)	9,920
Physical Health	9,920
Mental Health	9,888
Educational attainment	9,915
labor force status	9,920
Benefits	9,920
Saving information	7,178
Therapy	7,299
Linked Biological Relatives	
Biological parent genotype linked	550
Biological child genotype linked	392
Biological sibling genotype linked	223

Notes: This table reports sample sizes for key variables after applying standard genetic quality control (QC) protocols. "Full genetic sample (post-QC)" refers to individuals with valid genomic data. "Linked" categories indicate successful matching of genotype records between the primary respondent and their biological relatives within the dataset.

B Additional Empirical Results

B.1 Descriptive Statistics

Table 9 compares men below and above the median of the depression polygenic index on health, labor market and background characteristics. The health and labor market gaps are large and precisely estimated, while the parental background variables are balanced, which is the comparison Section 3 relies on.

Table 9: Descriptive Statistics and Tests of Differences by Genetic Risk for Depressive Symptoms

	High Genetic Risk		Low Genetic Risk		
Variable	Mean	(Std. Dev.)	Mean	(Std. Dev.)	Diff. (p-value)
Health Outcomes					
Depression (Prob.)	0.23	(0.42)	0.12	(0.32)	0.00
Poor Physical Health (Prob.)	0.18	(0.38)	0.13	(0.33)	0.00
Disability (Prob.)	0.05	(0.22)	0.02	(0.13)	0.00
Therapy (Prob.)	0.02	(0.15)	0.01	(0.11)	0.00
Economic Outcomes					
Annual Pay (£)	33,888	(18,942)	36,493	(21,051)	0.00
High Occ. (Prob.)	0.19	(0.39)	0.24	(0.43)	0.00
Total Assets (£)	18,825	(51,119)	24,342	(101,924)	0.10
Employed (Prob.)	0.89	(0.32)	0.94	(0.24)	0.00
Benefit Receipt (Prob.)	0.28	(0.45)	0.22	(0.42)	0.00
Annual Benefits (£)	3,197	(3,455)	2,765	(3,041)	0.00
Demographics & Human Capital					
College (Prob.)	0.27	(0.45)	0.32	(0.47)	0.00
Father High Occ. (Prob.)	0.18	(0.38)	0.20	(0.40)	0.31
Mother's Education (Yrs)	10.90	(1.92)	11.00	(2.01)	0.26
Age	42.81	(8.33)	42.74	(8.19)	0.55
Observations	9,770		10,574		

Notes: Standard deviations are reported in parentheses. The final column reports p-values from a two-sided t-test for the difference in means between individuals with an depressive symptoms PGI above (High Genetic Risk) and below (Low Genetic Risk) the median. Parental education is mapped to years based on highest qualification. High occupation indicators are binary for managerial roles (SOC90 < 20). Sample sizes for parental variables are smaller ($N \approx 1,500$–$2,500$) due to retrospective reporting. All variables are for UKHLS men aged 25–55.

B.2 Descriptive Employment and Mental-Health Transitions

Table 10 shows that both legs of this relationship are present in our own data. We estimate a dynamic logit for each transition on genotyped men aged 25–55, controlling for lagged earnings, demographics, survey wave, the genetic principal components, and within-person time averages of the time-varying regressors, so that the comparisons are net of permanent individual heterogeneity as well as of income. Panel A reports the effect of last year's employment on this year's mental health. Men who were employed are 9.2 percentage points less likely to become depressed (s.e. 1.8) and 18.0 percentage points more likely to recover if they were depressed (s.e. 2.8). Panel B reports the reverse. Men who were depressed are 1.2 percentage points more likely to leave employment (s.e. 0.3) and 10.2 percentage points less likely to re-enter it (s.e. 2.1).

These magnitudes are associations, not treatment effects, and the recovery and re-entry rows in particular are estimated on the smaller at-risk samples. They nonetheless describe a feedback loop of the size the literature above would lead one to expect, and they do so holding both permanent heterogeneity and past income fixed. Section 3.5 returns to each leg with event-study evidence.

Table 10: The Two-Way Relationship Between Employment and Mental Health

Transition	Effect on annual transition prob. (pp)	Observations at risk
Panel A: Effect of being employed at $t-1$ on mental health transitions		
Onset (healthy → depressed)	-2.15***	67,789
	(0.71)	
Recovery (depressed → healthy)	3.97***	13,787
	(1.24)	
Panel B: Effect of being depressed at $t-1$ on employment transitions		
Exit (employed → non-employed)	1.09***	73,534
	(0.17)	
Entry (non-employed → employed)	-3.79***	8,042
	(0.50)	

Notes: Each entry is an average marginal effect, in percentage points, of the lagged state named in the panel heading on the transition named in the row, from a dynamic logit on UKHLS men aged 25–55 observed for at least five waves. Every row is signed to read as the effect on the transition named, so a positive entry in Panel B row one means depression raises the probability of leaving employment. Marginal effects are averaged over the men at risk of each transition. Controls: lagged earnings, so these are comparisons at a given past income, plus number of children, a quadratic in age and wave fixed effects, and within-person time averages of the time-varying regressors, so the comparisons are net of permanent heterogeneity. Unlike Table 3 this table is estimated on the full panel rather than the genotyped subsample, since it has no genetic regressor; the genotype restriction leaves only 544 within-person employment switches against 2,772 depression switches, too few to sign Panel A. Depression is GHQ-12 caseness ≥ 4. Standard errors, clustered at the individual level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Dynamics by Genetic Risk: Alternative Specifications

Table 11 repeats the main transition table at an alternative GHQ-12 caseness threshold of 2, which counts mild symptoms and above, instead of the threshold of 4 used throughout the paper. The estimates are almost unchanged. Depression onset is 3.64 percentage points higher for high-risk men at the lower threshold against 3.51 percentage points in the main table, and the trap exit differential is −26.14 percentage points against −24.44. The choice of caseness threshold is therefore not driving the genetic gradient in either the entry or the persistence margin.

Table 11: Dynamics by Genetic Risk: Alternative GHQ-12 Threshold (Caseness ≥ 2)

Annual transition probability (%)	Low genetic risk (1)	High genetic risk (2)	Difference (2)−(1)
Panel A: Mental health dynamics			
Onset (healthy → depressed)	16.29	19.93	3.64***
	(0.70)	(0.83)	(1.09)
Recovery (depressed → healthy)	55.61	42.51	-13.10***
	(1.62)	(1.37)	(2.12)
Panel B: Employment dynamics			
Exit (employed → non-employed)	1.58	2.26	0.68**
	(0.18)	(0.21)	(0.28)
Entry (non-employed → employed)	38.40	25.07	-13.33***
	(3.83)	(2.21)	(4.42)
Panel C: "Depression trap" dynamics			
Entry into trap	1.22	1.86	0.64***
	(0.16)	(0.18)	(0.24)
Exit from trap	61.09	34.95	-26.14***
	(5.78)	(3.18)	(6.60)

Notes: Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit on genotyped UKHLS men aged 25–55 observed for at least five waves. Genetic risk is a median split of the depression polygenic index. The model is estimated separately within each risk group and both fits averaged over the same pooled covariate distribution, so column (3) is a difference in behaviour, not composition; its standard error is the square root of the sum of the two squared standard errors, valid because the subsamples are disjoint. Each cell conditions on the origin state only, so entries are marginal with respect to the other lagged state. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 2; the "depression trap" is non-employment jointly with caseness ≥ 2. Standard errors clustered on the individual. All panels use 15,206 person-waves on 1,534 individuals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Conditioning on latent types

The main table conditions on lagged earnings, demographics, wave and the genetic principal components, but not on the permanent heterogeneity the structural model carries as the latent type x_i. Since types govern productivity and health trajectories, a natural question is whether the genetic gradient is instead a gradient in type. Table 12 adds indicators for the five types of Appendix F to the specification, holding the sample fixed at the 1,534 men who have a type assignment.

Column (4) is the specification to read. The types are estimated on moments residualized on a depression polygenic index, so conditioning on them is only clean when the sample is split on that same index; the check is otherwise contaminated by construction, since the type dummies then absorb part of the genetic signal they were built to exclude. On that split the type dummies change essentially nothing. Five of the six margins land between 92 and 121 percent of their uncontrolled values, only trap exit moves materially, from −21.30 to −17.90 percentage points, and every estimate remains significant at the five percent level or better. The implied trap spell is 1.80 years at low genetic risk against 2.66 at high, compared with 1.72 against 2.72 without the type dummies.

Column (2) shows what the contaminated version of the same check looks like: the persistence margins shed about a fifth, with recovery falling from −15.31 to −12.11 and trap exit from −24.44 to −19.06. That attenuation is the mechanical consequence of conditioning on a variable built from person-average mental health that has been purged of a correlate of the split variable rather than of the split variable itself. It is reported here because the distinction between the two columns is the substance of the exercise, not because the column (2) attenuation should be read as evidence of confounding.

The residual caveat applies to both columns. Types are a five-point discretization of person-mean mental health, physical health and wages, and only the linear projection on the polygenic index is removed when they are constructed, so a nonlinear residue of "conditioning on a consequence of genetic risk" survives. That residue biases against finding a gradient, so the estimates in columns (2) and (4) are if anything conservative. A correlated random effects version of the same table is the limiting case of this logic and is uninformative for exactly that reason: within-person time averages absorb all between-person variation, which is the only variation a fixed genetic endowment has, and every one of the six transitions is nulled. Five type dummies do not, which is what makes them informative.

Table 12: Dynamics by Genetic Risk, Conditioning on Latent Types

	Main index (Table 3)		Type-clustering index (alternative depression PGI)	
Difference in annual transition probability, high − low risk (pp)	Baseline (1)	+ types (2)	Baseline (3)	+ types (4)
Panel A: Mental health dynamics				
Onset (healthy → depressed)	3.51***	3.75***	6.28***	6.56***
	(0.78)	(0.70)	(0.75)	(0.65)
Recovery (depressed → healthy)	-15.31***	-12.11***	-14.54***	-13.36***
	(2.72)	(2.20)	(2.79)	(2.19)
Panel B: Employment dynamics				
Exit (employed → non-employed)	0.71**	0.76***	0.81***	0.98***
	(0.28)	(0.28)	(0.26)	(0.27)
Entry (non-employed → employed)	-13.70***	-11.65***	-9.12*	-9.20**
	(4.47)	(4.11)	(4.67)	(4.22)
Panel C: "Depression trap" dynamics				
Entry into trap	0.69***	0.63***	0.98***	1.02***
	(0.23)	(0.23)	(0.21)	(0.21)
Exit from trap	-24.44***	-19.06***	-21.30***	-17.90**
	(7.68)	(6.96)	(8.19)	(6.99)
Expected trap spell, low risk (years)	1.66	1.77	1.72	1.80
Expected trap spell, high risk (years)	2.80	2.68	2.72	2.66
Latent type dummies	No	Yes	No	Yes

Notes: Each entry is the difference in the average predicted annual transition probability between high- and low-genetic-risk men, in percentage points, constructed exactly as in Table 3: a dynamic logit estimated separately within each risk group, both fits averaged over the same pooled at-risk covariate distribution, with SE(diff) = $\sqrt{se_0^2 + se_1^2}$. Columns (2) and (4) add indicators for the five latent types x_i estimated by the clustering procedure of Appendix F and treated as a fixed state variable in the structural model. Columns (1)–(2) split on the depressive-symptoms index used throughout the paper; columns (3)–(4) split on the alternative depression index that the type clustering residualises its moments against, so that in column (4) the type dummies are orthogonal to the split variable by construction. Column (4) is therefore the cleaner comparison: the type moments are purged of the column (3)–(4) index but not of the column (1)–(2) index, with which they correlate 0.79, so column (2) mechanically absorbs part of the genetic signal that column (4) does not. All four columns use the same 15,206 person-waves on 1,534 men, namely the Table 3 sample restricted to men with a type assignment; the two median splits agree on 78.9 percent of them. Expected trap spells are the reciprocal of the exit probability. Controls otherwise as in Table 3. Standard errors clustered on the individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Women

Table 13 repeats the same table for women. The construction is identical except for the sex filter and the labor-force states admitted: women's sample adds looking after home and family, because on the men's state set those person-waves are deleted rather than counted as non-employment, and that is where most of women's non-employment sits.

Every one of the six gradients holds with the same sign and at conventional significance. Women at high genetic risk are 4.7 percentage points more likely to become depressed in a given year and 8.7 points less likely to recover; 0.9 points more likely to leave employment and 3.8 points less likely to enter it; and 1.7 points more likely to enter the depression trap. Entry into the trap is in fact more strongly graded for women than for men, at 1.65 percentage points against 0.69.

Exit is not. The women's exit differential is −7.66 points against the men's −24.44, and it is the one estimate in this appendix that moves materially with a specification choice: on the men's narrower state set it falls to −1.65 (s.e. 5.62) and is indistinguishable from zero. Since the paper's mechanism is about persistence rather than incidence, this is the most consequential difference between the sexes, and the sensitivity is itself informative. For men, leaving employment while depressed is close to unambiguous, and the state the model calls the depression trap is the state the data record. For women a large share of the same transitions run into and out of family care, which for many is a response to a birth rather than a symptom of illness. Appendix C shows that this is not a hypothetical concern: the arrival of a first child moves women's employment by more than thirty percentage points while moving their mental health no more than it moves men's.

Table 13: Mental Health, Employment and "Depression Trap" Dynamics by Genetic Risk: Women

Annual transition probability (%)	Women: Low genetic risk (1)	Women: High genetic risk (2)	Women: Difference (2)−(1)	Men: Difference (3)
Panel A: Mental health dynamics				
Onset (healthy → depressed)	14.03	18.73	4.71***	3.51***
	(0.51)	(0.60)	(0.78)	(0.78)
Recovery (depressed → healthy)	54.85	46.11	-8.74***	-15.31***
	(1.46)	(1.15)	(1.85)	(2.72)
Panel B: Employment dynamics				
Exit (employed → non-employed)	3.48	4.34	0.86***	0.71**
	(0.22)	(0.24)	(0.33)	(0.28)
Entry (non-employed → employed)	21.09	17.29	-3.81**	-13.70***
	(1.30)	(0.91)	(1.59)	(4.47)
Panel C: "Depression trap" dynamics				
Entry into trap	2.66	4.32	1.65***	0.69***
	(0.20)	(0.22)	(0.30)	(0.23)
Exit from trap	49.89	42.22	-7.66**	-24.44***
	(3.24)	(2.07)	(3.84)	(7.68)

Notes: The construction is identical to Table 3, which reports the men's version, except for the sex filter and the set of labour-force states admitted. Each entry is an average predicted annual transition probability, in percentage points, from a dynamic logit estimated separately within each genetic-risk group and averaged over the same pooled covariate distribution, so the difference is one of behaviour rather than composition. The women's sample admits employment, unemployment, long-term sickness *and looking after home and family*; the men's table uses the first three only. That difference is not cosmetic: family care is where most of women's non-employment sits, and on the men's state set it is deleted from the sample rather than counted as non-employment. On that narrower definition the women's trap-exit differential is -1.65 (s.e. 5.62) rather than the -7.66 reported here. Column (3) reproduces the men's difference from Table 3 for comparison. Genetic risk is a median split of the depression polygenic index. Controls: lagged earnings, number of children, a quadratic in age, wave fixed effects and the ten genetic principal components. Depression is GHQ-12 caseness ≥ 4; the "depression trap" is non-employment jointly with caseness ≥ 4. Standard errors clustered on the individual. The women's estimates use 22,549 person-waves on 2,284 women. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 The Genetic Penalty: Continuous Index and All Outcomes

The main text reports the earnings penalty per standard deviation of the depression polygenic index. Tables 14 and 15 report the same exercises with the index split at the median, which is the form used by the descriptive table, the transition table and the gene-by-environment table, and add the outcomes the main table omits.

Nothing of substance turns on the choice. The two ladders attribute 20 and 21 percent of the earnings association to education, 5 and 6 percent to first occupation, and both leave the same qualitative ordering, with contemporaneous labor force status the largest of the life-cycle blocks on the median split. The by-outcome table tells the same story in larger units: men above the median of genetic risk are 11.2 percentage points more likely to meet the threshold for probable major depression and 4.4 points less likely to be employed, with over half of the labor-supply effects running through contemporaneous mental health. The two outcomes the main table omits behave as the argument implies. Unemployment rises by 1.2 percentage points, well under the 1.9-point rise in long-term sickness, confirming that the withdrawal is into ill health rather than into job search; and the professional-occupation gap of 1.8 points is not attenuated at all by conditioning on current mental health, as one would expect of an outcome largely settled beforehand.

Table 14: The Genetic Earnings Penalty and Its Channels (Binary High/Low Split)

Mediators included	Effect on labor income (s.d.)	Share of baseline (%)	Explained by this block (%)
Baseline	-0.1369***	100.0	–
	(0.0408)		
+ Education	-0.1082***	79.0	21.0
	(0.0359)		
+ First occupation	-0.0999***	73.0	6.0
	(0.0352)		
+ Labor force status	-0.0500	36.5	36.4
	(0.0330)		
+ Current occupation	-0.0256	18.7	17.9
	(0.0291)		

Notes: Each row is the coefficient on genetic risk in a regression of standardized gross labor income, adding one block of mediators at a time. Genetic risk is a binary indicator for being above the median of the depression polygenic index, so each coefficient is the gap between high- and low-risk men. Appendix B reports the continuous per-standard-deviation version. All specifications control for the first ten genetic principal components and for age, survey-wave and interview-year fixed effects. Column (2) expresses each coefficient as a percentage of the baseline, column (3) the share absorbed by the block added in that row. Contemporaneous mental health is deliberately excluded: the blocks above are predetermined relative to current earnings whereas mental health is a contemporaneous outcome of genetic liability, so conditioning on it here would net out part of the effect being decomposed; its share is reported in Table 2. Standard errors clustered on the individual. N = 19,995 person-waves. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15: Which Margin Carries the Genetic Penalty, and How Much Runs Through Mental Health (Binary Split, All Outcomes)

Outcome	High vs low genetic risk	Share explained by mental health (%)	95% CI
Depressed (pp)	11.150***	–	
	(0.991)		
Employed (pp)	-4.367***	55	[38, 103]
	(0.918)		
Long-term sick or disabled (pp)	2.991***	53	[40, 101]
	(0.652)		
Labor income (s.d.)	-0.137***	35	[21, 67]
	(0.041)		
Log gross pay, if employed (log pts.)	-0.050**	15	[1, 68]
	(0.024)		
Holds a degree (pp)	-2.819	≈ 0	
	(2.306)		
Unemployed	1.376**	58	[30, 294]
	(0.571)		
Professional/managerial	-1.622	≈ 0	
	(1.751)		
GHQ-12 distress (check)	0.374***	–	
	(0.031)		

Notes: Column (1) is the effect of genetic risk on the outcome named,in the units given in the row label. Genetic risk is a binary indicator for being above the median of the depression polygenic index, so each coefficient is the gap between high- and low-risk men. Column (2) is the percentage of that gap absorbed when contemporaneous GHQ-12 distress is added, with a cluster-bootstrap interval in column (3); ≈ 0 denotes no attenuation. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. The depression row has no share because it is the mediator itself. Appendix B reports every outcome and the continuous per-standard-deviation version. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Women

Tables 16 and 17 report the by-outcome table and the sequential decomposition for both sexes. Two conventions change relative to the main text, and both are forced by the comparison rather than chosen. First, the sample is the broad working-age one, admitting every activity status: the main text's filter keeps only employment, unemployment and long-term sickness, which applied to women deletes maternity and family care. Second, labor income is standardized once on the pooled sample rather than within each estimation sample, so that male and female coefficients are in common units. Men are re-estimated under both conventions, so the cost of each is visible; only the income rows move, and they move by exactly the ratio of the two standard deviations, 1.0606, which is a change of units and not of substance. Every other men's estimate in these tables reproduces Tables 1 and 2 to the last reported digit.

The penalty replicates, and on most margins it is larger for women. A standard deviation of genetic risk raises the probability of probable major depression by 8.3 percentage points for women against 6.6 for men, lowers employment by 4.0 points against 2.6, and raises long-term sickness by 2.5 points against 1.6. The mediation shares are close across sexes: about 37 percent of women's employment effect and 45 percent of the long-term sickness effect run through contemporaneous mental health, against 40 and 53 percent for men. The income penalty is the one margin that is smaller for women, at 0.078 standard deviations against 0.112, and conditional on working the two sexes lose the same 0.054 log points of gross pay.

Table 16: The Genetic Penalty by Outcome: Men and Women

	Men			Women		
Outcome	Est.	(s.e.)	% via MH	Est.	(s.e.)	% via MH
Probable major depression (GHQ≥4)	6.59***	(0.48)	—	8.31***	(0.44)	—
GHQ-12 distress (s.d.)	0.215***	(0.015)	—	0.269***	(0.013)	—
Employed	-2.61***	(0.76)	40	-4.01***	(0.64)	37
Long-term sick/disabled	1.61***	(0.32)	53	2.48***	(0.29)	45
Unemployed	1.02***	(0.27)	40	0.86***	(0.14)	47
Looking after home/family	0.37**	(0.15)	6	1.19***	(0.45)	32
Labour income (s.d.)	-0.112***	(0.022)	24	-0.078***	(0.012)	26
Log gross pay, if employed	-0.054***	(0.013)	7	-0.054***	(0.014)	4
Degree (predetermined)	-2.39**	(1.04)	≈ 0	-2.63***	(0.88)	16
Person-waves (max)		23,980			32,840	
Individuals		2,605			3,540	

Notes: Each cell is the coefficient on a one standard deviation increase in the depression polygenic index, from a separate regression of the row outcome on the index, the ten genetic principal components, and age, survey-wave and interview-year fixed effects. Binary outcomes are in percentage points. Standard errors, in parentheses, are clustered on the individual; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The sample is the *broad* working-age sample, which admits every activity status (`jbstat` 1–9), ages 25–55, pre-2020. The main text's Table 2 instead uses the narrower men-only filter (employed, unemployed, long-term sick), because that is the sample the structural model speaks to; applied to women that filter would delete maternity and looking after home and family, which is where women's non-employment sits. The men's column therefore differs slightly from Table 2. "% via MH" is the share of the estimate absorbed by conditioning on contemporaneous GHQ-12 distress and probable major depression; it is a descriptive sequential-control statistic, not causal mediation, and is left blank for the mental-health outcomes themselves. Labour income is standardised once on the pooled sample, so male and female coefficients are in common units.

The substantive difference is the destination of the withdrawal. Genetic risk moves women into caring for home and family by 1.19 percentage points per standard deviation against 0.37 for men, and about a third of that movement runs through mental health. Set against the 4.0-point employment effect, family care accounts for roughly thirty percent of women's extensive-margin penalty, a channel that is essentially absent for men.

The ladder in Table 17 shows the same thing from the earnings side. The women's penalty is more fully accounted for by observable channels than the men's, falling to 14 percent of its baseline against 33 percent, and the two blocks that do the work are different: education absorbs 38 percent for women against 19 percent for men, and contemporaneous labor force status a further 39 percent against 32. The final block, the number of children, is the one that behaves anomalously. It does not absorb the penalty for either sex; it *raises* it, by 0.7 percent for men and 3.5 percent for women. Appendix C takes up why.

Table 17: The Genetic Earnings Penalty and Its Channels: Men and Women

	Men		Women	
Mediators included	Effect on labor income (s.d.)	Explained by this block (%)	Effect on labor income (s.d.)	Explained by this block (%)
Baseline	-0.1118***	—	-0.0778***	—
	(0.0218)		(0.0120)	
+ Education	-0.0908***	18.8	-0.0481***	38.2
	(0.0197)		(0.0106)	
+ First occupation	-0.0860***	4.3	-0.0457***	3.0
	(0.0191)		(0.0106)	
+ Labor force status	-0.0504***	31.9	-0.0157*	38.7
	(0.0179)		(0.0090)	
+ Current occupation	-0.0364**	12.5	-0.0079	10.0
	(0.0158)		(0.0077)	
+ Children	-0.0372**	-0.7	-0.0106	-3.5
	(0.0158)		(0.0075)	
Person-waves	23,980		32,840	

Notes: The construction is that of Table 1, estimated separately by sex and with a sixth block added. Each row is the coefficient on a one standard deviation increase in the depression polygenic index in a regression of standardized gross labor income, adding one block of mediators at a time; every block enters as a set of fixed effects with an explicit missing category, so the estimation sample is held fixed down each column. All specifications control for the ten genetic principal components and for age, survey-wave and interview-year fixed effects, with standard errors clustered on the individual. As elsewhere in this appendix the sample is the broad working-age one, ages 25–55, pre-2020, and labor income is standardized once on the pooled sample so that the two sexes share units; the men's column therefore differs from Table 1, which standardizes within the men's narrower estimation sample. Contemporaneous mental health is excluded for the reason given there. The "+ Children" block is the number of own children in the household, top-coded at three; it is reported last because it is the block that differs by sex, and it *raises* the women's coefficient rather than absorbing it. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 The Redundancy Event Study: Specifications and Robustness

This appendix reports the specification behind the redundancy design of Section 3.5 and the robustness checks the text summarizes.

Redundancy is dated from the reason-for-job-ending questions in the BHPS job-history files and the UKHLS individual response files, taking the first calendar year in which a man reports that a job ended because the employer eliminated the position. The sample is men aged 25–55, excluding the retired, balanced on the window $k \in [-3, +3]$ around the event, which yields 591 treated men. Because staggered adoption with heterogeneous effects biases two-way fixed-effects estimators (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021), we use the interaction-weighted estimator of Sun and Abraham (2021). It estimates the saturated regression

$$Y_{it} = \gamma_t + \sum_{e} \sum_{k \neq -1} \delta_{e,k} \, \mathbf{1}\{E_i = e\} \cdot \mathbf{1}\{t - E_i = k\} + \mathbf{X}'_{it}\beta + \varepsilon_{it}, \tag{25}$$

where E_i is the year of i's redundancy and γ_t are interview-year effects, and then reports, for each event time k, the average of the cohort-specific $\delta_{e,k}$ weighted by cohort shares among units observed at k. This averaging is what removes the contamination of already-treated cohorts serving as controls. Identification comes from comparing the redundant to the not-yet-redundant. Controls $\mathbf{X}_{it}$ are age and its square, with education fixed effects; standard errors are clustered on the individual. For the mental-health outcomes, and for every outcome within the smaller genetic subsamples, cohorts are pooled into two-year bins so that each event time is populated; the employment and earnings estimates use the unbinned job-end year. The distinction is not cosmetic. Binning displaces the event date by up to a year, which is immaterial for an outcome that moves slowly and fatal for one that jumps: applying it to employment as well would replace the −20.2 point impact with +2.7, as the penultimate row of Panel A of Table 19 shows.

Mental health is flat before the event on the GHQ-12 caseness scale. The raw employment rate among the treated is 91.1 percent three years before it, 93.8 percent two years before and 94.8 percent in the year before, so employment is at its maximum immediately before the job ends, which is the opposite of what selection of men already withdrawing from work would produce. The raw GHQ-12 caseness path is 1.57, 1.76 and 1.98 over the same three years against 2.54 at the event, but once age and calendar time are absorbed the estimated pre-event coefficients are −0.004 (s.e. 0.221) and −0.180 (s.e. 0.328): the level rise is accounted for by ageing and period effects rather than by a differential trend. Mental health measured three years before a redundancy, which precedes any plausible announcement window, does not predict it. The binary indicator for poor mental health is less well behaved than the count on which it is built: its pre-event coefficients average −4.6 points, and a joint Wald test that they are zero returns $p = 0.06$, against $p = 0.09$ for employment. Neither rejects at conventional levels and neither is comfortable, and the estimates reported below on samples where those pre-trends are measured over more men are correspondingly smaller.

Any remaining contamination is confined to the year immediately before the event. Redundancies are normally announced months before the job ends, so distress recorded at the last pre-event interview is already a response to a job loss the man knows is coming. Treating that year as already treated leaves the mental-health impact where it was, at 9.0 points against a reference at $k = -2$ and 10.2 against $k = -3$, compared with 9.6 at $k = -1$, while attenuating the employment impact to −17.5 and −14.7 points. The asymmetry is itself informative: employment had begun to fall before the job formally ended, which is what an announcement window produces, whereas the distress is concentrated at the event.

Genetic risk does not predict redundancy. Table 18 reports seven specifications, all estimated on the frame the event study itself uses: genotyped men aged 25–55. They cover five questions: whether the index predicts ever being made redundant, the annual hazard of a first redundancy in the at-risk set, the time to it in a proportional-hazard model, its calendar year and the age at which it arrives, and which men survive into the balanced estimation sample. None attains significance at five percent, and the smallest p-value is 0.065, on the probability of being observed at all seven event times.

What makes this more than a bare null is the last panel of the same table. The same index predicts a job ending for *health* reasons at 1.6 percentage points per standard deviation, with $p = 0.004$. Genetic liability loads heavily on the separation margin that mental health ought to govern and not at all on the one it ought not, so the null on redundancy reflects an absence of selection rather than an index too weak to detect it. Within the treated, high- and low-risk men do differ in the *level* of mental health, earnings and employment before the event; they must, since that association is this paper's central finding, and the largest normalized difference in levels is 0.46. The event study differences those levels out, and what it requires instead is that the two groups were not moving differently before the event. They were not: the largest normalized difference in the change from $k = -3$ to $k = -1$ is 0.31, and a joint test that the pre-event coefficients differ by genetic risk returns $p = 0.37$ for employment and $p = 0.89$ for mental health. This is what licenses the gene-by-environment comparison in Table 4, in which the treated sample is split by genetic risk: differences in the response cannot be differences in who is selected into treatment.

Table 18: Is There Selection into Redundancy by Genotype?

Outcome	Estimate	(s.e.)	N
Ever made redundant			
Cross-sectional LPM	0.516	(0.920)	2,401
Annual hazard			
Panel LPM, year and education FE	0.027	(0.102)	21,248
Time to redundancy			
Cox proportional hazard	0.018	(0.055)	2,190
Timing among the redundant			
Age at redundancy	0.695	(0.465)	473
Calendar year of redundancy	-0.157	(0.374)	578
Selection into the estimation sample			
Observed at all seven event times	3.054*	(1.655)	578
Still observed two years after the event	0.328	(1.966)	578
Other reasons a job ended			
Ever better job	0.129	(0.963)	2,401
Ever dismissed	-0.171	(0.433)	2,401
Ever health	-1.565***	(0.540)	2,401
Ever temp ended	-0.378	(0.649)	2,401

Notes: Each row is the coefficient on the standardised depressive-symptoms polygenic index from a separate regression of the row outcome, estimated on genotyped men aged 25–55: the same index and the same frame used everywhere else in the paper, including the event studies whose robustness Table 19 reports. The index is oriented towards well-being, so a higher value is *lower* genetic risk. All specifications control for the first ten genetic principal components and for education; cross-sectional models add birth year, and panel models add age, its square and interview-year fixed effects, with standard errors clustered on the individual. Units are percentage points per standard deviation of the index, except the year and age of the redundancy (years) and the Cox model (log hazard ratio). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The contrast in the last panel is the point. Genetic risk does not predict whether a man is made redundant, when it happens, or who survives into the estimation sample, but it does predict a job ending for health reasons, which is the separation margin mental health should govern. A null everywhere would be consistent with the index being noise; the contrast shows it carries signal that is absent from redundancy specifically. High- and low-risk treated men differ in the *level* of mental health, earnings and employment before the event, as the paper's central finding requires, but not in their pre-event trends, which is what identification needs.

Table 19: Robustness of the Redundancy Design: Estimator, Definition and Falsification

Specification	GHQ-12 caseness at $k = 0$ (pts)	(s.e.)	Employment at $k = 0$ (pp)	(s.e.)	Treated men
Panel A: Estimator, comparison group and anticipation					
Sun–Abraham (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Balanced window (former baseline)	0.74	(0.16)	-20.18	(2.02)	591
Two-year binned cohorts, mental health	0.45	(0.13)	-21.83	(1.03)	3,339
Callaway–Sant'Anna	0.60	(0.10)	-23.09	(1.13)	3,515
Borusyak–Jaravel–Spiess imputation	0.49	(0.07)	-24.85	(1.07)	3,515
Pooled dynamic TWFE	0.63	(0.08)	-22.28	(1.02)	3,515
Never-redundant controls, SA with unit FE	0.60	(0.08)	-21.47	(1.09)	3,339
Never-redundant controls, Callaway–Sant'Anna	0.63	(0.08)	-22.67	(1.04)	3,515
Never-redundant controls, TWFE with unit FE	0.62	(0.08)	-22.08	(1.06)	3,515
One year of anticipation (base $k = -2$)	0.73	(0.09)	-18.81	(1.12)	3,339
Two years of anticipation (base $k = -3$)	0.83	(0.09)	-18.19	(1.18)	3,339
Two-year binned cohorts, both outcomes	0.45	(0.13)	5.02	(1.67)	3,211
Varying (not universal) base period	0.60	(0.10)	-23.09	(1.13)	3,515
Panel B: Definition and dating of the event					
First redundancy, job-end year (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Dated on the interview year instead	0.31	(0.09)	-34.45	(1.03)	3,358
Reported job-end year only, no fallback	0.64	(0.08)	-21.79	(1.03)	3,325
Last redundancy rather than the first	0.55	(0.08)	-22.51	(1.04)	3,369
UKHLS reports only	0.63	(0.11)	-20.32	(1.27)	2,081
BHPS reports only	0.60	(0.12)	-23.17	(1.67)	1,403
"Most recent job ended" question only	0.69	(0.09)	-25.91	(1.18)	2,780
Employed in the year before the event	0.68	(0.09)	-28.57	(1.05)	2,098
Employed at $k = -3$ and $k = -1$	0.69	(0.12)	-25.85	(1.37)	1,189
Panel C: Falsification: other reasons a job ended					
Redundancy, same window (comparison)	0.59	(0.09)	-22.38	(1.15)	1,953
Left for a better job (voluntary)	-0.25	(0.10)	0.80	(1.28)	1,012
Dismissed or sacked (for cause)	-0.25	(0.46)	-35.15	(6.63)	103
Temporary job ended (anticipated)	0.16	(0.26)	-5.93	(3.91)	240
Job ended for health reasons	1.08	(0.32)	-23.70	(3.89)	234

Notes: Each row is a separate pair of event studies: the effect of redundancy on GHQ-12 caseness (0–12 points) and on employment (percentage points), in the year the job ends relative to the year before. The baseline is Sun and Abraham (2021) on men aged 25–55, retirees excluded, identified from men not yet made redundant, controlling for age and its square with interview-year and education fixed effects. Cohorts are unbinned throughout; the two binning rows show why, since binning misaligns the event date by up to a year and destroys the employment estimate. Panel A's never-redundant rows add individual fixed effects, without which the comparison is of employment *levels* between differently composed samples. Panel C re-estimates the design on other recorded reasons for a job ending. Standard errors clustered on the individual, in parentheses.

Table 20: Robustness of the Redundancy Design: Window, Attrition, Sample and Outcomes

Specification	GHQ-12 caseness at $k = 0$ (pts)	(s.e.)	Employment at $k = 0$ (pp)	(s.e.)	Treated men
Panel A: Event window, panel balance and attrition					
Balanced $k \in [-3, +3]$ (baseline)	0.74	(0.16)	-20.18	(2.02)	591
Observed at $k = -1$ and $k = 0$ only	0.59	(0.09)	-22.38	(1.15)	1,953
Unbalanced, any observation in window	0.64	(0.08)	-21.83	(1.03)	3,339
Balanced $k \in [-2, +2]$	0.64	(0.13)	-20.94	(1.54)	982
Balanced $k \in [-1, +1]$	0.54	(0.10)	-20.76	(1.24)	1,613
Unbalanced $k \in [-5, +5]$	0.63	(0.08)	-21.66	(1.03)	3,535
Inverse-probability weighted for balance	0.77	(0.16)	-17.68	(2.33)	572
Multiple imputation for unobserved cells	0.59	(0.09)	-22.14	(1.15)	1,953
Panel B: Sample					
Men 25–55 (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Men 25–64	0.56	(0.08)	-22.72	(0.94)	3,769
Men 30–55	0.65	(0.09)	-21.96	(1.10)	2,866
Men 25–45	0.67	(0.10)	-20.26	(1.27)	2,394
Men 25–55, retirees retained	0.64	(0.08)	-22.13	(1.03)	3,341
Women 25–55	0.54	(0.10)	-22.45	(1.17)	2,803
Men and women 25–55	0.60	(0.07)	-22.07	(0.77)	6,141
Men 40–55	0.67	(0.12)	-24.83	(1.40)	1,946
BHPS era, redundancy before 2009	0.60	(0.12)	-22.74	(1.66)	1,417
UKHLS era, redundancy from 2009	0.65	(0.11)	-20.61	(1.31)	1,922
Panel C: Definition of the outcomes					
GHQ-12 caseness count; employee only (baseline)	0.64	(0.08)	-21.83	(1.03)	3,339
Indicator for caseness ≥ 2 (mild symptoms)	7.57	(1.31)	-21.83	(1.03)	3,339
Indicator for caseness ≥ 4 (probable major depression)	7.19	(1.16)	-21.83	(1.03)	3,339
Employment including self-employment	0.64	(0.08)	-20.07	(0.91)	3,339
Unemployment rather than employment	0.64	(0.08)	18.71	(0.87)	3,339
GHQ-12 Likert score (0–36) rather than caseness	0.86	(0.15)	-21.83	(1.03)	3,339

Notes: A continuation of Table 19, on the same specification and column definitions. Panel A varies the event window. The balance filter, used as the baseline until this revision, keeps only 591 of 3,515 treated men; it selects mildly on outcomes (raw within-person employment change −21.2 against −23.4 points for the men it drops) but costs six-sevenths of the sample, so the unbalanced window is now the baseline. The imputation row completes the seven-cell grid by predictive mean matching with twenty imputations, pooled by Rubin's rules; the weighting row weights by the inverse probability of complete observation. Panel B varies the sample and Panel C the outcome definitions, one at a time, so the mental-health and employment columns move independently. Standard errors are in parentheses.

Tables 19 and 20 collect the robustness checks. The employment impact is negative and significant in every one of the forty-two specifications reported, with a median of −20.9 points and a range from −32.5 to −14.7. It is unmoved by the estimator, by adding men never made redundant as an explicit comparison group, by relaxing the balance requirement, by the age window, by sex, or by survey era. Under the relative-magnitudes sensitivity analysis of Rambachan and Roth (2023) it remains signed for post-period violations of parallel trends more than five times the largest pre-period violation; randomization inference permuting the redundancy date over four hundred draws gives $p < 0.001$; and assigning placebo event dates to men never made redundant returns −0.15 points (s.e. 0.77).

The mental-health impact is positive in every specification of the battery and significant in almost all of them. Replacing the interaction-weighted estimator with the group-time estimator of Callaway and Sant'Anna (2021), the imputation estimator of Borusyak et al. (2024), or two-way fixed effects gives 0.41 to 0.53 points against a baseline of 0.74; admitting men never made redundant as controls gives 0.55 to 0.57; relaxing the balance filter, or imputing the event-time cells it discards, gives 0.54 to 0.77. The range across the whole battery is 0.41 to 0.88, with the single exception of dating the event by the interview year rather than the reported job-end year, which gives 0.20 and is a measurement failure rather than a robustness result.

Two diagnostics deserve to be stated plainly. The pre-event coefficients are not jointly zero (Wald $p = 0.013$), and under Rambachan and Roth (2023) the confidence set admits zero at a low breakdown value on the balanced sample, though not on the unbalanced one, where more pre-periods are available. Against that, the pre-event path runs in the conservative direction. Caseness is lower two and three years out than in the reference year, so distress was rising into the job loss, and treating the reference year as already treated raises the impact to 0.76 and then 0.80. The two tests that speak most directly to whether the design is measuring anything real both pass: a placebo event assigned to men never made redundant returns 0.07 points (s.e. 0.08), and randomization inference gives $p = 0.12$ on a test the permutation scheme makes conservative.

The balance requirement is the largest selection device in the design and Table 20 is explicit about it: of the 3,515 treated men with any observation in the window, 1,953 are observed on both sides of the event and only 591 at all seven event times. The filter turns out to select mildly. The raw within-person change from the year before the event to the year of it is −21.2 points of employment among the men it keeps against −23.4 among those it drops, and +0.61 against +0.63 points of GHQ-12 caseness, so it is not retaining the men whose outcomes moved least. Panel exit does rise from about 14 percent per wave before the event to between 19 and 22 percent after it, however, so the missingness is not ignorable. Imputing the discarded cells by predictive mean matching, with twenty imputations pooled by Rubin's rules, returns −22.2 points for employment and +5.9 for mental health.

Panel C of Table 19 is the design's own falsification test. Both surveys record *why* a job ended on a common code frame, so the identical event study can be run on separations that are voluntary, for cause, foreseen, or reverse-causal by construction, among men who never report a redundancy anywhere in the panel. A voluntary move to a better job produces no employment loss and no deterioration in mental health at all, at −3.1 points (s.e. 1.9). A job that was always temporary, and whose ending is therefore foreseen from the day it begins, produces a much smaller employment fall and no mental-health response. A job ended for health reasons produces +14.0 points, more than twice what a redundancy produces on a comparable employment fall. The design is measuring involuntary job loss rather than job change, and where the separation genuinely is driven by health the estimate inflates exactly as it should. The dismissal row is reported for completeness but is too imprecise to interpret, since its confidence interval contains both zero and the redundancy estimate.

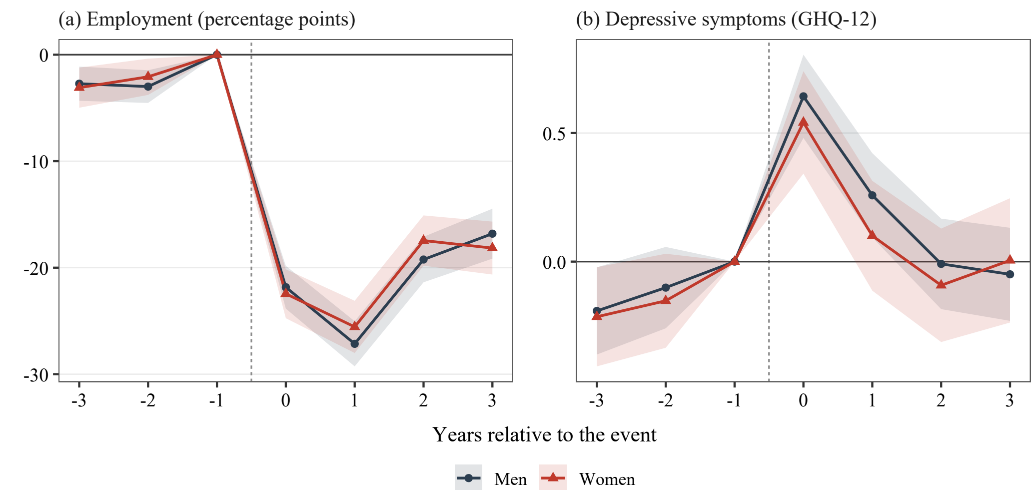
Finally, the event study can be read as the reduced form and first stage of an instrumental-variables specification in which the redundancy switch instruments employment in a mental-health equation, with individual and interview-year fixed effects. Being employed lowers GHQ-12 caseness by 3.14 points (s.e. 0.35) on the baseline window and 2.68 points (s.e. 0.37) when the sample is restricted to men observed at both $k = -1$ and $k = 0$, against a within-person least-squares association of 1.24; the first-stage F

statistics are 433 and 374. That the instrumented estimate is more than twice the least-squares one is what the selection story predicts, since the men who lose work for reasons unrelated to their health are not those whose non-employment is least costly. Instrumenting log labour income in place of employment gives 1.65 points (s.e. 0.53) per log point, an order of magnitude too small to account for the employment effect, so the mental-health cost of a redundancy is not simply priced by the income it destroys. Two caveats attach. Using the post-event event-time indicators as separate instruments makes the model overidentified, and the Sargan test rejects ($p = 5 \times 10^{-5}$), which is what a fading effect implies: the implied Wald ratio falls from −0.48 in the year of the event to −0.13 a year later, so no single ratio summarizes the window and the static estimate should be read as dominated by the impact year. And the exclusion restriction, that a redundancy moves mental health only through employment, is strong. We report the specification as a bound on the non-pecuniary value of work rather than as a preferred estimate.

Women

The design transfers to women without qualification, which is worth establishing before the next subsection reports that the other design does not. Figure 11 re-estimates it on women, changing only the sex filter: the same interaction-weighted estimator, the same unbalanced ±3-year window, the same controls and fixed effects, and no cohort binning. Employment falls by 22.4 percentage points on impact for women (s.e. 1.2) against 21.8 for men, and GHQ-12 caseness rises by 0.54 points (s.e. 0.10) against 0.64. The pre-event paths are close to identical: −3.1 and −2.1 points of employment for women against −2.7 and −3.0 for men, and −0.21 and −0.15 points of caseness against −0.19 and −0.10. Pooling the sexes gives −22.1 points and 0.61 points of caseness, close to both. There is no case here for excluding women.

Figure 11: The Redundancy Event Study by Sex



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both panels use the interaction-weighted estimator of Sun and Abraham (2021) with controls for age and its square and interview-year and education fixed effects, on 3,339 treated men and 2,803 treated women aged 25–55. The specification is that of the main text in every respect other than the sex filter; the men's series is the one plotted in panels (a) and (b) of Figure 1. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse.

B.6 The Recurrence Event Study: Specifications and Robustness

This appendix reports the specification behind the recurrence design of Section 3.5 and the robustness checks the text summarizes.

An episode is a crossing of the GHQ-12 caseness threshold, $C_{it} = \mathbf{1}\{\text{caseness}_{it} \geq 4\}$, so that an onset is $O_{it} = \mathbf{1}\{C_{it} = 1,\ C_{i,t-1} = 0\}$. First onset is a man's first such crossing and a recurrence is his second, after an intervening spell below the threshold. Employment is the work-history extensive margin, which is the labor-supply object of the structural model. The sample is men aged 25–55 with a valid GHQ-12 score, matching the age range of the redundancy design, and the event window is $k \in [-3, +4]$ for the main-text specification and $k \in [-4, +5]$ for the timing-only variant below.

The main-text estimates come from the interaction-weighted estimator of Sun and Abraham (2021), applied as in equation (25) with E_i the wave of i's recurrence, wave fixed effects, controls for age and its square, and standard errors clustered on the individual. The pre-event employment path under this specification is flat, at −0.41 (s.e. 0.73) and −0.03 (s.e. 0.72) three and two waves before the event.

The substantive design decision is the comparison group, and we report both choices. The main-text specification compares a man who recurs to men who have not yet recurred and to men who never recur. An alternative restricts the sample to men who eventually recur, so that identification comes only from variation in *when* the recurrence happens. For that variant we use the group-time average treatment effects of Callaway and Sant'Anna (2021),

$$ATT(g,t) = \mathbb{E}[\,Y_{it}(g) - Y_{it}(\infty) \mid G_i = g\,]\,, \tag{26}$$

where G_i is the wave of i's recurrence, with a universal base period at $g - 1$ and the not-yet-recurred comparison set $C_{gt} = \{i : G_i > \max(t,\ g-1)\}$. Requiring a control to be untreated at the base wave as well as at t matters here: a control treated between $g-1$ and t would have a contaminated base difference. Inference is by multiplier bootstrap with 999 draws, clustered on the individual, and the comparison pool exceeds 1,500 men at every event time. The two give 2.7 and 2.4 percentage points respectively, so the headline magnitude does not turn on the choice.

The choice does matter for one heterogeneity exercise, and we flag it. Never-recurrers are not simply untreated recurrers: roughly half of first-onset cases never recur, and the recurrence-prone and single-episode courses are distinct clinical entities (Monroe and Harkness, 2022). Holding the treated group fixed at men who reached full remission and varying only the control group, the impact estimate moves from −0.8 percentage points when the controls are never-recurrers to +0.3 when men with residual chronic symptoms are also demoted into the control pool. Since the treated group is unchanged, that movement is attributable entirely to the comparison group. Under the timing-only design the sensitivity disappears, which is why the gene-by-environment split in Table 4 uses it.

Table 21: Robustness of the Recurrence Design

Specification	Impact on employment at $k = 0$ (pp)	(s.e.)	Treated men
Panel A: Estimator and comparison group			
Sun–Abraham (main text)	-2.96	(0.76)	3,577
Callaway–Sant'Anna, timing only	-2.64	(0.58)	3,586
Sun–Abraham, timing only	-2.14	(0.61)	3,586
Borusyak–Jaravel–Spiess imputation	-3.37	(0.48)	3,586
One year of anticipation (base $k = -2$)	-3.71	(0.68)	3,586
Panel B: Definition of a recurrence			
Any recurrence (primary)	-2.64	(0.58)	3,586
Well ≥ 2 waves before recurrence	-3.41	(0.74)	2,105
Well ≥ 3 waves before recurrence	-4.02	(0.94)	1,357
Near symptom-free, ≥ 2 waves	-2.24	(0.99)	1,191
Symptom-free, ≥ 1 wave	-2.39	(0.84)	1,631
Symptom-free, ≥ 2 waves	-1.05	(1.24)	646
Panel C: Caseness threshold defining an episode			
Caseness ≥ 3	-2.08	(0.52)	4,292
Caseness ≥ 4 (baseline)	-2.64	(0.58)	3,586
Caseness ≥ 5	-2.48	(0.73)	3,026
Caseness ≥ 6	-2.18	(0.73)	2,523
Panel D: Attrition, reverse causality and external anchoring			
Multiple imputation for panel exit	-2.38	(0.63)	3,586
Employed and stable before the event	-7.93	(0.54)	2,456
Anchored on a doctor's diagnosis	-2.01	(1.03)	1,204
Panel E: Sample			
Men 25–55	-2.41	(0.65)	3,121
Men 30–60	-3.02	(0.56)	3,146
Men, BHPS era (to 2008)	-2.42	(0.79)	1,197
Men, UKHLS era (from 2009)	-2.08	(0.80)	2,293
Women 25–60	-2.42	(0.49)	6,365

Notes: Effect of a depressive recurrence on the probability of employment in the year of the event, in percentage points, relative to the year before. Panel A's first row is the main-text specification, Sun and Abraham (2021). Rows labelled "timing only" restrict the sample to men who eventually recur, so identification comes from *when* a recurrence happens rather than from any comparison with men who never recur; that is the specification used in Panels B–E, estimated by Callaway and Sant'Anna (2021) on men aged 25–60 with not-yet-recurred controls, a universal base period, and a 999-draw multiplier bootstrap clustered on the individual. Panel B varies how deep and how long a recovery must be before a subsequent crossing counts as a recurrence; the primary definition is adopted for its flattest pre-event path and largest sample, not its magnitude, and is among the smaller estimates. Standard errors in parentheses. Every estimate is significant at five percent except the symptom-free, ≥ 2 wave definition, whose 646 men are the smallest sample in the table.

Table 21 collects the robustness checks. The impact estimate is negative and significant under all three leading estimators for staggered designs, across six definitions of what counts as a recurrence, and under alternative caseness thresholds from three to six. The primary "any recurrence" definition is adopted because it has the flattest pre-event path and the largest sample, not because it produces the largest effect; it is among the smaller estimates. Every definition displays the same modest pre-event elevation of about one percentage point at $k = -2$. This is a base-period artifact rather than a trend: the event is dated to a wave that is mechanically symptom-free, and the first-onset design, which conditions on no prior recovery at all, displays an identical pre-event elevation of +1.4 points. Tightening the required depth or duration of recovery raises rather than removes it, while costing power.

Under the relative-magnitudes sensitivity analysis of Rambachan and Roth (2023), the impact remains signed for post-period violations of parallel trends as large as the largest observed pre-period violation ($\bar{M} = 1$), with the robust confidence interval reaching zero only beyond that point. Randomization inference that permutes the recurrence date within the pool of men who ever recur gives $p < 0.001$ over 300 draws.

Attrition deserves a direct answer, because panel exit roughly triples after a recurrence, from 3.2 to 9.2 percent per wave, and on a balanced panel the estimate falls to −0.8 (s.e. 0.5). This is selection, not correction. The impact estimate $\theta(0)$ cannot itself be biased by attrition, because it is measured at the event wave, where every recurrer is present by construction: recording a recurrence requires a well wave at $k = -1$ and a case wave at $k = 0$, whereas attrition occurs at $k \geq 1$. Moreover, the men who leave are precisely those whose employment fell hardest. Raw within-person employment falls by 1.7 points among the men retained to $k + 2$ and by 7.0 points among those who attrit. Imputing the missing observations by multiple imputation with twenty draws, conditioning on pre-event employment, the GHQ-12 trajectory and age, and pooling by Rubin's rules, returns −2.4 (s.e. 0.6), essentially the headline estimate.

Finally, the design replicates when the illness is anchored on clinical rather than survey data. Restricting to individuals ever told by a doctor that they have depression, dating the illness by the recalled age at first diagnosis, and taking the event to be the first GHQ-12 onset after that age, the impact is −2.0 percentage points (s.e. 1.0) with a pre-event path flattened to +0.2 once pre-event employment is controlled for.

Selection into recurrence by genetic risk

Table 18 defends the redundancy design by showing that no polygenic index predicts being made redundant. Recurrence admits no such defense, and we do not attempt one. Genetic risk plainly predicts who relapses: among men at risk, a one standard deviation increase in genetic risk raises the probability of a recurrence at the next wave by 2.8 percentage points on a base of 16.5, and by 2.0 points conditional on residual symptoms, time since remission and the number of prior episodes. Recurrence hazards are 20.1 percent in the highest-risk tercile against 14.5 in the lowest, and the recurrer sample sits 0.19 standard deviations up the genetic-risk distribution.

What the comparison in Table 4 requires is weaker than the absence of selection: that selection does not itself produce the difference in the employment response. A recurrence is a collider of genetic liability and environmental stress, so conditioning on one having occurred induces a negative association between the two among recurrers, since high-risk men need a smaller shock to relapse. If the employment damage travels with the shock rather than with the episode, that association biases the contrast. The relevant questions are its size and its sign.

Calibrating a liability-threshold model to the observed recurrence hazard and its genetic gradient answers both. The induced association is 0.09 standard deviations of environmental shock per standard deviation of genetic risk. Assuming generously that a one standard deviation larger shock doubles the employment damage, that translates into a spurious contrast of 0.6 percentage points against the 7.0

observed, and in the direction that makes high-risk men look *better*, so the estimate is conservative rather than inflated. A simulation that imposes no genetic moderation at all and selects into recurrence exactly as the data do recovers a spurious contrast of 0.3 points.

The signature such a collider would leave is also absent. Among recurrers, genetic risk predicts none of the environmental triggers we observe. Across redundancy, dismissal, unemployment and marital dissolution the largest t statistic is 0.82. Nor does it operate through episode severity, which is near-identical across the two groups: the GHQ-12 rows of Panel B of Table 4 differ by less than a fifth of a point. Reweighting recurrers back to the at-risk population by a clinical recurrence propensity, stratifying on that propensity, and conditioning directly on episode severity each move the per-standard-deviation estimate by less than half a percentage point. Finally, the employment cost does not sort on how likely a man was to relapse in the first place: across quintiles of predicted recurrence probability the impact estimate is flat, with a slope t statistic of 0.36. Selection into recurrence by genotype is real, but it is not what Table 4 is measuring.

Figure 12 plots the full event-study paths by genetic group for both designs, redundancy in panel (a) and recurrence in panel (b), behind the year-of-impact estimates in Table 4.

Women: the design fails

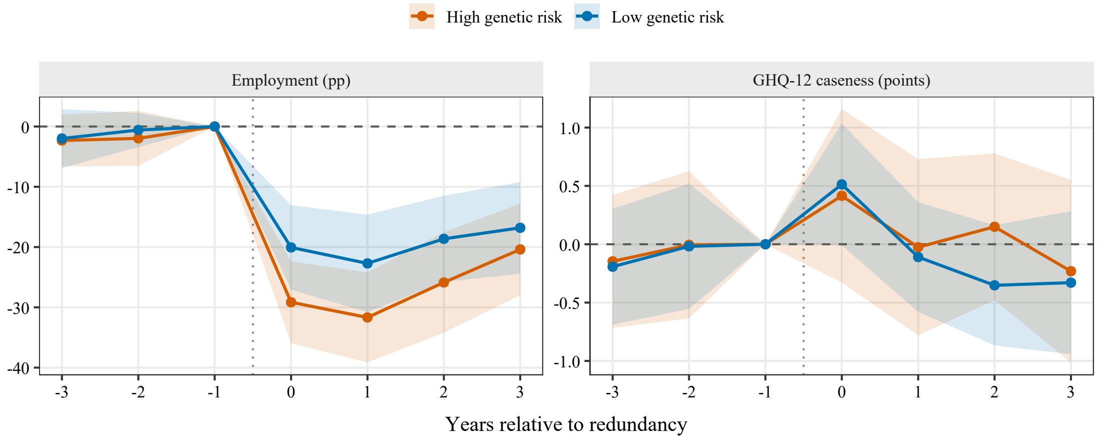
The redundancy design transfers to women (Appendix B.5). This one does not, and the failure is visible in the pre-event path rather than in the impact. Estimated on women, employment is *rising* into the event, by 1.74 percentage points (s.e. 0.71) three waves before it and 2.30 (s.e. 0.70) two waves before, both significantly different from zero at the five percent level. The impact is then a precise null, at −0.67 (s.e. 0.71), against −2.96 (s.e. 0.76) for men estimated on the same specification. The identifying assumption fails before the estimate can be read, so we do not report the estimate as an effect.

This is not a power problem. Women contribute 6,352 recurrences against men's 3,577, so if anything the women's design is the better powered of the two. Nor is it a feature of the recurrence definition: the first-onset design shows the same pattern for women more strongly still, with pre-event coefficients of +3.20 and +2.80.

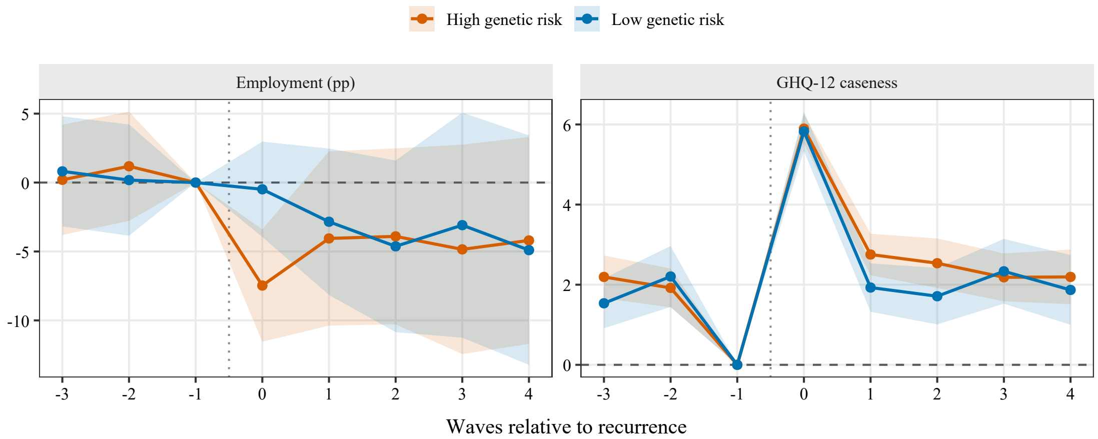
The likely explanation is fertility, and it is specific enough to check. For women in this age range a substantial share of depressive episodes fall near the employment interruptions around childbearing, and Appendix C shows those interruptions to be very large: the arrival of a first child moves women's employment by more than thirty percentage points and their mental health modestly in the adverse direction. An event that lowers employment and raises symptoms at the same time will, in a design that dates treatment by the symptom crossing, place the employment decline at or just before the event and so generate exactly the upward-sloping pre-trend observed. Recurrence timing is therefore not independent of the employment path for women in the way the design requires, and since the paper identifies the mental-health-to-employment leg of the trap from recurrence timing, a mixed-sex estimate of that leg would rest on a design valid for one sex only.

Figure 12: Event-Study Paths by Genetic Risk

(a) Redundancy



(b) Recurrence of a depressive episode



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted), estimated separately for men above and below the median of the depression polygenic index; bands are 95 percent confidence intervals with standard errors clustered on the individual. Both panels use the interaction-weighted estimator of Sun and Abraham (2021) with the controls and fixed effects of the corresponding main-text specification (Section 3.5; Appendices B.5 and B.6). Within each panel the left plot is the work-history employment indicator and the right plot GHQ-12 caseness (0–12, higher is worse). Panel (a) is estimated on the full event window rather than the trimmed window behind Table 4, so its impact coefficients differ marginally from the table's. The two genetic groups track each other closely before the event in both designs; they diverge only in employment, and only after it.

B.7 Mediation and Mental Health

This subsection collects the mediation evidence behind the claim in Section 3 that the genetic penalty operates through mental health and, in turn, through withdrawal into long-term sickness. Table 22 reports the first link, regressing standardized GHQ-12 distress on an above-median polygenic index and adding parental background and the educational-attainment and physical-health indices in turn; the coefficient is stable across all seven specifications. Figure 13 takes the second link apart, tracing the coefficient on genetic risk in an earnings regression as labor force status enters first as a block and then component by component. The attenuation is concentrated in long-term sickness and disability, and very little of it comes from unemployment, which is the pattern the structural model is built to reproduce.

Table 23 reports the descriptive dynamics underlying Section 3.5 in a correlated random effects logit, with within-person time averages of the time-varying covariates included as Mundlak terms to proxy permanent heterogeneity. Column (1) models depression and column (2) employment. Both legs of the trap survive conditioning on permanent heterogeneity and past income, which is what motivates identifying each of them from a shock in the main text.

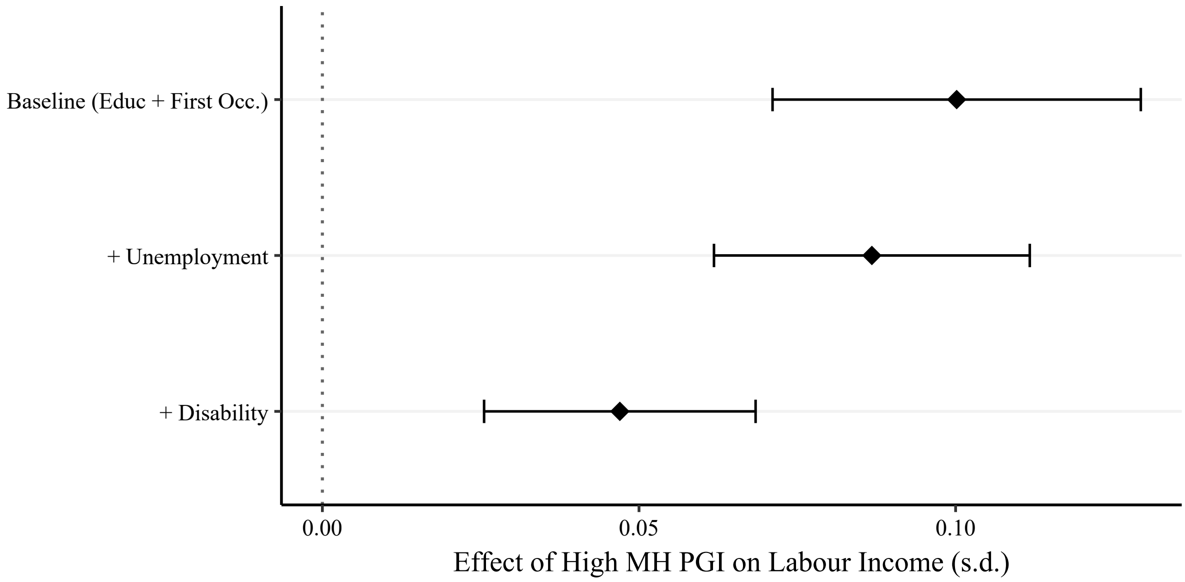
Two caveats attach to the first two of these. Their standard errors are clustered on the age fixed effect rather than on the individual, which is not the appropriate level for this design, and they use the median-split rather than the continuous index. Both are flagged in the notes and neither is used for inference anywhere in the paper.

Table 22: Mental Health PGI and Mental Health

Dependent Variable:	Mental Health (standardized)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
High MH PGI	0.31 (0.01)	0.30 (0.01)	0.30 (0.01)	0.30 (0.01)	0.30 (0.01)	0.28 (0.01)	0.29 (0.01)
Controls							
High EA PGI		✓	✓	✓	✓	✓	✓
Parental Occupation			✓	✓	✓	✓	✓
Parental Education			✓	✓	✓	✓	✓
Education				✓	✓	✓	✓
Occupation: First Job					✓	✓	✓
Employment						✓	✓
Current Occupation							✓
Observations	57,243	57,243	57,243	57,091	57,091	38,822	38,822

Notes: This table reports OLS regressions of standardized mental health (GHQ) on an indicator for having a mental health PGI above the median. All specifications include age, survey wave, and year fixed effects, as well as sex and the first 10 genetic principal components. Missing-value indicators are included for parental occupation and parental education. *Standard errors in this table, reported in parentheses, are clustered on the age fixed effect (`age_dv`) rather than on the individual. With roughly thirty age cells and repeated observations of the same people, this is not the appropriate level of clustering for this design; the estimates are retained here pending regeneration and should be read as provisional.*

Figure 13: Genetic Risk and Earnings: Decomposing the Role of Labor Force Status



Notes: This figure decomposes the role of current labor force status in accounting for the association between high mental health genetic risk and earnings. Each point reports the estimated coefficient on an indicator for being above-median in the mental health polygenic index (High MH PGI) across specifications that sequentially incorporate labor force status and its components. The decomposition shows that the attenuation attributed to labor force status is driven primarily by long-term sickness or disability rather than shorter-term unemployment. *Standard errors in this figure are clustered on the age fixed effect (`age_dv`) rather than on the individual. With roughly thirty age cells and repeated observations of the same men, this is not the appropriate level of clustering for this design; the estimates are retained here pending regeneration and should be read as provisional.*

Table 23: Descriptive Dynamics: Correlated Random Effects Logit Estimates

	Depression (t)	Employment (t)
	(1)	(2)
Employed (t-1)	-0.31	2.44
	(0.10)	(0.12)
Depression (t)		-1.54
		(0.09)
Depression (t-1)	0.48	-0.35
	(0.04)	(0.10)
Earnings (t-1, £10k)	0.56	-1.03
	(0.18)	(0.57)
Baseline Mental Health (Mean)	0.76	0.20
	(0.01)	(0.02)
Baseline Employment (Mean)	0.43	7.10
	(0.12)	(0.22)
Baseline Earnings (Mean)	-0.33	3.13
	(0.21)	(0.76)
Age	0.04	0.16
	(0.03)	(0.06)
Age^2	-0.00	-0.00
	(0.00)	(0.00)
Number of Children	-0.01	-0.01
	(0.01)	(0.03)
Fixed Effects		
Survey Wave	✓	✓
Year	✓	✓
Observations	46,681	46,681
Pseudo R^2	0.27	0.69

Notes: This table reports coefficients from a Mundlak-style correlated random effects (CRE) logit model. "Baseline" terms are within-person time averages of time-varying covariates, included to proxy individual unobserved heterogeneity. Column (1) models depression at time t and column (2) models employment at time t. Standard errors clustered at the individual level are in parentheses.

C Fertility and the Restriction to Men

The preceding appendix reports each of the paper's empirical exercises for women alongside men. The results divide cleanly. The descriptive evidence generalizes and is often stronger for women: the genetic penalty is larger on every extensive margin (Appendix B.4), every transition gradient replicates (Appendix B.3), and the redundancy design transfers with almost identical coefficients (Appendix B.5). But the recurrence design fails for women (Appendix B.6); the trap-persistence gradient, which is the object the structural model is built around, is a third the size and sensitive to whether family care counts as non-employment; and about thirty percent of women's employment penalty runs into family care rather than into disability.

Those three problems have one common cause, and this appendix sets it out, because it is also the reason we do not simply re-estimate the model of Section 4 on women. Fertility is not a nuisance to be conditioned out of the women's regressions. It is predicted by the same genetic risk that drives health and labor supply, it is the proximate cause of the labor-supply transitions the trap is meant to describe, and conditioning on it makes the penalty larger rather than smaller. A model of women would therefore have to treat fertility as a choice determined jointly with health and employment, which is a strictly larger object than the one estimated here.

C.1 Genetic Risk Predicts Fertility

Table 24 reports three margins. The timing of a first birth is not detectably affected: the annual hazard of a first birth among childless women moves by -0.18 percentage points per standard deviation of genetic risk on a base of 6.8, a precise null, and the age at first birth is unmoved. Completed fertility is a different matter. By ages 40–45, a standard deviation of genetic risk is associated with 0.09 fewer children in the household (s.e. 0.026) and a 3.2 percentage point higher probability of childlessness (s.e. 1.06), on a base of 26.7 percent. Neither association is present for men, whose corresponding estimates are 0.02 children (s.e. 0.030) and 1.5 points (s.e. 1.32). Genetic risk for depression predicts how many children a woman has and does not predict how many a man has.

Table 24: Genetic Risk for Depression and Fertility

		Women		Men	
Outcome	Mean	Est.	(s.e.)	Est.	(s.e.)
First-birth hazard, childless 20–45 (pp)	6.82	-0.182	(0.309)	-0.229	(0.309)
First birth observed in window (pp)	19.47	-1.156*	(0.695)	-1.148	(0.816)
Children in household, max over window	1.45	-0.009	(0.023)	0.023	(0.025)
Children in household at ages 40–45	1.32	-0.094***	(0.026)	-0.020	(0.030)
Childless at ages 40–45 (pp)	26.70	3.201***	(1.063)	1.520	(1.315)

Notes: Each cell is the coefficient on a one standard deviation increase in the depression polygenic index, from a separate regression. "Mean" is the sample mean of the outcome for women. The first row is a person-wave linear probability model for the annual hazard of a first birth, estimated on those currently childless and aged 20–45, with age and survey-wave fixed effects and standard errors clustered on the individual. The remaining rows are person-level regressions with heteroskedasticity-robust standard errors, controlling for the ages at which the individual enters and leaves the observation window where applicable. Every specification controls for the ten genetic principal components. Fertility is measured by `nchild_dv`, the number of own children in the household, so a transition from zero to one is a proxy for a first birth rather than a birth record, and the ages 40–45 rows are a proxy for completed fertility rather than a measure of it. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C.2 Why It Cannot Be Conditioned Away

That fertility is itself an outcome of genetic risk is what makes it impossible to handle as a control. Adding the number of children to the women's specifications does not attenuate the penalty; it slightly amplifies it, moving the employment estimate from -4.01 to -4.21 percentage points and the family-care estimate from 1.19 to 1.39. The same is visible as the final rung of the earnings ladder in Table 17, where the children block raises the coefficient by 3.5 percent for women and 0.7 percent for men rather than absorbing any of it.

A confounder is absorbed on conditioning. A variable that is partly a consequence of the treatment need not be, and this one is not. The direction is what one would expect if higher genetic risk lowers completed fertility and lower fertility raises employment: conditioning on children then removes a channel that was partially *offsetting* the penalty, so the remaining estimate grows. Whatever the exact accounting, the implication for modelling is the same in either direction. Fertility cannot be held fixed without conditioning on an outcome.

C.3 The Size of the Interference

Figure 14 shows why this matters for the trap in particular, by applying the paper's own event-study machinery to the arrival of a first child. Women's employment falls by 35.7 percentage points on impact (s.e. 1.0) and is still 15.9 points below baseline four years later. Men's does not move, at −1.8 points (s.e. 0.6). The result is not an artifact of the estimator: the person-fixed-effects form and the Kleven et al. (2019) child-penalty form give −36.1 and −37.6 points for women at impact. Women's employment does rise into the first birth, by about ten points over the two preceding years, which is a familiar feature of that literature and reflects the life-cycle stage at which first births occur; the fall at the event is more than three times that movement, and the men's series provides the within-design control.

Panel (b) is the part that bears directly on the depression trap. Mental health deteriorates after a first birth for *both* sexes, and by almost the same amount: GHQ-12 caseness rises by 0.15 points for women and 0.12 for men at the event, and the two series are within a tenth of a point of one another throughout. The birth is therefore an event that moves women's employment by thirty-six points and their mental health by about as much as it moves men's.

That combination is exactly what makes the trap hard to read for women. A woman observed non-employed and above the caseness threshold in the years after a birth enters the data in the same cell as a man who has lost his job and become depressed, and the transition tables cannot tell them apart. For the man the joint state is the mechanism the model describes. For the woman a large part of the employment component was produced by the birth, and the modest accompanying rise in symptoms is no larger than the one men experience without leaving work. This is why the women's trap-exit gradient moves by a factor of five depending on whether family-care spells are counted (Appendix B.3), and why we do not read the women's transition estimates as establishing that the trap operates for them as it does for men.

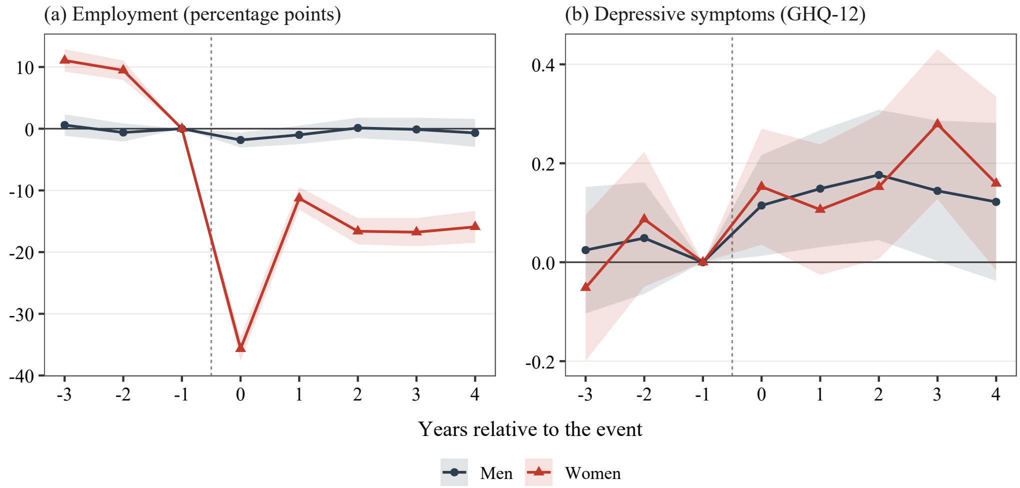
C.4 What Would Be Required

Extending the model to women is therefore not a matter of re-estimating it on a different sample. It would need a fertility decision entering the state space, with the timing and number of births chosen jointly with labor supply and treatment; a home-production or family-care state distinct from both employment and disability, since that is where the genetic penalty sends women; and a reconsidered identification strategy for the mental-health-to-employment leg, since recurrence timing is not available. Each is feasible and none is a variation on the present model. We regard this as the natural next paper rather than a robustness check on this one.

D Clinical Background: Depression and the Channels to Labor Supply

To understand why depression acts as such a potent barrier to labor supply, it is necessary to examine its clinical manifestations. The diagnostic criteria for Major Depressive Disorder require five or more

Figure 14: Employment and Mental Health Around a First Birth, by Sex



Notes: Event-study coefficients relative to the year before the event ($k = -1$, omitted); bands are 95 percent confidence intervals with standard errors clustered on the individual. Both panels use the interaction-weighted estimator of Sun and Abraham (2021) with controls for age and its square and survey-wave fixed effects, on 3,983 men and 4,281 women aged 20–50 in panel (a). A first birth is dated as the first transition from no own children in the household to one, so it is a proxy for a birth rather than a birth record. As in the redundancy design, identification comes from variation in the timing of the event among those who experience it: people who never have a first child inside the panel are not a counterfactual for those who do, and admitting them as controls produces pre-event coefficients of +25 points for women, which is composition rather than a pre-trend. Depressive symptoms are measured by the GHQ-12 caseness score, which runs from 0 to 12, with higher values worse.

symptoms over the same two-week period, at least one of them either persistently depressed mood or anhedonia; the remainder cover sleep, guilt, energy, concentration, appetite, psychomotor disturbance and suicidality (American Psychiatric Association, 2022). Our measure is not a diagnostic interview but the GHQ-12, whose twelve items and caseness scoring are set out in Table 25. The correspondence is close enough for our purposes: the items on concentration, sleep, strain, unhappiness, confidence and worthlessness map directly onto the diagnostic domains that the channels below turn on.

Table 25: The GHQ-12 Items and the Caseness Score

	Item: "Have you recently. . . "	**Direction**
1	been able to concentrate on whatever you're doing?	Positive
2	lost much sleep over worry?	Negative
3	felt that you were playing a useful part in things?	Positive
4	felt capable of making decisions about things?	Positive
5	felt constantly under strain?	Negative
6	felt you couldn't overcome your difficulties?	Negative
7	been able to enjoy your normal day-to-day activities?	Positive
8	been able to face up to your problems?	Positive
9	been feeling unhappy and depressed?	Negative
10	been losing confidence in yourself?	Negative
11	been thinking of yourself as a worthless person?	Negative
12	been feeling reasonably happy, all things considered?	Positive

Notes: Each item is answered on a four-point scale. Positively worded items run from "better than usual" to "much less than usual"; negatively worded items from "not at all" to "much more than usual". The *caseness* score (`scghq2_dv`) recodes each item to one if the respondent's answer falls in either of the two more distressed categories and zero otherwise, then sums the twelve indicators, giving an integer from 0 to 12. This is the measure used throughout the paper: a threshold of two or more denotes mild depressive symptoms and four or more probable major depressive disorder (Anjara et al., 2020). The alternative Likert scoring (`scghq1_dv`) instead sums the raw responses coded 0–3, giving a score from 0 to 36. The GHQ-12 is a screening instrument for common mental disorder rather than a diagnostic interview: it identifies probable cases, and its caseness threshold has been validated against clinical diagnosis of depression (Wojujutari et al., 2024).

D.1 Mechanisms of Labor Supply Reduction

The symptoms of depression are fundamentally inimical to productive labor, acting through three primary channels: reduced marginal productivity, increased disutility of effort, and the erosion of physical health capital.

Productivity and the wage channel. Depression impairs productivity directly through cognitive dysfunction. Clinical symptoms such as diminished concentration and excessive guilt capture attention and facilitate rumination, distorting an individual's beliefs regarding their own abilities (Ridley et al., 2020; Gotlib and Joormann, 2010). These impairments manifest as "presenteeism", where individuals remain physically present at work but exhibit diminished output per hour (Mall et al., 2015). In a structural framework, this cognitive tax acts as a negative shock to human capital, reducing the wage offer. Furthermore, external barriers such as employer stigma and discrimination exacerbate this channel, as others may underestimate the productivity of individuals with mental health conditions, hindering job retention and wage growth (Ridley et al., 2020).

Disutility and the participation channel. Beyond hourly productivity, depression reduces the overall desire and ability to engage in the labor market. Symptoms like fatigue and anhedonia significantly increase the non-pecuniary cost of effort. This is compounded by a "pessimism about the returns to effort" (De Quidt and Haushofer, 2016), where depressed individuals struggle with delayed gratification and reduced motivation. Job search requires high upfront effort for uncertain future rewards, which makes it particularly taxing in this psychological state. Functional impairments like sleep disturbances and psychomotor retardation further restrict the ability to adhere to rigid work schedules, often precipitating the transitions into long-term sickness or disability observed in our descriptive analysis.

Physical health spillovers. Finally, depression reduces labor supply through its negative impact on physical health. Clinical evidence suggests that chronic mental distress triggers systemic inflammatory responses (Dantzer et al., 2008), which accelerate physiological ageing and cardiovascular decay (Prince et al., 2007). This "weathering" effect leads to higher rates of chronic physical conditions (Katon, 2011), which in turn impairs productivity and labor supply (Currie and Madrian, 1999).

We incorporate these pathways in our structural model, allowing mental health to affect not only contemporaneous wage offers and the cost of working but also an individual's physical health trajectory. This multi-channel approach is essential for capturing the "depression trap," where initial psychological shocks lead to a self-reinforcing cycle of non-employment and physical-mental co-morbidity.

E Robustness: Within-Family Identification

A central challenge in identifying causal genetic effects in the genoeconomics literature is the correlation between an individual's genotype and their developmental environment. Genetic variants may have a 'direct genetic effect', a causal effect on their mental health. However, 'genetic risk for depression' may instead reflect their environmental upbringing. In the literature, this is referred to as an 'indirect genetic effect' and occurs for two main reasons (Young et al., 2022). First, genetic variants may be more or less concentrated in certain areas of social classes. Norman ancestry remains a strong predictor of wealth in the UK a millennium following the conquest (Clark and Cummins, 2015). While we mitigate this in our primary analysis by controlling for the first 10 principal components of the genome (Price et al., 2006), residual confounding ma persist. Second, 'genetic nurture' may occur as an individual's genotype is inherited from their parents and thus mechanically correlated with their genetic predispositions (Kong et al., 2018). A parent with a high genetic predisposition for depression may affect their child's risk of mental health issues through their interactions or the home environment they provide.

Many of the key conclusions of this paper do not require us to disentangle direct from indirect genetic effects. Even if our genetic associations purely capture indirect genetic effects, genetic variation remains a measurable, ex-ante predictor of depression risk, the persistence of the 'depression trap' and labor market outcomes. It may still be useful as a diagnostic tool in providing targeted mental health care, and doing so may be cost-effective in generating increased labor supply and reducing benefit payments. Nonetheless, it may suggest that rich socioeconomic indicators could be just as effective in targeting mental health interventions, and may have different implications regarding the design of mental health policy. If our results are closer to direct genetic effects, individuals in the same family may differ significantly in their risk profile and need for mental health care. We therefore aim to assess whether our key results reflect direct genetic effects by exploiting variation within family lineages.

E.1 Identification Strategy: Family vs. Sibling Designs

While family fixed effects in a between-sibling comparison provide the standard "lottery" of alleles within a shared household, the molecular genetic sample in the UKHLS is subject to significant attrition when restricted to genotyped sibling pairs. We identify only 153 siblings observed during working age with valid labor force status. When restricted to male sibling pairs, the resulting sample size is insufficient for meaningful statistical inference.

To maintain statistical power while adhering to causal identification standards, we expand our strategy to include all biological pedigrees. We identify 998 individuals connected by direct parent-child or sibling links, grouped into 538 unique family units. When we filter on all-male pedigrees, we are left with 418 men in 334 families. We identify the genetic effect by including family fixed effects, which purge the estimates of shared dynastic shocks and persistent social background factors. While variation between generations in a pedigree is influenced by assortative mating and environmental shifts, we include birth year fixed effects to capture cohort effects, and the stability of coefficients across these units provides a robust diagnostic for whether genetic nurture is a first-order driver of our results.

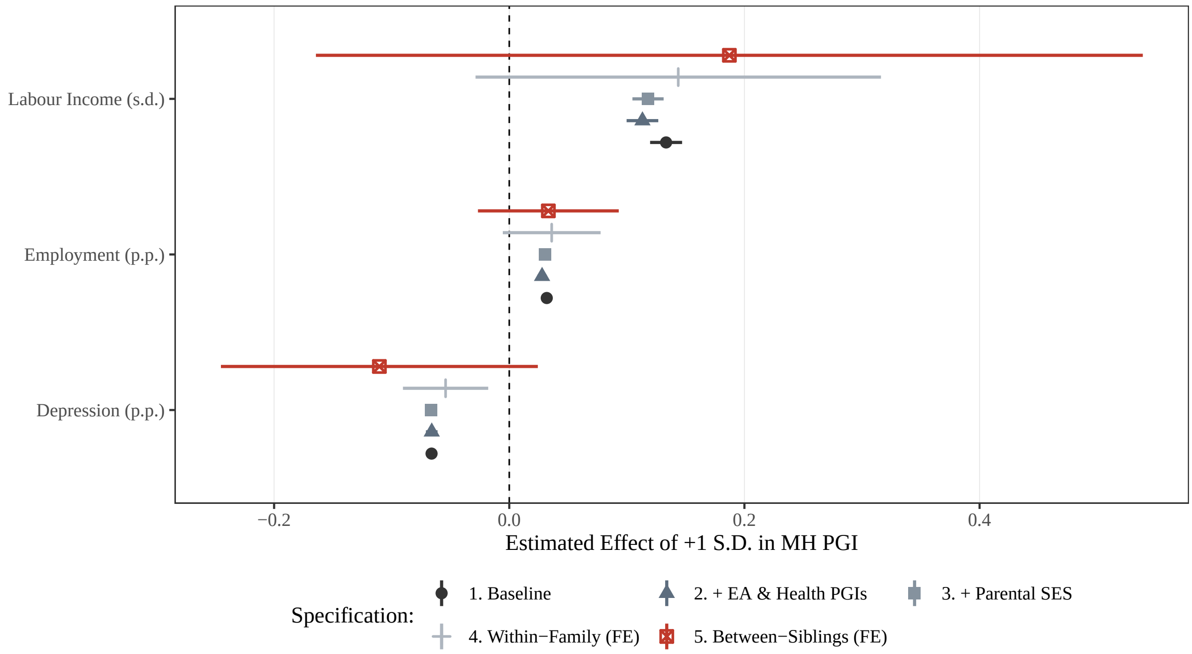
In these restrictive models, we transition from the binary measure used in our structural model to a continuous PGI to capture the smaller identifying variation within lineages. Furthermore, as the identifying variation within families is limited, we pool the sample and control for sex-genotype interactions to maximize power while maintaining a focus on the male "genetic penalty" established in Section 3. For employment and labor income, we report within-family results using the all-male family units.

E.2 Results and Coefficient Stability

Figure 15 displays the stability of the point estimates for our primary outcomes: labor income, employment, and clinical depression. To formally evaluate this stability, we perform a Z-test for the equality of coefficients between our socioeconomic mediation models (β_{med}) and these within-family models (β_{WF}).

As shown in Table 26, we fail to reject the hypothesis that the coefficients are statistically equivalent across all outcomes ($p > 0.50$). The point estimates for employment and income are especially stable, which suggests that the genetic penalty is rooted in individual genetic liability rather than purely environmental transmission or dynastic sorting. The point estimates in Figure 15 are also generally quite robust to to inclusion of controls for socioeconomic background or EA or Health PGIs. For our main analyses, we therefore rely on between-family variation in MH PGIs for identification, which enables more comprehensive mediation analysis and dynamics. Given the robustness across specification, we generally interpret these as direct genetic effects. Okbay et al. (2022) test wellbeing, neuroticism and depression-related PGIs, comparing population-based estimates to within-family estimates with much larger sample sizes, and generally find consistent results.

Figure 15: Stability of MH PGI Estimates Across Specifications



Notes: This figure compares point estimates and 95% confidence intervals for the effect of a one standard deviation increase in the mental health polygenic index (MH PGI) across specifications. The "Socioeconomic Controls" specification includes parental occupation and parental education indicators. The "Within-Family" specification includes family fixed effects, identifying the association within extended family networks. Income is standardized; employment, disability, and depression are in probability units, with depression defined using GHQ-12 caseness (≥ 4). All models control for the first 10 genetic principal components and include survey-wave fixed effects; age enters as a quadratic (age and age^2) rather than as fixed effects, and no calendar-year fixed effects are included. All between-sibling specification controls for sex and sex interacted with the MH PGI, and report the estimated coefficient for men. Within-family estimates are done comparing men for all outcomes except depression, which also includes sex and sex interacted with the MH PGI as a control. Standard errors are clustered at the household level in the socioeconomic-controls specifications and at the family level in the within-family specifications.

Table 26: Tests for Equality of Coefficients: Socioeconomic vs. Pedigree Specifications

Outcome	Socioeconomic Controls (β_{SES})	Family FE (β_{WF})	Z-stat	p-value (Diff)
Employment	0.03	0.04	0.12	0.91
Depression	-0.07	-0.05	0.67	0.50
Income (s.d.)	0.13	0.14	0.14	0.89

Notes: This table tests whether coefficient estimates differ across two specifications: one controlling for socioeconomic characteristics and one using family fixed effects (pedigree specification). The Z-statistic tests the null hypothesis $\beta_{SES} = \beta_{WF}$ for each outcome, and the final column reports the corresponding two-sided p-value.

F Estimation of individual heterogeneity with k-means clustering

Individuals have types x that take values on a discrete set $x \in \{1, ..., K\}$. We estimate these types by building upon the approach of Bonhomme et al. (2022) (BLM) (see also Bonhomme et al. (2019). BLM estimate individual heterogeneity classes by k-means clustering on relevant moments of outcome variables, with moments calculated at an individual level. Worker have types $x \in \{1, ..., K\}$, there are individuals $i \in \{1, .., n\}$ and $\mathbf{m}_i$ is a vector of M moments of individual i's outcomes. Workers are partitioned into K classes by finding the partition that solves

$$\min_{\{x_i\} \in \{1,...,K\}^N} \sum_{i=1}^{N} \|\mathbf{m}_i - \widetilde{\mathbf{m}}(x_i)\|^2, \tag{27}$$

The consistency of BLM's approach relies on an injective map between individual unobserved heterogeneity and the vector of moments m_i. In our application, productivity, mental health and physical health are the primary sources of individual heterogeneity. We therefore compute individual moments m using average wages, mental health status and physical health status.

In aiming to capture individual heterogeneity in productivity, we face an additional identification challenge. For a subset of individuals in our sample, we do not observe realized wages. These are the persistently non-employed, who we do not observe working, and retirees who we do not observe during their working lives. We address this challenge by generating a 'wage potential' (w_i^*) for each individual based on their ex-ante characteristics. We first estimate a log-wage regression on the subsample of employed workers[8]:

$$w_{it} = \gamma_{c(i)} + f(Age_{it}) + \delta_e Educ_i + \delta_o Occ_i + \delta_{pgi} PGI_i + \epsilon_{it} \tag{28}$$

where $\gamma_{c(i)}$ and $f(Age_{it})$ capture cohort fixed effects and a third-degree age polynomial, respectively. $Educ_i$ and Occ_i represent educational attainment and first recorded occupation, while PGI_i includes indices for educational attainment and depressive symptoms. We calculate w_i^* for all individuals by predicting their log-wage at a fixed reference age (40) and anchoring them to a common cohort (1970):

$$w^i = \hat{\gamma}1970 + \hat{f}(40) + \hat{\delta}_e Educ_i + \hat{\delta} o Occ_i + \hat{\delta} pgi PGI_i \tag{29}$$

[8]We filter out workers earning less than £1500 per year

As we do not want our types to absorb variation attributable to genetic risk for depression, we residualize our vector of moments m_i on the depressive symptoms PGI. As in Jolivet and Postel-Vinay (2025), the data requires us to make an additional modification to the original BLM technique. In the UKHLS, individuals start the data observation period with differing ages, and we observe individuals for differing lengths of time. Estimating Equation 27 would lead to clustering based on cohort and age effects in additional to individual heterogeneity. We resolve this by first residualising physical and mental health on birth-year dummies to remove cohort effects, and then adapting the partitioning to account for individual's ages:

$$\min_{\{x_i\}\in\{1,\dots,K\}^N} \sum_{i=1}^{N} \left[\sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2 \right] \tag{30}$$

where Age_i is the average age of an individual through the periods observed and $\mathbf{g}(1,\cdot), \dots g(K,\cdot)$ are second-degree polynomials in age, aimed at capturing within-class moment differences due to age. The age polynomial is not included for the productivity moment, as it already accounts for age-effects. This approach mirrors Jolivet and Postel-Vinay (2025), who do not include cohort effects as they use the same observation window for all individuals.

F.1 Algorithm for the classification step

We present a simple iterative algorithm based on the one that has been developed by Bonhomme and Manresa (2015), which was in turn adapted by Jolivet and Postel-Vinay (2025) to implement their group fixed effect approach. Since this setting shares many features with theirs, we can use a similar, though slightly modified, algorithm for our classification step. The number of classes is set to $K \geq 1$.

1. Set $s = 0$ and let $\hat{k}_i^{(0)}$ be an initial partition.
2. In each class $k^{(s)}$ with at least two elements, regress by OLS each of the $M-1$ individual moments (excluding w^i) on a constant, T, T^2. Set the $g^{(s)}$ function parameters for this class to the resulting estimates. In each class with only one individual, set the $g^{(s)}$ slope parameters to zero and the constant to the value of the individual's moment.
3. If at least one class is empty:
   (a) Compute the "distance"

   $$\sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2$$

   for each observation.
   (b) For each $k \in \{1, \dots, K\}$, if class $k^{(s)}$ is empty, assign to $k^{(s)}$ the observation with the highest distance and reset this distance to 0.
   (c) Do step 2 and replace the g_H function parameters with the new values.

4. Update the partition as follows:

$$k_i^{(s+1)} = \underset{k=1,\ldots,K}{\text{argmin}} \sum_{H\in\{MH,PH\}} \|\bar{m}_{H,i} - g_H(x_i, Age_i)\|^2 + \|w_i^* - \bar{w}(x_i)\|^2$$

5. If no observations changes class at Step 4 then stop. Otherwise, set $s = s + 1$ and go to Step 2.

As explained by Jolivet and Postel-Vinay (2025), this algorithm is a variant of one of the first heuristic clustering procedures, referred to as H-means algorithms. More precisely, a H-means algorithm would not include Step 3 and could thus terminate with empty classes. The addition of Step 3 overcomes this issue and leads to what is sometimes labelled a H-means+ algorithm. See (Brusco and Steinley, 2007) for more information about clustering algorithms.

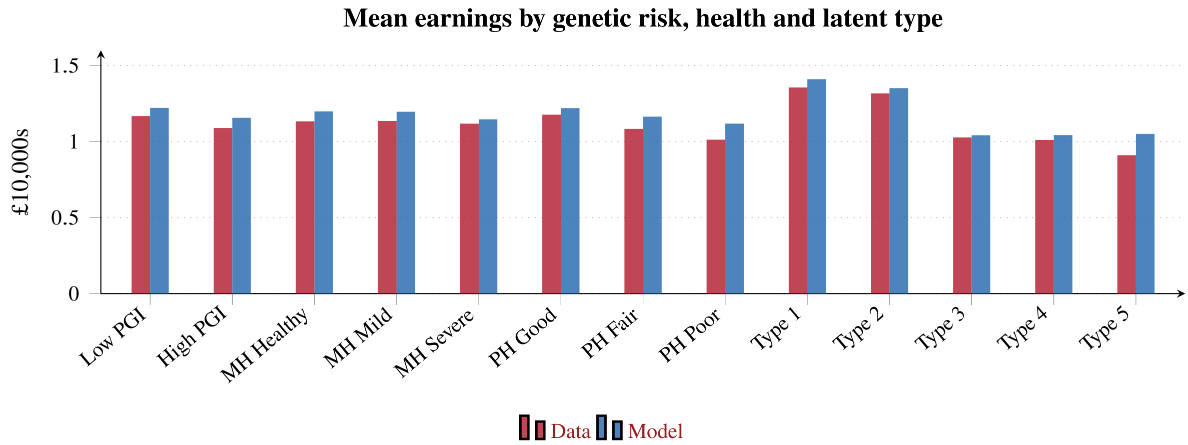
G Model Estimates

This appendix collects the estimates behind Sections 5 and 5.4. Table 27 reports the estimated mental and physical health transition processes, Table 28 the survival probabilities and the parameters taken from outside the model, Table 29 the external calibration of benefits, pensions, taxes and therapy costs, and Table 30 the parameters estimated by simulated method of moments. The remaining model-fit panels appear first.

G.1 Model Fit: Additional Targeted Moments

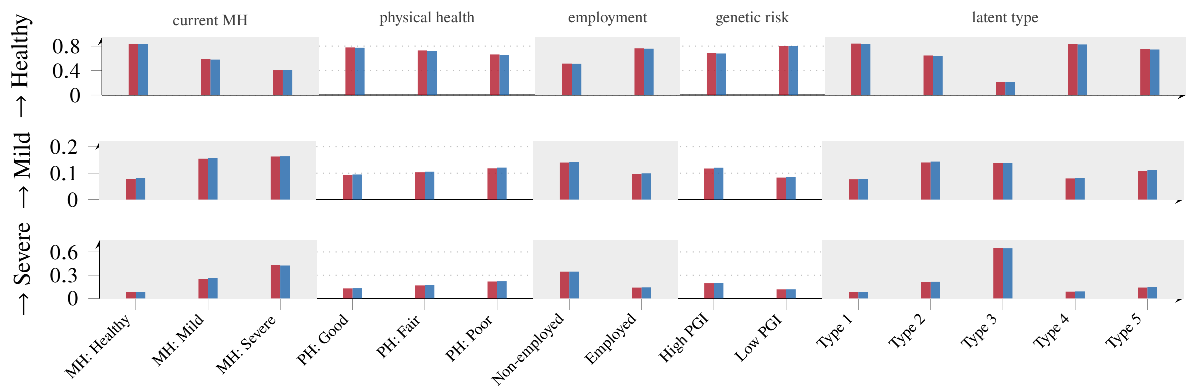
Figures 16 and 17 report the two targeted blocks omitted from Figures 5 and 6 in the main text. The first is the cross-section of mean earnings by genetic-risk group, health state and latent type, which identifies the coefficients of the wage equation. The second is the full set of one-year mental-health transition probabilities, all 45 cells, which identifies the ordered logit governing the process the therapy technology acts on.

Figure 16: Model Fit: Mean Earnings by Genetic Risk, Health and Latent Type



Notes: Targeted moments, continued from Figure 5. Data in red with solid filled markers, the estimated model in blue with dashed open markers. Mean gross labour income in 2015 £10,000s, by depression polygenic-index group, by mental and physical health state, and by latent permanent type.

Figure 17: Model Fit: One-Year Mental-Health Transition Probabilities, All 45 Cells



Notes: Targeted moments, continued from Figure 6. Red bars are the data, blue the model, as in Figure 6. Every one-year transition probability used in estimation, grouped by conditioning variable and ordered within group by originating state. Rows are destination states; each vertical pair of bars is one cell. Largest absolute error across the 45 cells: 0.013.

Table 27: Estimated Health Transition Parameters

Mental Health Transition Process (μ)		*Physical Health Transition Process (π)*	
Parameter	Estimate	Parameter	Estimate
$\kappa_{mh,1}$: Threshold 1 (Severe $\rightarrow$ Mild)	-0.63	$\kappa_{ph,1}$: Threshold 1 (Poor $\rightarrow$ Average)	1.18
$\kappa_{mh,2}$: Threshold 2 (Mild $\rightarrow$ Healthy)	-0.24	$\kappa_{ph,2}$: Threshold 2 (Average $\rightarrow$ Healthy)	2.76
$\mu_{1,1}$: Lagged Mental Health (Mild)	0.84	$\pi_{1,1}$: Lagged Physical Health (Average)	1.10
$\mu_{1,2}$: Lagged Mental Health (Healthy)	2.05	$\pi_{1,2}$: Lagged Physical Health (Healthy)	2.18
μ_3: Employment (n_t)	0.16	$\pi_{2,1}$: Lagged Mental Health (Mild)	0.25
$\mu_{4,1}$: Lagged Physical Health (Average)	-0.19	$\pi_{2,2}$: Lagged Mental Health (Healthy)	0.40
$\mu_{4,2}$: Lagged Physical Health (Healthy)	-0.31	π_3: Employment (n_t)	0.12
$\mu_{5,g}$: Genetic risk (g_i)	0.49	$\pi_{4,g}$: Genetic risk (g_i)	-0.34
$\mu_{5,t}$: Age (linear)	0.06	$\pi_{4,t}$: Age (linear)	-0.00
μ_{5,t^2}: Age (squared/100)	-0.07	π_{4,t^2}: Age (squared/100)	0.02
$\mu_{5,x,1}$: Health type 1	-0.16	$\pi_{4,x,1}$: Health type 1	2.65
$\mu_{5,x,2}$: Health type 2	-0.02	$\pi_{4,x,2}$: Health type 2	0.86
$\mu_{5,x,3}$: Health type 3	-1.59	$\pi_{4,x,3}$: Health type 3	1.00
$\mu_{5,x,4}$: Health type 4	0.28	$\pi_{4,x,4}$: Health type 4	1.74

Notes: This table reports point estimates from standard (proportional-odds) ordered logit health transition models: each covariate has a single coefficient, common across both thresholds, while $\kappa_{mh,1}, \kappa_{mh,2}$ and $\kappa_{ph,1}, \kappa_{ph,2}$ are the threshold-specific cutpoints. Coefficients enter the latent health indices for mental health (mh^*) and physical health (ph^*); positive values indicate a higher probability of transitioning to a better health state. Health type 5 is the omitted (reference) category for health-type indicators. Coefficient subscripts follow the numbering used in the model equations 4 and 5. *qy: Placeholder estimates: these single-coefficient values were obtained by averaging our earlier two-threshold estimates and should be replaced with a direct re-estimation of the proportional-odds model.*

Table 28: Survival Probability Parameters and Calibration Summary

Panel A: Logit Model (Pr(Death))			Panel B: UK Life Table Calibration		
Variable	Estimate	(S.E.)	Age	Shift (α_t)	Target (f)
ξ_0: Intercept	-28.55	(22.07)	65	0.10	1.25%
ξ_1: MH Healthy	49.64	(24.09)	70	0.36	2.15%
ξ_2: MH Mild	27.67	(32.27)	75	0.38	3.46%
ξ_3: PH Healthy	-16.20	(25.15)	80	0.38	5.98%
ξ_4: PH Average	-7.64	(25.48)			
ξ_5: Age	0.68	(0.61)			
ξ_6: Age^2/100	-0.43	(0.42)			
Interactions					
MH Healthy $\times$ Age	-1.42	(0.66)			
MH Healthy $\times$ Age^2/100	0.99	(0.45)			

Notes: Panel A reports logit coefficients for the probability of death in the next wave, estimated on the retired UKHLS sample, with standard errors in parentheses. Panel B summarizes the life-table calibration: α_t is an age-specific shift applied to the latent index so that model-implied mortality matches the target rate f at each listed age.

Table 29: External Calibration: Benefits, Pensions, Taxes, and Therapy Costs

Panel A: Benefit Schedule ($b(mh, ph)$)				Panel B: Retirement & Therapy	
MH State	PH: Healthy	Average	Poor	Variable	Value (£)
Healthy	5,128	6,179	8,589	Pension (£)	
Mild	5,651	6,627	9,268	Type 1	23,438
Severe	5,889	8,433	10,126	Type 2	19,437
				Type 3	15,067
Values in £ per annum				Type 4	15,980
				Type 5	14,951
Panel C: Progressive Income Tax Schedule ($\tau(w)$)				Therapy Cost	
0% rate (personal allowance)			£0–£6,745	Unit Cost (£)	97.82
20% basic rate			£6,745–£37,400	Sessions	8.60
40% higher rate			$> \$37,400$	Private Share	0.26
				Total (P_τ)	218.63

Notes: This table summarizes external inputs used to calibrate the model. Panel A reports mean annual benefit receipt by mental health state (mh) and physical health state (ph). Panel B reports mean annual pension income by unobserved type and the components used to calibrate the out-of-pocket therapy cost P_τ. Panel C reports the progressive income tax schedule applied to labor earnings. All pecuniary amounts (benefits, pensions, and therapy costs) are expressed in 2015 £.

Table 30: Estimated Structural Parameters

Parameter	Estimate
Preferences	
β: Discount factor	0.95
γ: Consumption share of utility	0.53
Wage Process	
β_1: Wage intercept	-0.06
β_g: Genetic return	0.07
$\beta_{2,1}$: Type 1 return	0.45
$\beta_{2,2}$: Type 2 return	0.52
$\beta_{2,3}$: Type 3 return	0.21
$\beta_{2,4}$: Type 4 return	0.15
$\beta_{3,1}$: Mental health return (Good)	0.15
$\beta_{3,2}$: Mental health return (Average)	0.14
$\beta_{4,1}$: Physical health return (Good)	0.21
$\beta_{4,2}$: Physical health return (Average)	0.02
β_5: Age (linear)	0.04
β_6: Age (squared)	-0.07
ρ: Wage persistence (AR1)	0.89
σ_η: Wage shock std. dev.	0.41
Disutility of Labor	
$\delta_{MH,Good}$: Disutility of labor (Good MH)	0.02
$\delta_{MH,Avg}$: Disutility of labor (Average MH)	0.05
$\delta_{MH,Bad}$: Disutility of labor (Severe MH)	0.08
$\delta_{PH,Good}$: Disutility of labor (Good PH)	0.03
$\delta_{PH,Avg}$: Disutility of labor (Average PH)	0.06
$\delta_{PH,Bad}$: Disutility of labor (Severe PH)	0.07
Treatment Costs	
ψ_{emp}: Therapy utility cost (Employed)	0.03
ψ_{unemp}: Therapy utility cost (Non-employed)	0.02

Notes: This table reports point estimates from the structural model. Health and labor-type returns are measured relative to omitted baseline categories in the log wage equation.

H Additional Counterfactual Results

H.1 Recovery from the Depression Trap: Conditioning on the Pre-Shock State

Figure 18 repeats the experiment of Section 6.2 on two sub-populations defined by the state an agent occupied before the shock. The top row keeps agents who were employed; the bottom row keeps agents who were in healthy mental health. Between them the two restrictions remove the obvious composition reading of Figure 9, under which the high-risk group diverges because more of its members were already detached from work, or already unwell, when the shock arrived.

Neither restriction narrows the divergence. Ten years after the shock, unemployment among the pre-shock employed stands 29.9 percent above baseline for the high-risk group against 7.6 percent for the low-risk group, and among those in healthy mental health beforehand 31.9 against 7.4 percent. Both are wider than the 23.9 against 6.0 percent of the unconditional sample. Trap prevalence behaves the same way, 31.0 against 8.6 percent and 31.5 against 10.2 percent respectively. The clinical series are as in the main figure: severe depression is back within a few percent of baseline by the fourth year in every cell, and within one percent by the tenth.

H.2 Priority Access to Therapy: Additional Detail

Solving for the market-clearing allocation

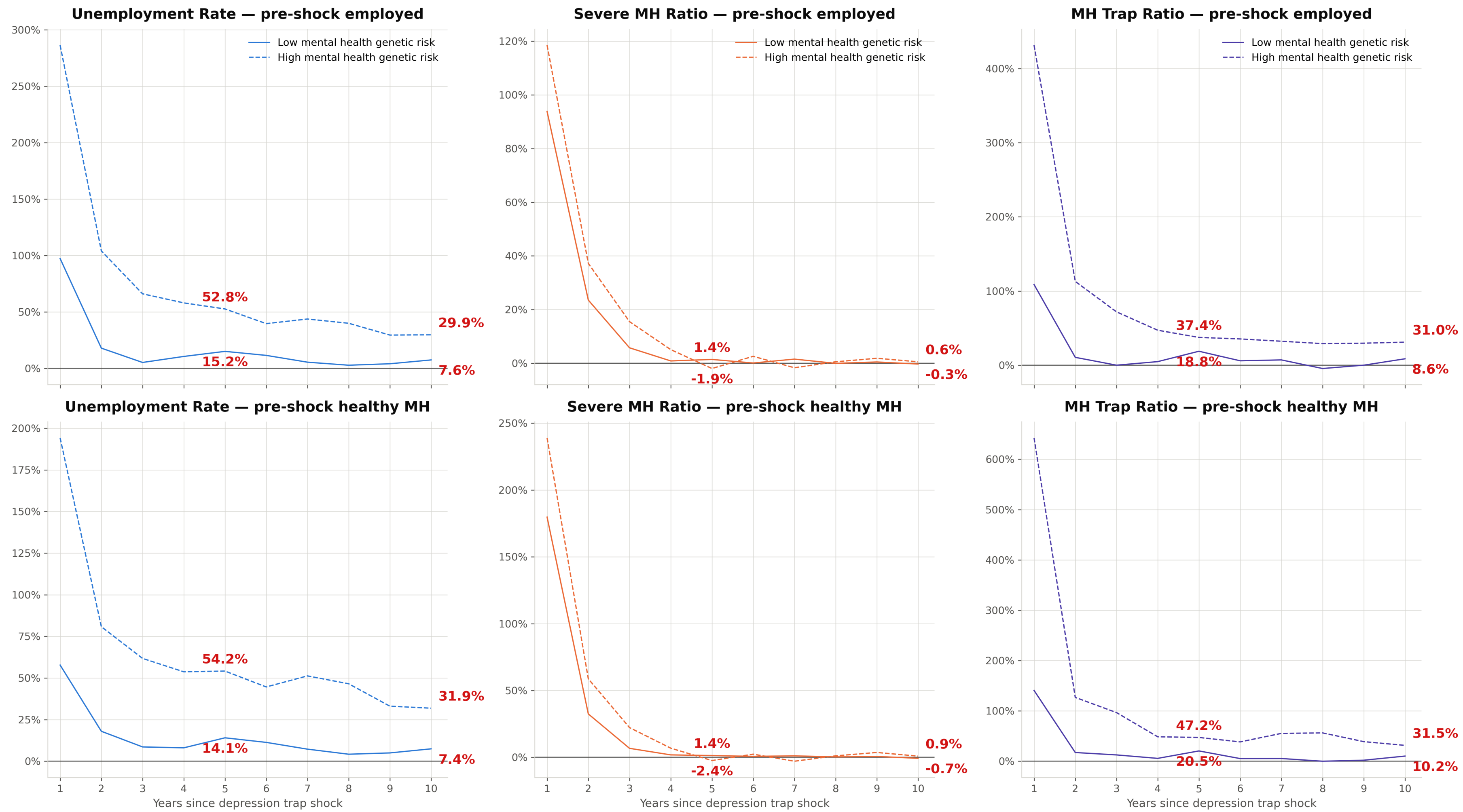
Under fixed capacity the treatment probabilities solve a fixed point. Writing s_g for the population share of group g and $a_g(\cdot)$ for its equilibrium application rate, the clearing condition is

$$s_L\, a_L\, \lambda_L \;+\; s_H\, a_H\, \frac{\lambda_L}{1-X} \;=\; \bar{u}, \tag{31}$$

where $\bar{u}$ is benchmark treatments per working-age person-year and Equation (21) has been substituted for λ_H, which leaves λ_L as the only unknown. Applications respond to λ only weakly: the application rule reduces to $V_t^1 > V_t^0$, since $\lambda V_t^1 + (1-\lambda)V_t^0 > V_t^0$ if and only if $V_t^1 > V_t^0$ for any $\lambda > 0$, so λ affects behavior only through the continuation value.

We solve (31) by a Newton step on the (near-linear) uptake schedule, re-solving the value function and re-simulating at each iteration, with tolerance 10^{-4} on realised take-up and a maintained bracket with bisection fallback. Two features of the problem require care. First, at large X the tie $\lambda_H = \lambda_L/(1-X)$ saturates at one during the iteration, and while it is saturated a lower λ_L barely moves total take-up, because almost all of it comes from the pinned high-risk term; a naive multiplicative update crawls and terminates with λ_H stuck at the ceiling. The Newton step anticipates the cap by solving the interior and capped cases separately and taking the interior solution when it is feasible. Second, the iteration must target *realised* take-up rather than $a(\lambda) \times \lambda$: because the simulation uses common random numbers, the realised share of therapy draws falling below λ is a fixed number slightly under λ, so an update built on the predicted product converges to a point at which realised take-up misses the benchmark. With both

Figure 18: Recovery from the Depression Trap, Conditioning on the Pre-Shock State



Notes: This figure extends Figure 9. The experiment is identical: starting from the empirical initial distribution and keeping individuals first observed between ages 25–45, every agent is placed simultaneously in severe depression and in non-employment at the start of year 1, and then simulated for ten years with no further intervention. The top row conditions on agents employed in the year before the shock and the bottom row on agents in healthy mental health in the year before it. Each series is the difference between the shocked and benchmark economies as a percentage of that group's own benchmark level in the same year. High genetic risk denotes the group with the lower polygenic index. Labelled values are the fifth and tenth years after the shock.

corrections every scenario clears to within 6×10^{-5} of benchmark take-up in three or four iterations, and no scenario terminates with λ_H at the ceiling.

Fiscal accounting

The fiscal balance is discounted tax revenue less benefit expenditure less therapy provision cost, per person, discounted to $t = 1$ at the model's interest rate $r = 0.046$:

$$\text{NPV} \;=\; \sum_t \frac{1}{(1+r)^{t-1}} \Big[\tau_t - b_t - c_t \Big], \tag{32}$$

where τ_t is tax paid on gross labor earnings under the schedule reported in Table 29, b_t is benefit received, and c_t is the public cost of any therapy delivered in period t. Following the convention used elsewhere in the paper, c_t is the market price of a course less the patient's out-of-pocket contribution: a course costs $8.6 \times \$97.8 = \$841$ at 2015 prices, against out-of-pocket contributions of £353 for the employed and nil for the non-employed, so the state pays £488 and £841 respectively. Tax, benefits and provision are all discounted on the same schedule and over the same working-age window. Pensions are excluded: they depend only on permanent type and are therefore identical across scenarios, cancelling in every difference reported. Under fixed capacity total treatments are unchanged, so the provision term very nearly cancels; it is nonetheless computed from the simulated panels rather than assumed away, because the employed/non-employed composition of who is treated shifts and the two carry different unit costs.

Why the ceiling on the elastic arm no longer binds

Since $\lambda_H \leq 1$, holding λ_L at its estimated value 0.6885 would cap the achievable wait gap at $1 - 0.6885 = 31.2\%$: an elastic arm defined by $\lambda_H = \lambda_L/(1 - X)$ with λ_L fixed cannot deliver a target beyond that, and above it the realised gap stalls below the stated one. The design used here avoids the problem because the elastic arm does not compute λ_H from the benchmark λ_L; it inherits the value found under fixed capacity, where clearing has already driven λ_L down. At $X = 80\%$, for instance, clearing yields $\lambda_L = 0.195$ and hence $\lambda_H = 0.977$, comfortably interior. The cost of this is that the elastic arm's realised wait gap is smaller than its target, since low-risk waits are left unchanged: 6.7%, 14.3%, 20.4% and 29.5% for targets of 20, 40, 60 and 80%. Both are reported in Table 6.

Simulation detail

Every scenario draws 10,000 individuals with replacement from the initial distribution, weighted by the survey weight, with a common random seed across all scenarios so that comparisons are paired on identical agents. Each draw carries the individual's full entry bundle jointly, so that genetic risk, permanent type, entry age, mental and physical health, assets and human capital keep their empirical joint distribution; the persistent wage component is drawn independently from the stationary distribution of its

Table 31: Priority access by genetic risk, post-COVID initial distribution

Wait gap	λ_L	λ_H	Take-up	Δ Unemp. (pp)	Δ Severe MH (pp)	Δ MH trap (pp)	Δ NPV (£/person)
Panel A: fixed capacity (reallocation, no incremental cost)							
Benchmark	0.6885	0.6885	0.01485	—	—	—	—
20%	0.5813	0.7266	0.01485	−0.013	−0.008	−0.010	+11.8
40%	0.4600	0.7666	0.01485	−0.016	−0.002	−0.012	+12.5
60%	0.3251	0.8127	0.01485	−0.023	−0.003	−0.010	+19.7
80%	0.1730	0.8651	0.01485	−0.025	−0.008	−0.010	+23.6
Panel B: elastic supply (net of therapy provision cost)							
20%	0.6885	0.7266	0.01532	−0.010	−0.022	−0.009	+7.5
40%	0.6885	0.7666	0.01587	−0.013	−0.031	−0.012	+6.7
60%	0.6885	0.8127	0.01644	−0.022	−0.044	−0.018	+11.6
80%	0.6885	0.8651	0.01707	−0.030	−0.067	−0.022	+16.6

Notes. As Table 6, for the post-COVID initial distribution. Panel B's realised wait gaps are 5.2%, 10.2%, 15.3%, 20.4%. The four intensities cut the high-risk expected wait from a benchmark of 5.4 months to 4.5, 3.7, 2.8, 1.9 months respectively, identically in both panels. Benchmark unemployment 6.77%; severe depression 14.50%; MH trap 3.43%; fiscal balance £54,845 per person; therapy provision £55.0 per person. The post-COVID benchmark has higher unemployment and lower therapy take-up than the pre-COVID benchmark, leaving more participation margin for the policy to act on; returns are correspondingly larger and are positive at every intensity in both supply regimes.

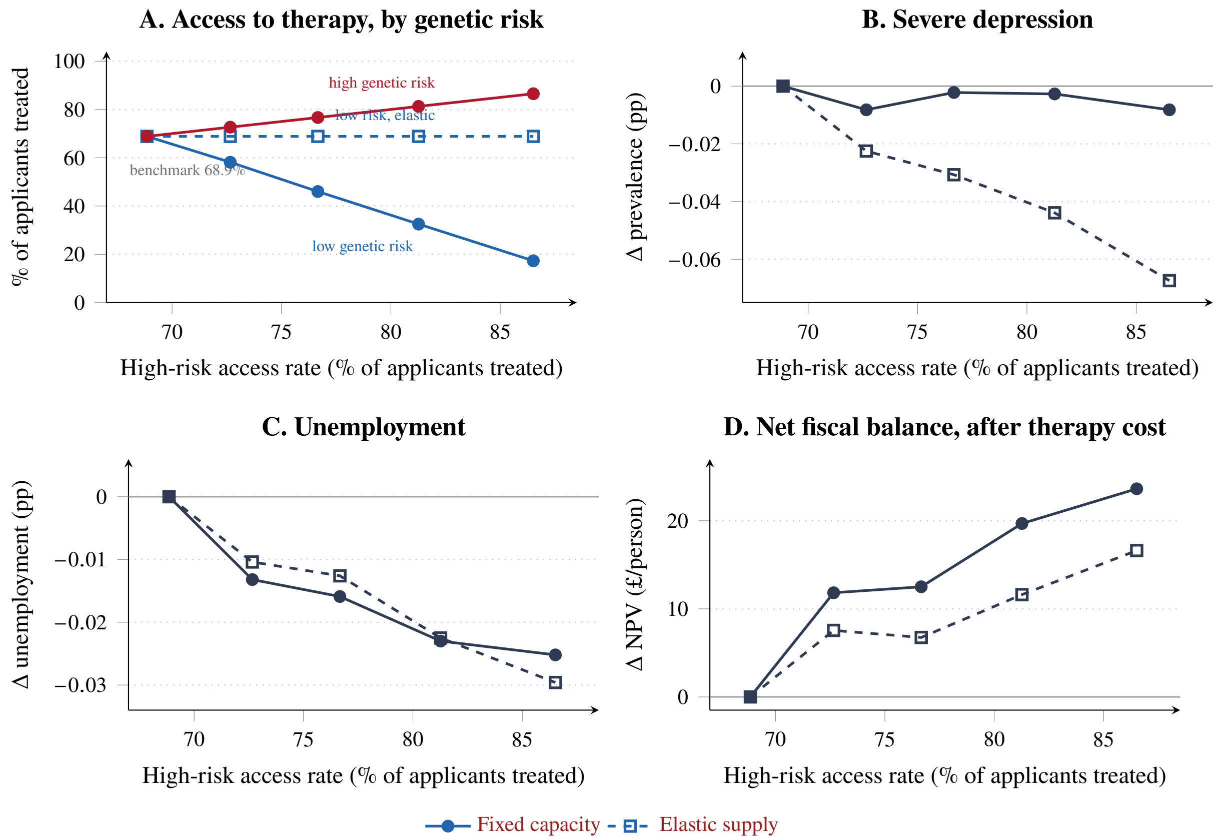
AR(1). Value functions are re-solved for every parameter configuration, including at each iteration of the market-clearing loop.

Robustness: the post-COVID initial distribution

Table 31 repeats the exercise using an initial distribution constructed from post-pandemic data. Fixed capacity still dominates elastic supply at every intensity, so the ranking of the two regimes is unchanged. The magnitudes are larger throughout, however, and more importantly the *source* of the gain changes. The post-COVID benchmark has lower employment (93.23% against 93.53%) and lower therapy take-up (0.0149 against 0.0182), leaving more participation margin for the policy to act on. At $X = 80\%$ the high-risk group contributes £16.62 of the £23.63 fixed-capacity total, against £1.36 of £19.07 pre-COVID. Under elastic supply, where the high-risk contribution is the entire return, the post-COVID policy returns £16.6 per person against £1.4 pre-COVID. Pre-COVID the fixed-capacity policy looks attractive largely because it rations low-risk access; post-COVID it is attractive because treating the high-risk group pays for itself. Figure 19 traces the four outcomes against the high-risk access rate, and Figure 20 disaggregates the fixed-capacity effects by genetic group: the small aggregate change in severe depression is the net of two large offsetting group effects, a fall of 0.145 percentage points among the high-risk group against a rise of 0.111 points among the low-risk group at the highest intensity.

Two features of that distribution should be borne in mind. First, its mental-health composition is more polarised rather than uniformly worse: relative to the pre-COVID sample the severe share is 0.8 points higher but the mild share is 2.4 points lower, so the total non-healthy share is in fact slightly lower (23.1% against 24.7%). Age-standardising to the pre-COVID age structure leaves the severe share at

Figure 19: Priority access by genetic risk, post-COVID initial distribution

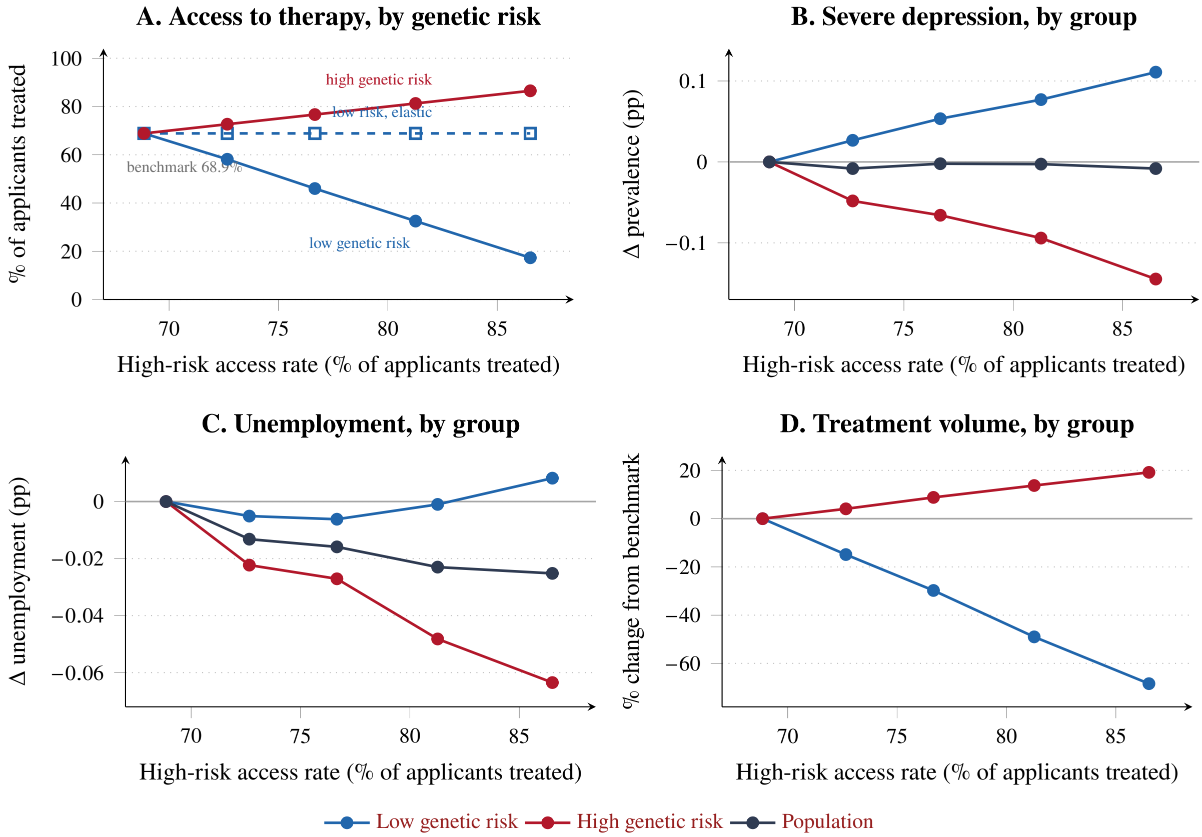


Notes: Post-COVID initial distribution; targeted policy, so only high genetic risk receives priority. The horizontal axis is the share of high-risk applicants treated in the period they apply, λ_H; both supply regimes deliver the same λ_H at each intensity, so it indexes policy intensity for both. The benchmark is $\lambda = 68.9\%$, calibrated to the 1.26 million treatments begun from 1.83 million referrals in NHS Talking Therapies in 2023–24. Panels B–D are anchored at the benchmark, where all effects are zero by construction. In panel A the high-risk series lies on the 45-degree line by construction; what the panel shows is how far fixed capacity must push low-risk access down to deliver a given high-risk access rate. Under fixed capacity the trade-off is close to linear: each additional percentage point of high-risk access costs 2.9 points of low-risk access, the ratio of the two groups' applicant volumes in the market-clearing condition. Under elastic supply low-risk access is unchanged by construction. Panels B–D are population aggregates in a single tone, with line style carrying the supply regime throughout. A negative entry in panel C means the policy moves people into work.

14.3% against a raw 14.6%, so the difference is not compositional in age. Second, the sample contains 605 distinct individuals against 1,662 in the pre-COVID file; since the simulation draws 10,000 agents with replacement, effective precision is governed by the smaller underlying sample and differences of one to two pounds between adjacent rows of Table 31 should not be over-interpreted.

A separate concern is that the post-COVID exercise picks up differential attrition rather than changed behavior. We model attrition with a logit specification and test whether its correlates differ between the COVID and the pre-COVID period. Table 32 reports the estimates: we fail to reject the null of no differential attrition with respect to genetic risk, or to observed mental and physical health.

Figure 20: Priority access under fixed capacity: who gains and who loses



Notes: Fixed-capacity regime only, post-COVID initial distribution. Same horizontal axis as Figure 19. Panels B–D disaggregate by genetic-risk group. The population aggregate in panels B and C is the residual of two large offsetting group effects: at the most intensive scenario severe-depression prevalence falls 0.145 pp among the high-risk group and rises 0.111 pp among the low-risk group, netting to −0.008 pp. Panel D reports the change in the number of treatment courses delivered to each group; because total capacity is fixed, the two series offset exactly in levels.

Table 32: Logistic Regression: Predictors of Attrition (COVID vs. Pre-COVID Comparison)

Variable	Estimate	Std. Error	z-value	Pr(> $\|z\|$)
Main Effects (COVID Wave 22→26)				
(Intercept)	0.21	0.20	1.05	0.29
Wealth (per £10,000)	-0.03	0.01	-3.10	0.00
Genetic Risk (High DEP G)	-0.11	0.10	-1.15	0.25
Health Type 4	0.06	0.17	0.36	0.72
Health Type 3	0.13	0.24	0.53	0.60
Health Type 2	-0.19	0.16	-1.19	0.23
Health Type 1	-0.41	0.19	-2.21	0.03
PH: Average	-0.25	0.15	-1.72	0.09
PH: Healthy	-0.24	0.15	-1.57	0.12
MH: Mild	-0.21	0.21	-0.98	0.33
MH: Healthy	-0.13	0.17	-0.77	0.44
Interactions (Pre-COVID Difference)				
Period: Pre-COVID (15→22)	-0.24	0.63	-0.38	0.70
Period × Wealth (£10,000)	-0.02	0.04	-0.46	0.65
Period × Genetic Risk	0.11	0.23	0.46	0.65
Period × Health Type 4	0.32	0.44	0.72	0.47
Period × Health Type 3	-0.95	0.63	-1.52	0.13
Period × Health Type 2	-0.08	0.46	-0.17	0.87
Period × Health Type 1	0.15	0.48	0.31	0.76
Period × PH: Average	0.14	0.53	0.26	0.80
Period × PH: Healthy	-0.20	0.53	-0.39	0.70
Period × MH: Mild	0.37	0.48	0.77	0.44
Period × MH: Healthy	0.17	0.36	0.47	0.64
Model Diagnostics	AIC: 3,011 | Null Deviance: 3,028.20 Residual Deviance: 2,967 | $n = 2,245$			

Notes: This table reports logit estimates for attrition. The main-effects block uses the COVID comparison period (Wave 22→26). The interaction block reports how coefficients differ in the pre-COVID period (Wave 15→22) relative to the COVID period. Health Type 5 is the omitted (reference) category for the health-type indicators. For physical health (PH) and mental health (MH), the omitted (reference) categories are the non-listed baseline groups (i.e., neither "Average" nor "Healthy" for PH, and neither "Mild" nor "Healthy" for MH). Reported values are coefficient estimates, standard errors, z-statistics, and two-sided p-values.

Table 33: Health and Labor-Market Effects of Expanding Therapy Capacity

				Δ severe depression	
Expansion	Δ unemployment	Δ severe depression	Δ trap	High risk	Low risk
5%	−0.009	−0.021	−0.008	−0.040	−0.005
10%	−0.011	−0.032	−0.010	−0.057	−0.010
15%	−0.019	−0.047	−0.015	−0.085	−0.014
20%	−0.024	−0.059	−0.018	−0.101	−0.022
30%	−0.031	−0.111	−0.030	−0.199	−0.035

Notes: Changes in shares of working-age person-years, in percentage points, under the untargeted capacity expansion of Section 6.4. The benchmark shares are 6.77 percent unemployed, 14.50 percent in severe depression and 3.43 percent in the depression trap, defined as non-employment and severe depression together. The last two columns split the change in severe depression by genetic-risk group; high genetic risk denotes the group with the lower polygenic index, since the index is oriented towards good mental health. The aggregate in column (2) is the population-share weighted average of the two group effects. The treatment probability λ is raised equally for both groups, so the difference between them reflects differential application rather than any targeting rule.

H.3 The Return to an Additional Therapist: Detail

This appendix reports the outcome side of the capacity expansion of Section 6.4, together with the conventions behind the per-therapist conversion and the limitations that attach to it.

Health and labor-market outcomes. Table 33 reports changes in shares of working-age person-years, in percentage points, against a benchmark of 6.77 percent unemployment, 14.50 percent severe depression and 3.43 percent in the depression trap. Every outcome improves monotonically in capacity, and the split by genetic risk shows the incidence of an intervention that targets nobody: severe depression falls between five and eight times as much among the high-risk group as among the low-risk group at every intensity, because high-risk individuals apply more often and so absorb a disproportionate share of the additional slots.

The per-therapist conversion. The initial distribution is a cross-section of the working-age population, scaled so that baseline annual course volume equals the NHS caseload. This implies 43.2 million people represented and one therapist per 3,923 people, or 25.5 therapists per 100,000. Writing $K = 3{,}923$ for that density and s for the fractional increase in supply, the per-person result converts as

$$\text{return per additional therapist} = \text{NPV}_{\text{person}} \times \frac{K}{s}. \tag{33}$$

The two scale factors collapse into the density ratio: multiplying by the population to obtain the system total, then dividing by the number of extra therapists. No separate population figure is therefore required.

Limitations. Four qualifications bear on how the figures should be read.

The therapist headcount is an assumption, not a measurement. The density K follows from an assumed annual caseload of 115 courses per therapist, and by (33) every per-therapist figure scales inversely with it. A published full-time-equivalent workforce statistic should replace it, and until it does the level of the per-therapist return is less firmly established than its ranking across scenarios, which is invariant to K.

Second, the discounting convention is conservative for a costing exercise. Discounting each individual's flows to their own age 25 is a lifetime-welfare convention. Much of the current working-age population is middle-aged, so gains arriving soon in calendar time are discounted by several decades. Discounting to the present instead raises the returns by a factor of between 1.1 and 1.6.

Third, the employment response rests on a dimension of the model that Section 5.4 flags as imperfectly matched. The estimated model returns the severely depressed to work somewhat too readily relative to the data, so the employment and fiscal channels here should be read as upper bounds on that margin. The mental-health effects are not subject to this concern.

Fourth, these are fiscal flows to the exchequer, not welfare. Because therapy is publicly funded, its provision cost does not enter household utility, so the welfare return to expansion exceeds the fiscal return reported here.

H.4 The Cost of Training a Therapist

The benefit–cost ratios in Table 7 divide the fiscal return by what it costs to produce a therapist. There is no single official figure for training a generic therapist, so we construct a transparent benchmark for the pathway that delivers most NHS Talking Therapies capacity: a High Intensity Cognitive Behavioural Therapy (HI-CBT) therapist entering through a mental-health nursing core profession. Other routes into the profession have different structures and different costs, among them clinical psychology, counselling, psychotherapy, occupational therapy and social work.

The pathway. The Nursing and Midwifery Council states that approved nursing courses take a minimum of three years (Nursing and Midwifery Council, 2024). HI-CBT programmes look for at least two years of post-qualification clinical experience (King's College London, 2026), and the specialist training itself is a one-year postgraduate diploma for which NHS England funds tuition on a regionally agreed basis together with salary support at starting Band 6 plus on-costs (NHS England, 2024a). The BABCP accreditation requirement of at least four years of relevant training across core-professional and CBT-specific components (British Association for Behavioural and Cognitive Psychotherapies, 2026) is consistent with, but does not mandate, this six-year timeline.

Core-profession training. Jones et al. (2022) estimate a total investment of £64,619 in training a nurse, an economic cost estimate, not a tuition figure. Rebasing on the ONS all-items CPI, whose annual 2022 index is 121.7 against 142.5 in June 2026 (Office for National Statistics, 2026a), gives $\$64{,}619 \times 142.5/121.7 = \$75{,}663$ in June 2026 prices. The source reports a total for the training period without a schedule by academic year, so we adopt a midpoint timing convention: the three-year investment is treated as occurring on average halfway through those three years, giving an average carrying period of 4.5 years to full qualification at the end of year six. At $r = 4.6$ percent,

$$FV_{\text{core}} = \$75{,}663 \times (1.046)^{4.5} = \$92{,}635.$$

Treating the whole core cost as occurring at the end of year three instead gives £86,592, and at the very start of year one £99,100.

The specialist year. The 2026/27 starting Band 6 salary is £39,959 (NHS Employers, 2026). Employer National Insurance is 15 percent of pay above a secondary threshold of approximately £5,000 (HM Revenue and Customs, 2026), or £5,244, and the total public pension cost is 23.78 percent of salary, or £9,502, made up of 14.38 percent remitted by the employer and a 9.4 percent central payment (NHS Business Services Authority, 2026). Total employment resource cost is therefore £54,705, to which a current PgDip fee benchmark of £7,000 (Teesside University, 2026) is added, giving £61,705 gross.

Counting all of that salary as training cost overstates the economic cost, because an HI-CBT trainee spends three days a week delivering clinical care and two at university (Central and North West London NHS Foundation Trust, 2026). Allocating two-fifths of the employment resource cost to training time gives

$$\$7{,}000 + 0.40 \times \$54{,}705 \ = \ \$28{,}882.$$

The 40 percent allocation follows the published two-day/three-day split, and is an analytical assumption with no standing as an NHS tariff.

Result. Adding the two components gives £121,517 at full qualification, which we round to £122,000. The timing bounds on the core component give £115,474 and £127,982, and we carry £128,000 as the conservative economic bound. Counting the whole of the final-year salary instead of the service-adjusted share gives £154,341 to £160,805; we carry £161,000 as a gross public-resource upper bound. We do not prefer that measure, since three days of each trainee week produce patient care, but the benefit–cost ratios in Table 7 exceed one even against it.

The scope of the calculation is worth stating. It capitalises spending to the date of qualification and does not value the therapist's subsequent career. The two post-qualification experience years are treated as productive employment, so their salary is excluded from training investment; only the opportunity cost of the earlier training capital continues to accrue over them.

H.5 Recovering Genetic Risk from Observables

Section 6.3 asks whether an individual's observed history is a sufficient statistic for the genetic risk on which the priority rule conditions. Table 34 reports the estimates behind the answer given there.

The predictor is a histogram gradient-boosting classifier trained on all observables available by the stated age: the number of observed years and the entry age; the employment rate, current employment status and number of employment transitions; therapy application and receipt rates and whether ever treated; shares of time in each mental-health state, the current state and the number of transitions into severe depression; the share of time in poor physical health and the current state; and the mean, most recent value and standard deviation of log earnings. The target is the binary depression polygenic-index group, whose population share is 0.47 to 0.48 across ages. All figures are five-fold cross-validated, with each individual contributing one observation, so the folds partition people and no individual appears in both training and test data.

Table 34: Recovering Genetic Risk from Observables: Simulated Panel

Age	AUC	(s.d. across folds)	Average precision
30	0.974	(0.013)	0.968
35	0.993	(0.003)	0.992
40	0.994	(0.002)	0.992
45	0.993	(0.003)	0.990
50	0.994	(0.001)	0.993
55	0.994	(0.002)	0.991
60	0.992	(0.001)	0.992

Notes: Simulated panel of 10,000 agents. A histogram gradient-boosting classifier predicts the binary depression polygenic-index group from the observables available by the stated age, listed in the text. All figures are five-fold cross-validated; each individual contributes one observation, so the folds partition people and no individual appears in both training and test data. The standard deviation is taken across folds. An individual enters the sample once observed at the evaluation age with at least three prior observations, so the number of individuals rises with age, from 862 at 30 to 9,198 at 60. The population share of the target group is 0.47 to 0.48 across ages, so a classifier with no information would score an AUC of 0.5 and an average precision of about 0.48.

An individual enters the sample once observed at the evaluation age with at least three prior observations, so the number of individuals rises with age, from 862 at age 30 to 9,198 at age 60; the wider dispersion across folds at age 30 reflects that smaller sample. As Section 6.3 notes, the exercise speaks to the model and not to the data: genetic risk enters the simulation through the wage equation and the mental-health transition process alone, so a long history identifies it far more sharply than it could in a population where genetic risk competes with everything the model leaves out.